

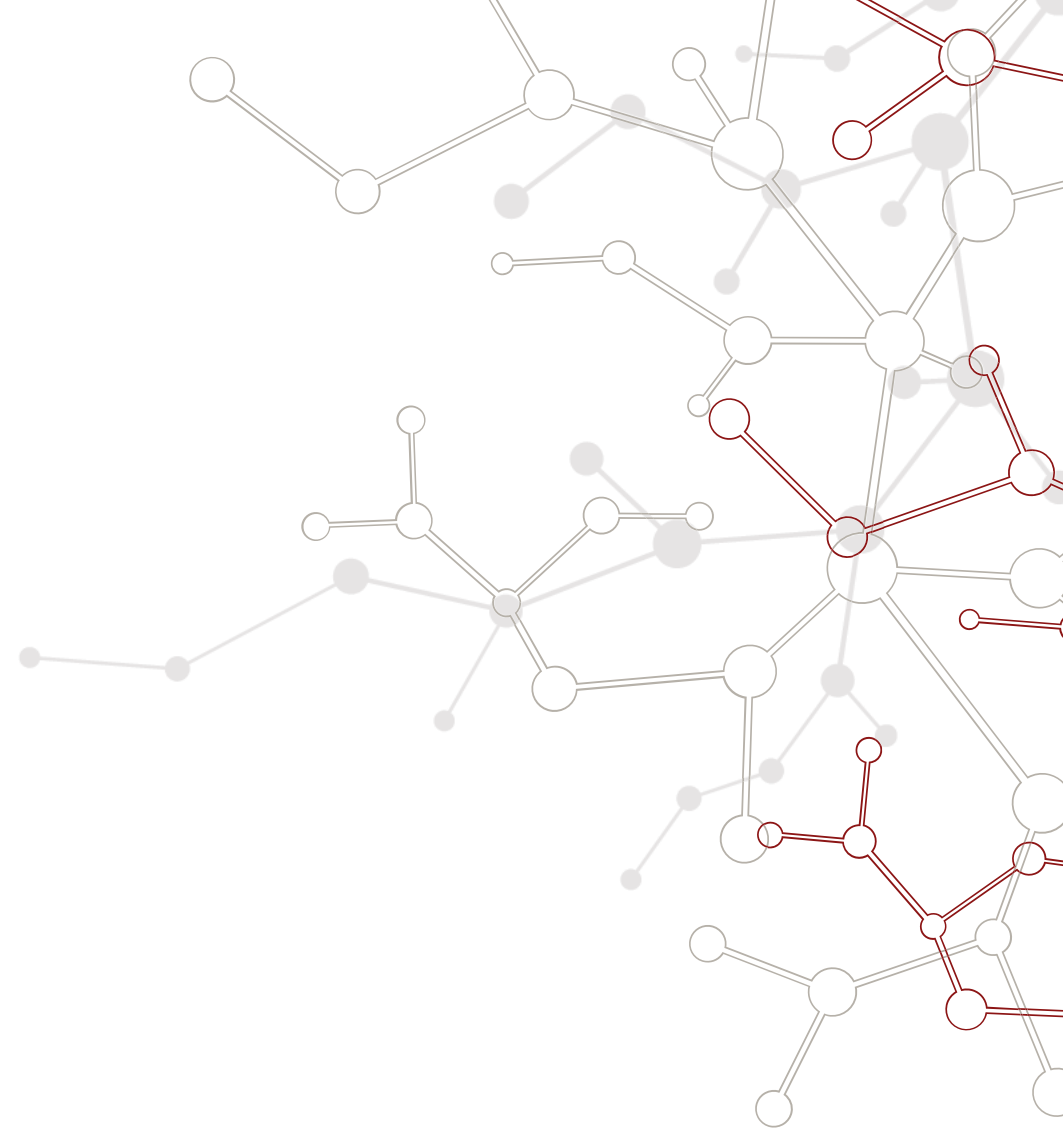
Free-electron lasers

River Robles

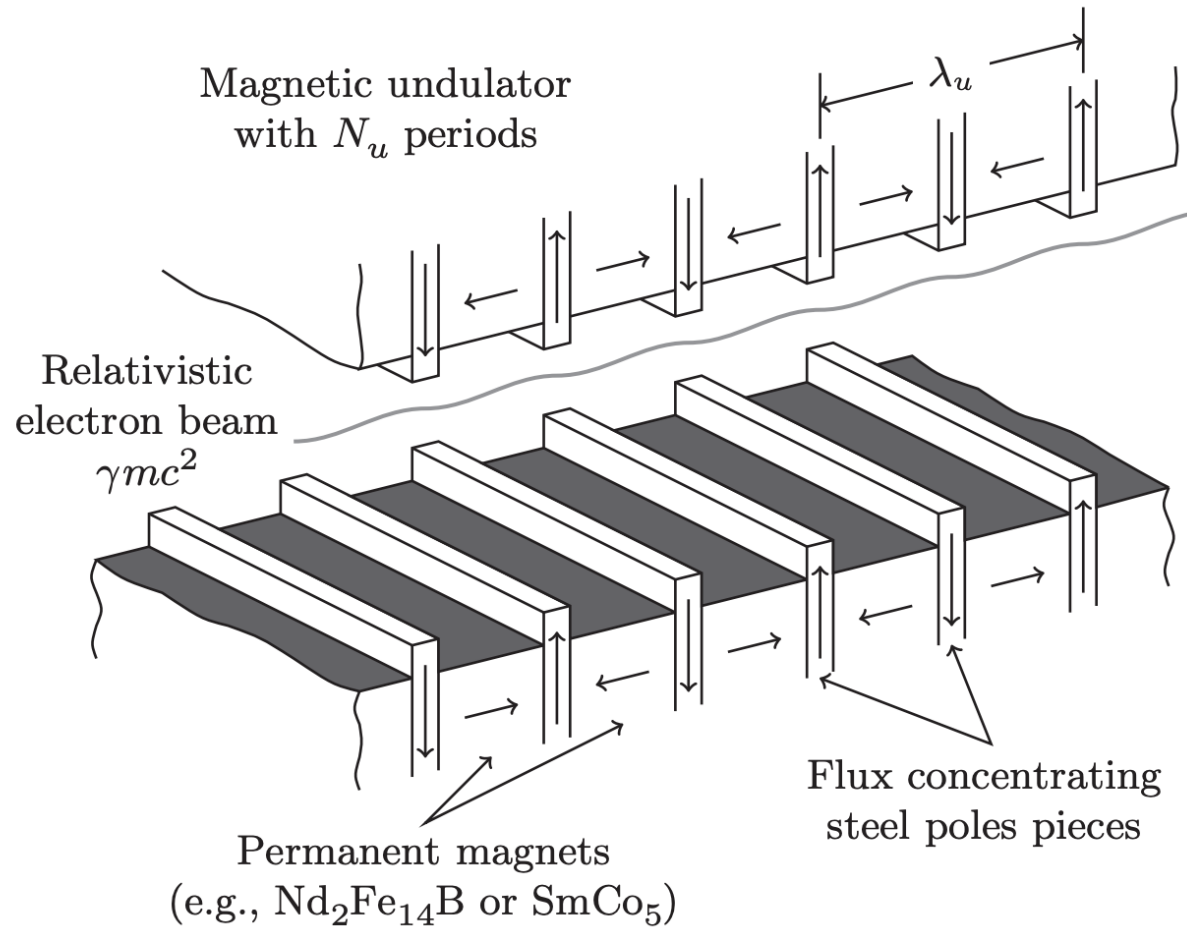
river.robles@fau.de

USPAS Accelerator Physics Guest Lecture

July 17th, 2026



Synchrotron radiation and undulators



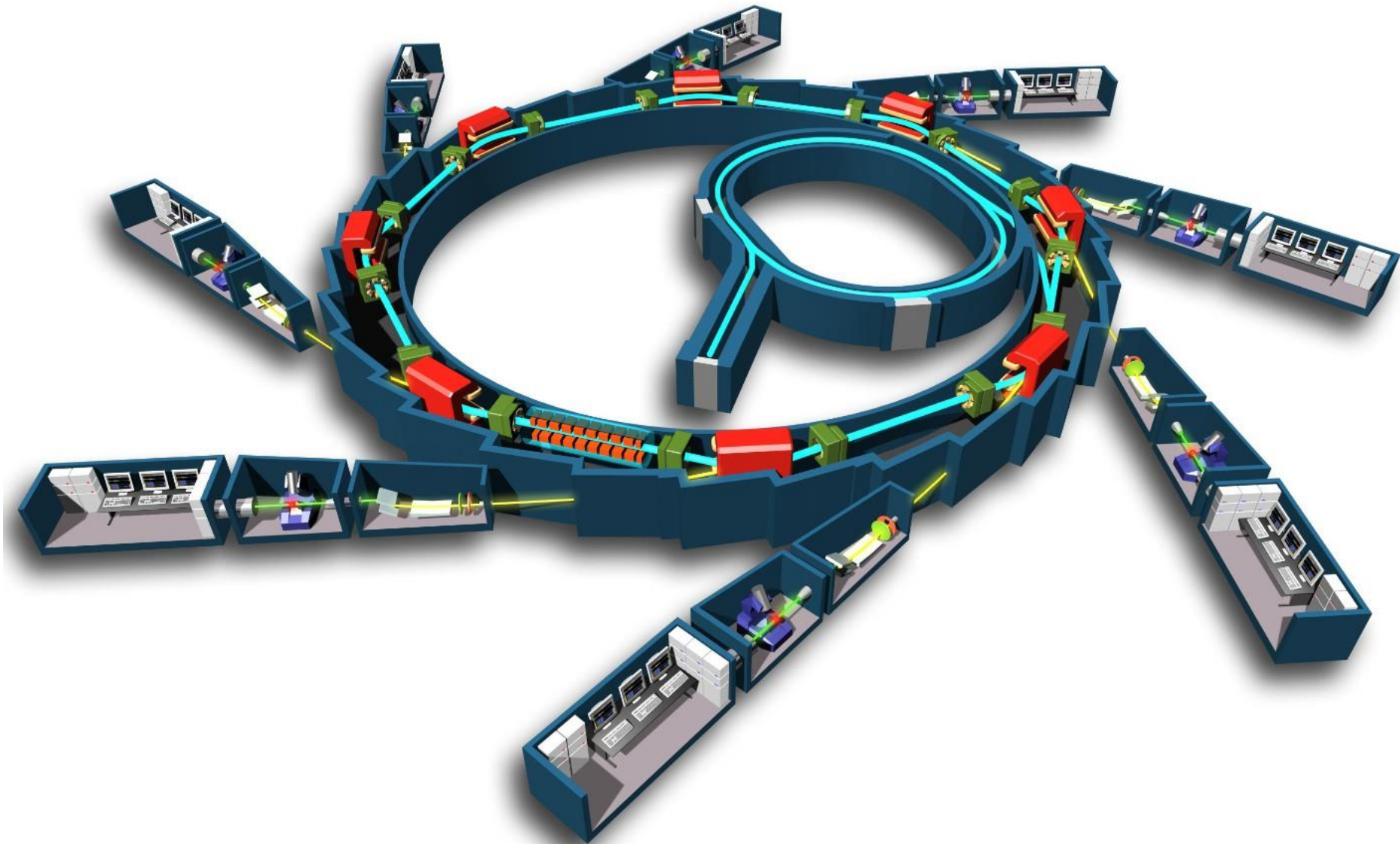
Planar* undulators generate sinusoidal magnetic fields

$$\vec{B} = -\hat{y}B_0 \sin(k_u z)$$

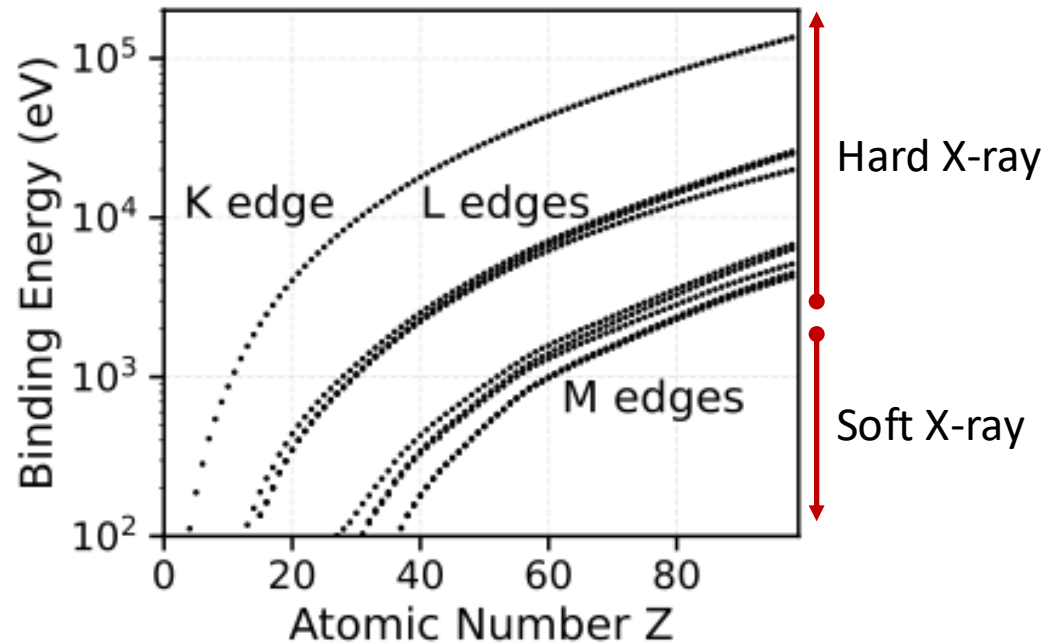
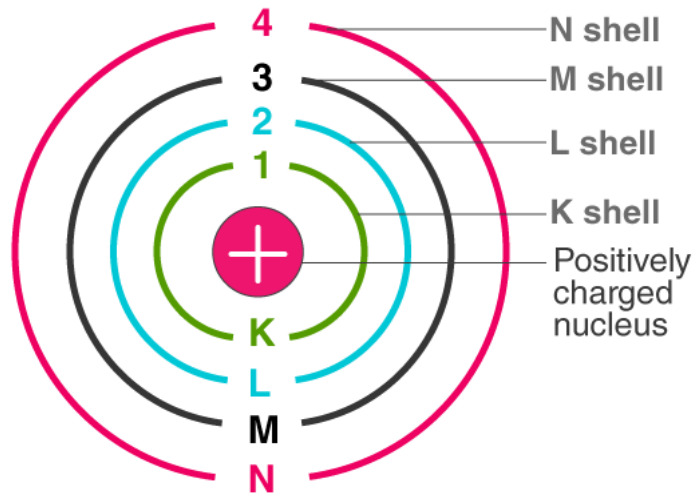
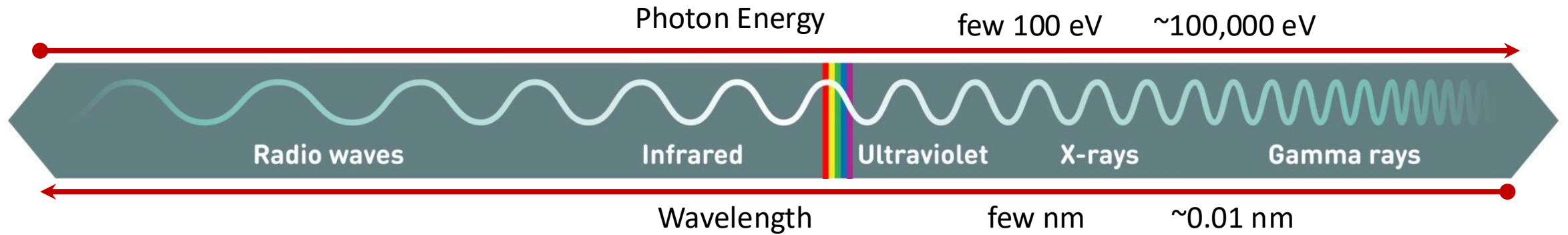
Electrons move on sinusoidal trajectories, emit radiation with typical wavelength:

$$\lambda_r = \frac{\lambda_u}{2\gamma^2} \left(1 + \frac{K^2}{2} \right)$$

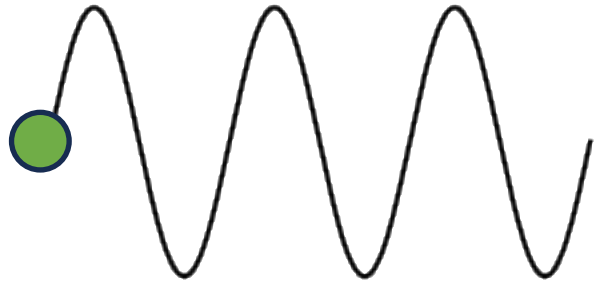
Synchrotrons as X-ray sources



X-rays



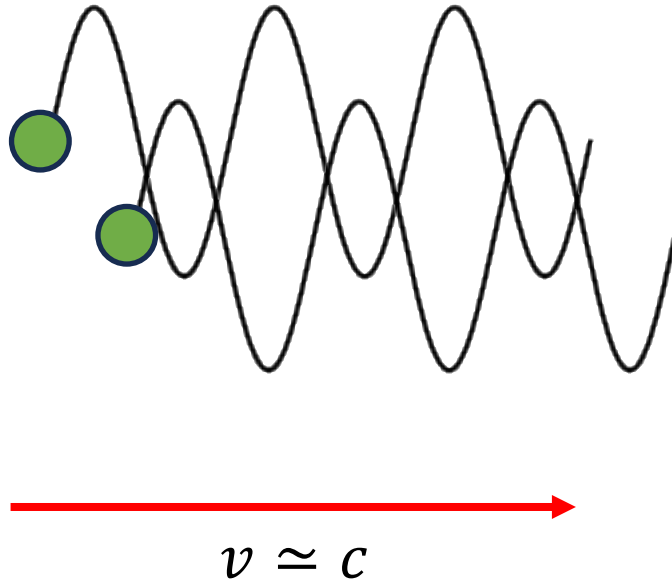
Radiation from multiple electrons



$$v \simeq c$$

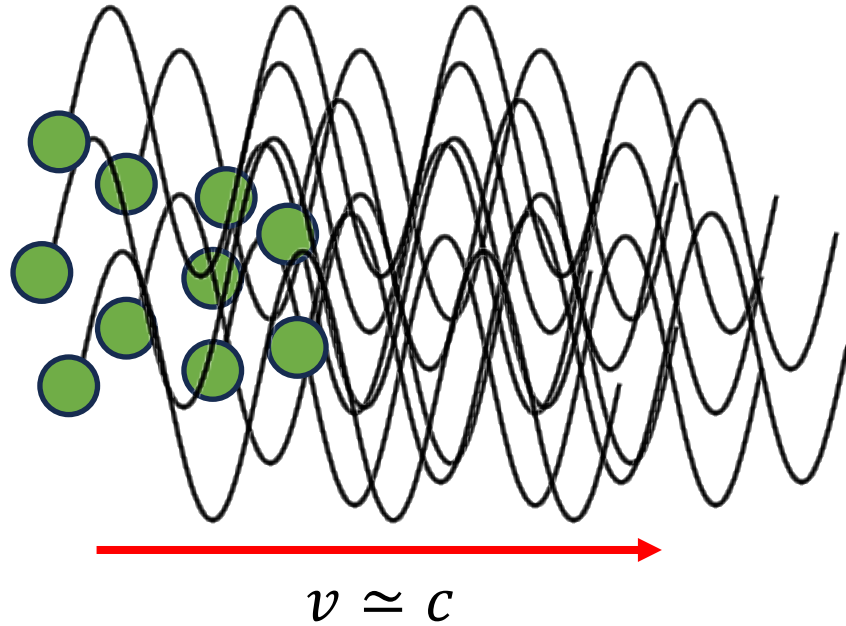
$$E_x(t) = E_0 \sin(\omega_X(t - t_1))$$

Radiation from multiple electrons



$$E_x(t) = E_0 \sin(\omega_X(t - t_1)) + E_0 \sin(\omega_X(t - t_2))$$

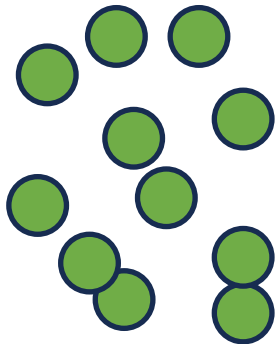
Radiation from multiple electrons



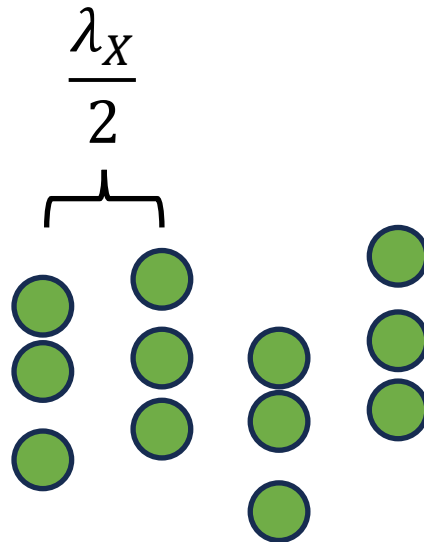
$$E_x(t) = E_0 \sum_{j=1} \sin(\omega_X(t - t_j))$$

Pop quiz: which beam do you want?

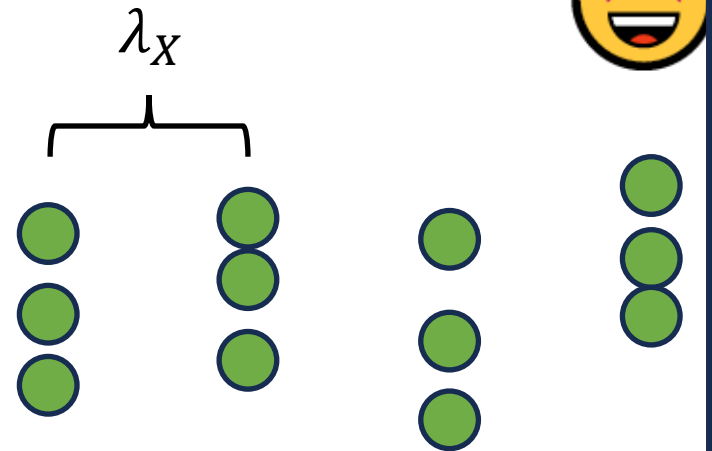
A



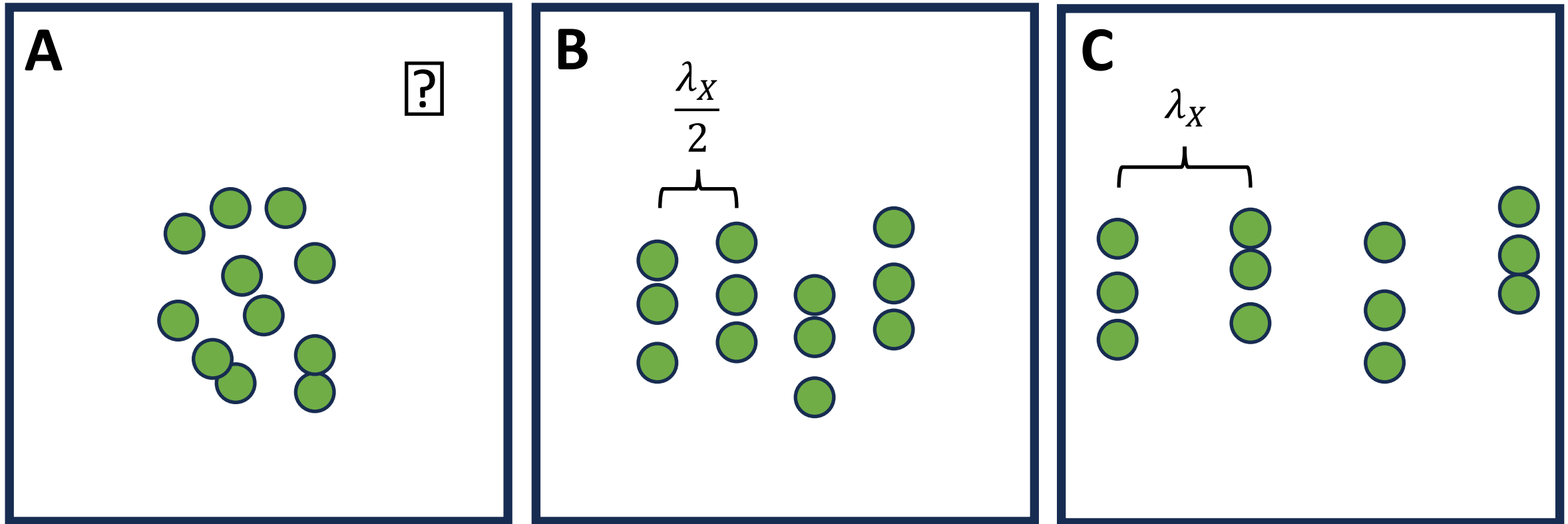
B



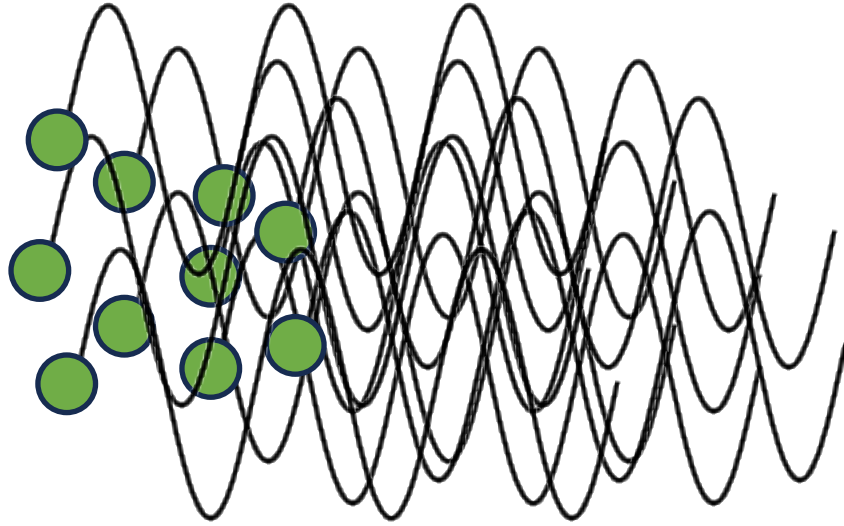
C



Pop quiz: which beam do you have?



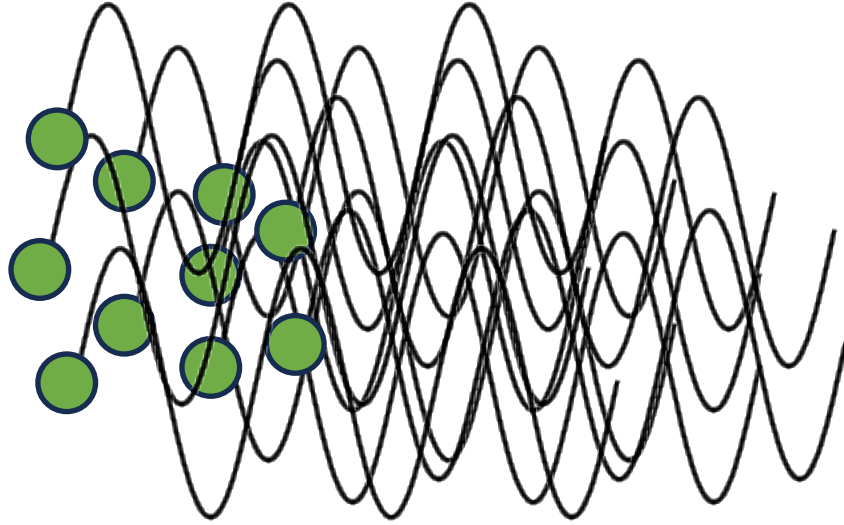
Typical synchrotron beams



$$E_x(t) = E_0 \sum_j \sin(\omega_X(t - t_j))$$

$$\langle E_x(t)^2 \rangle = E_0^2 \sum_{j,k} \left\langle \sin(\omega_X(t - t_j)) \sin(\omega_X(t - t_k)) \right\rangle = ?$$

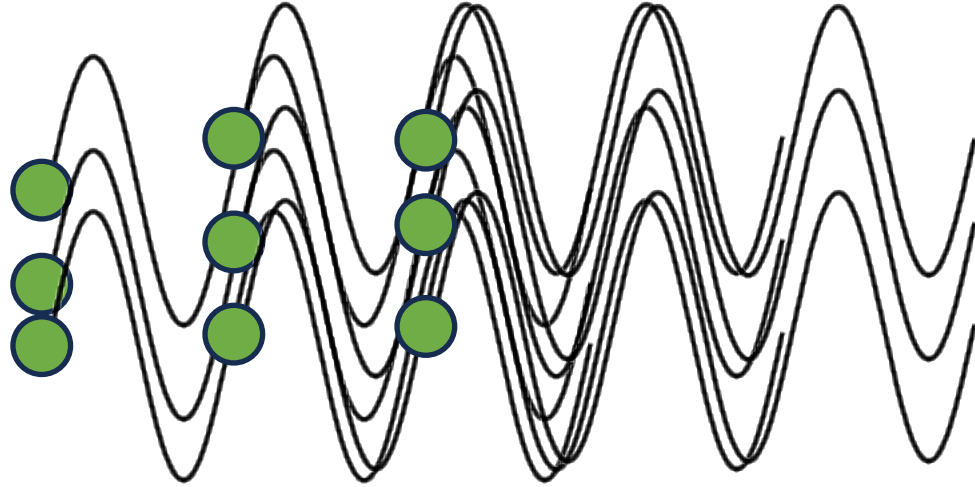
Typical synchrotron beams



$$E_x(t) = E_0 \sum_j \sin(\omega_X(t - t_j))$$

$$\begin{aligned} \langle E_x(t)^2 \rangle &= E_0^2 \sum_{j,k} \left\langle \sin(\omega_X(t - t_j)) \sin(\omega_X(t - t_k)) \right\rangle \\ &= E_0^2 \left[\sum_j \left\langle \sin^2(\omega_X(t - t_j)) \right\rangle + \sum_{j,k \neq j} \left\langle \sin(\omega_X(t - t_j)) \sin(\omega_X(t - t_k)) \right\rangle \right] \\ &= \frac{N_e E_0^2}{2} \end{aligned}$$

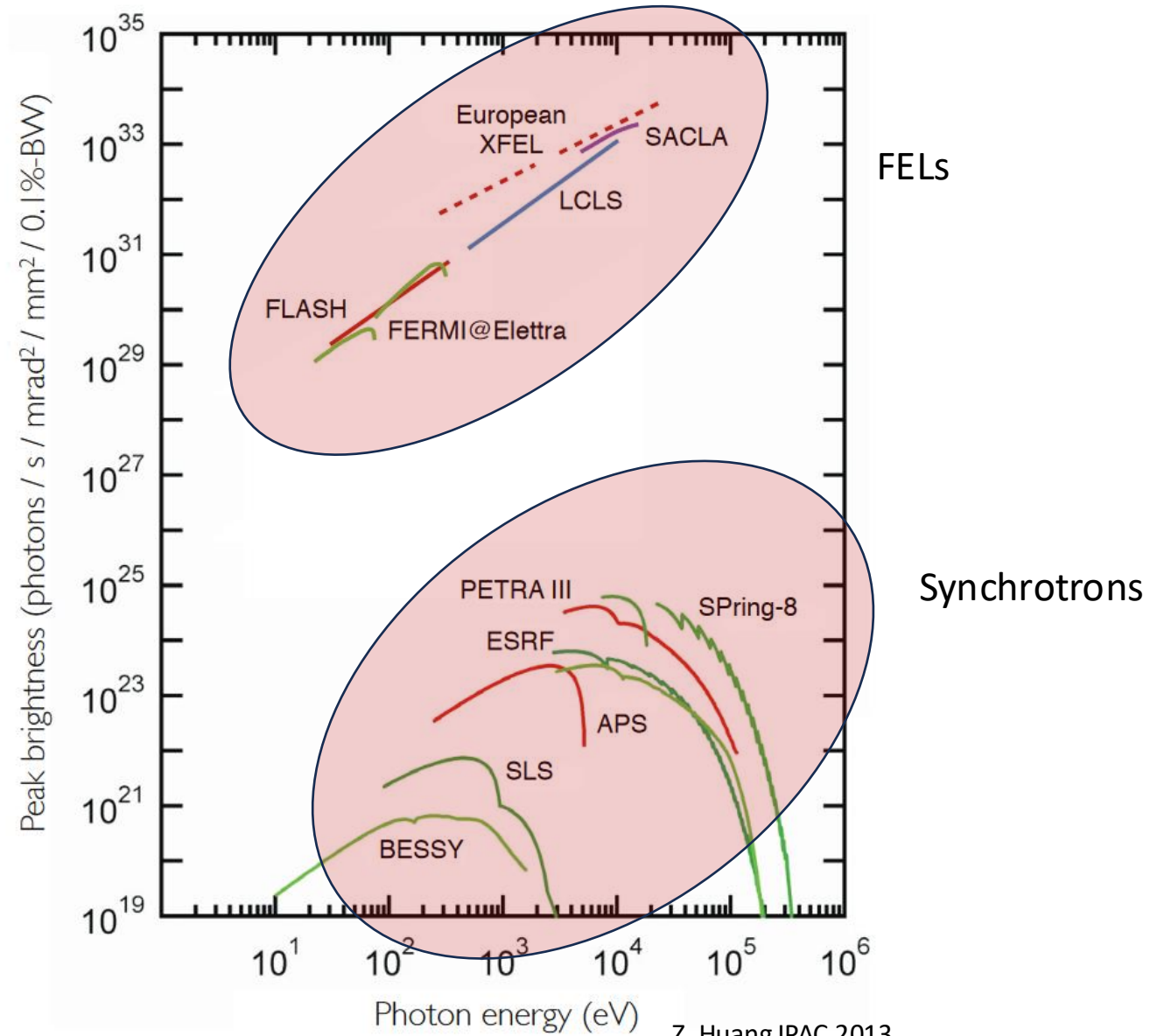
The ideal beam



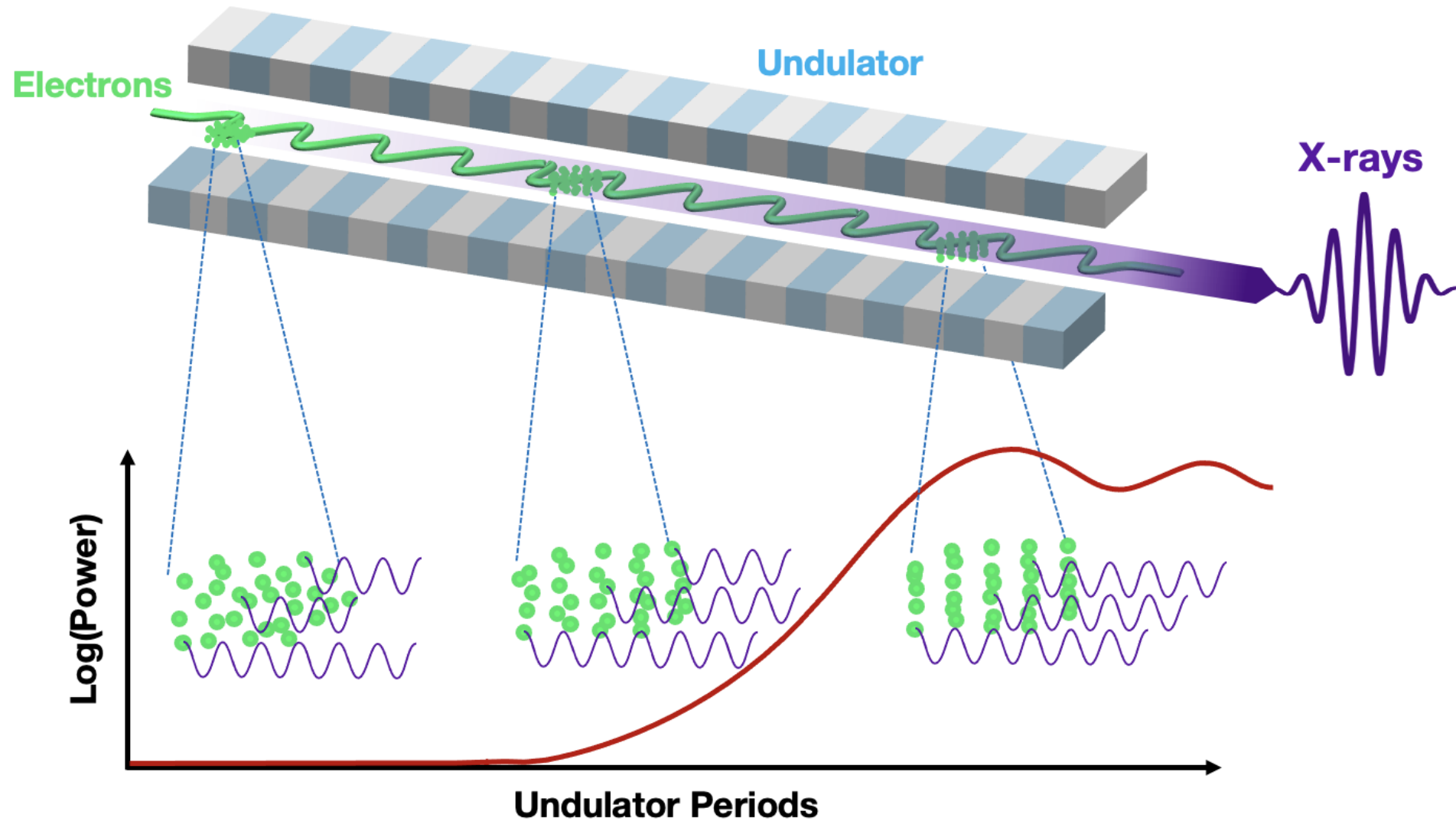
$$E_x(t) = E_0 \sum_j \sin(\omega_X(t - t_j))$$

$$\langle E_x(t)^2 \rangle = E_0^2 \sum_{j,k} \left\langle \sin(\omega_X(t - t_j)) \sin(\omega_X(t - t_k)) \right\rangle = \frac{N_e^2 E_0^2}{2}$$

Enter the FEL

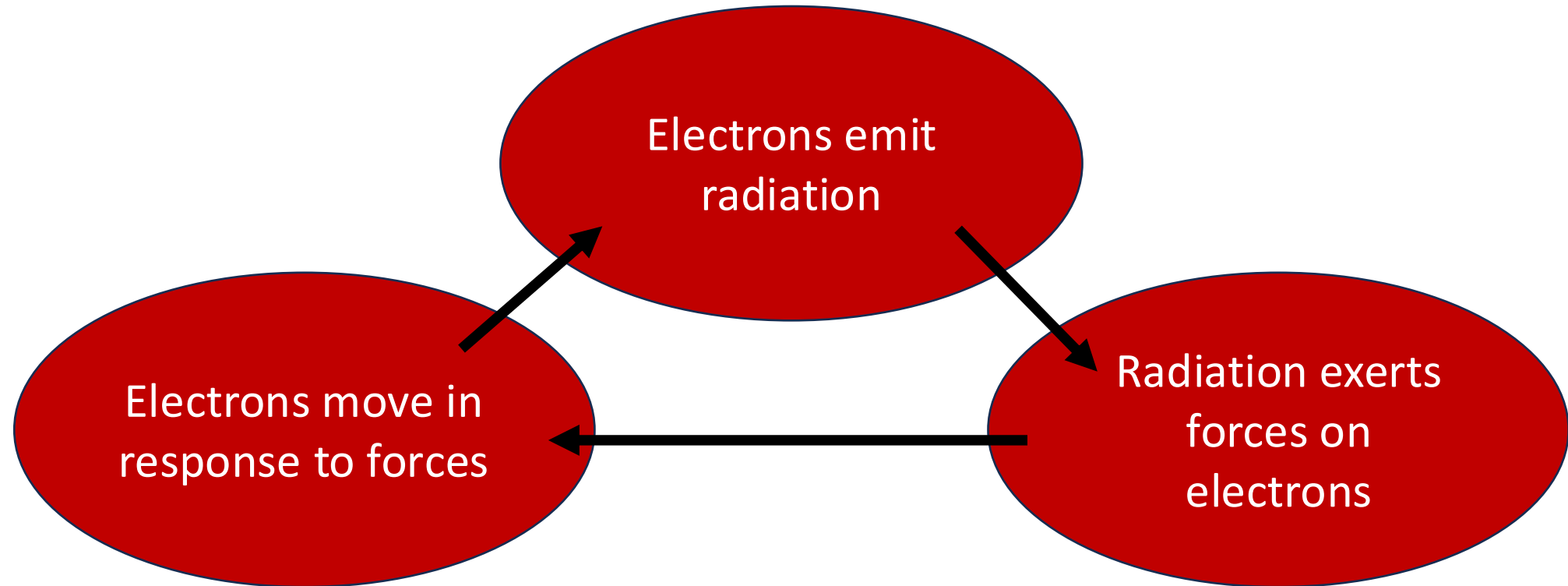


How does it work? (at a glance)



How does it work? (at a glance)

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \vec{A} = -\mu_0 \vec{J}$$



$$\frac{d\vec{x}}{dt} = \frac{\vec{p}}{\gamma m}$$

$$\frac{d\vec{p}}{dt} = \vec{F}$$

Outline

First half: FELs in theory

- Intro to FEL theory
- General FEL behavior
- Seeding and SASE
- Tapering
- 3D effects

Second half: FELs in practice

- XFELs around the world
- Modern XFEL R&D
- University-based FEL research
- Plasma-based FELs

FELs in Theory

Resources:

The primary textbook: Kim, Kwang-Je, Zhirong Huang, and Ryan Lindberg. *Synchrotron radiation and free-electron lasers*. Cambridge university press, 2017.

My derivations are detailed in my PhD thesis

Transverse motion in a planar undulator

$$\vec{B} = -\hat{y}B_0 \sin(k_u z)$$

$$\begin{aligned}\frac{dp_x}{dt} &= -e(\vec{v} \times \vec{B})_x \\ &= -eB_0 v_z \sin(k_u z)\end{aligned}$$

$$\rightarrow p_x(z) = Kmc \cos(k_u z)$$

$$K = \frac{eB_0}{k_u mc}$$

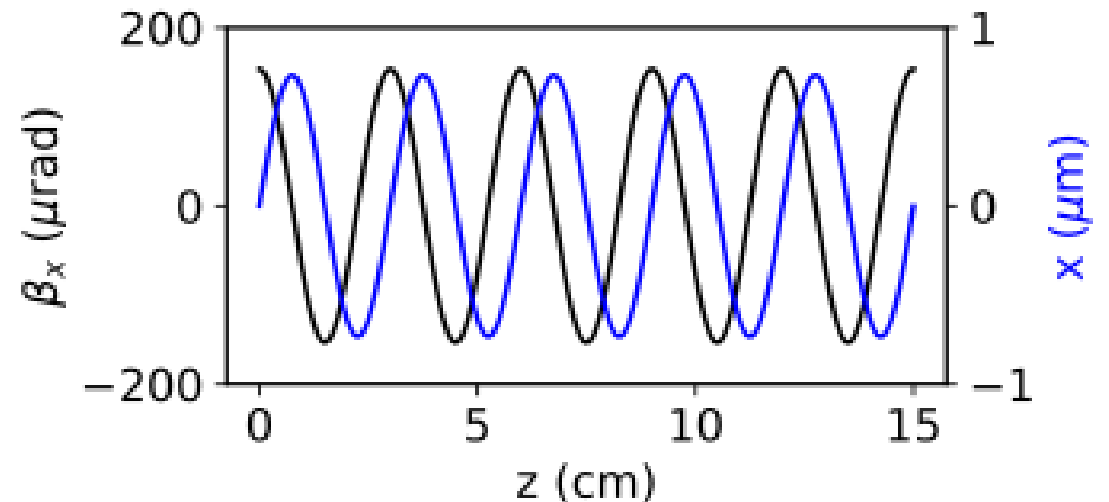
Typical values:

- $B_0 \sim 1$ Tesla
- $\lambda_u \sim$ few centimeters

$\longrightarrow K \sim 1 - 10$

$$\begin{aligned}\beta_x(z) &= \frac{K}{\gamma} \cos(k_u z) \\ x(z) &= \frac{K}{\gamma k_u} \sin(k_u z)\end{aligned}$$

$K = 3$, period = 3.0 cm, energy = 10 GeV



Longitudinal motion in a planar undulator

$$\beta_x(z) = \frac{K}{\gamma} \cos(k_u z)$$

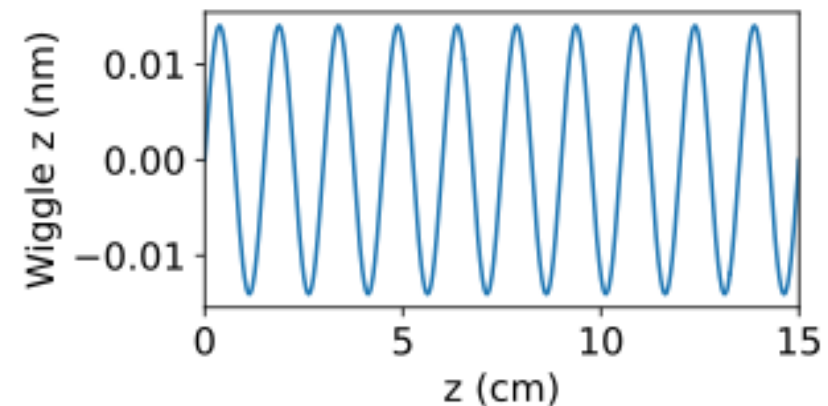
$$x(z) = \frac{K}{\gamma k_u} \sin(k_u z)$$

$$z_j(t) \simeq z_j(0) + c\bar{\beta}_z t$$

$$\bar{\beta}_z = 1 - \frac{1 + \frac{K^2}{2}}{2\gamma^2}$$

$$\begin{aligned} \beta_z &= \sqrt{1 - \frac{1}{\gamma^2} - \beta_x^2(z)} \\ &\simeq 1 - \frac{1 + K^2 \cos(k_u z)}{2\gamma^2} \\ &\simeq 1 - \frac{\left(1 + \frac{K^2}{2}\right)}{2\gamma^2} - \frac{K^2}{4\gamma^2} \cos(2k_u z) \end{aligned}$$

K= 3, period = 3.0 cm, energy = 10 GeV

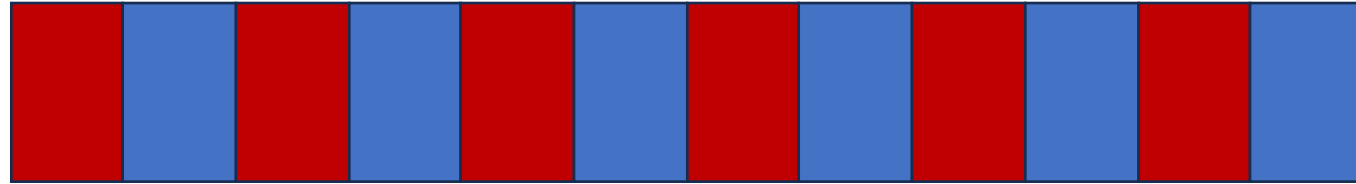


$$\lambda_r = 0.215 \text{ nm}$$

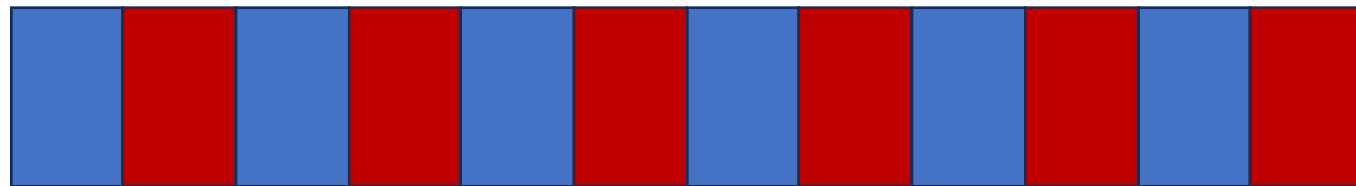
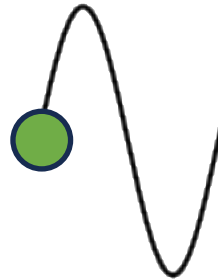
Slippage, phase matching, and the resonance condition

$$\beta_z \simeq 1 - \frac{\left(1 + \frac{K^2}{2}\right)}{2\gamma^2} < 1$$

In the time it takes the electron to travel λ_u , the radiation travels $\frac{\lambda_u}{2\gamma^2} \left(1 + \frac{K^2}{2}\right)$



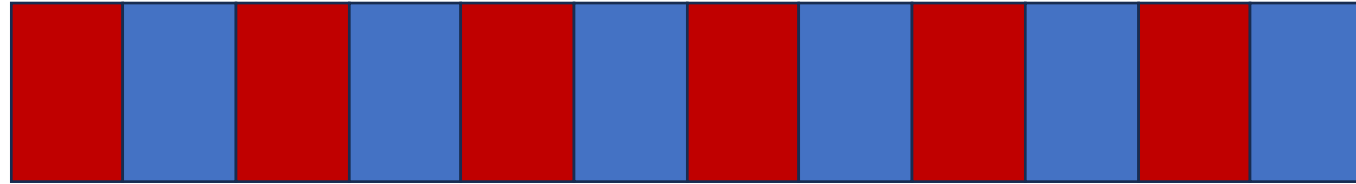
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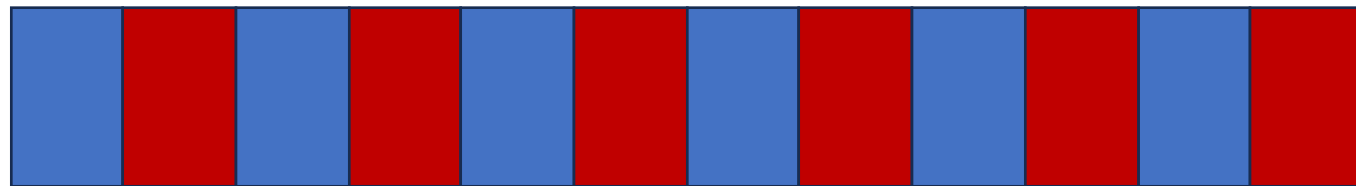
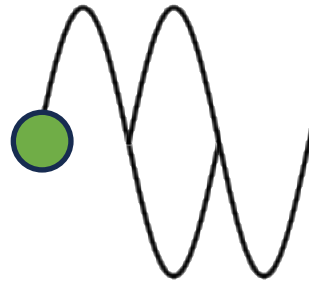
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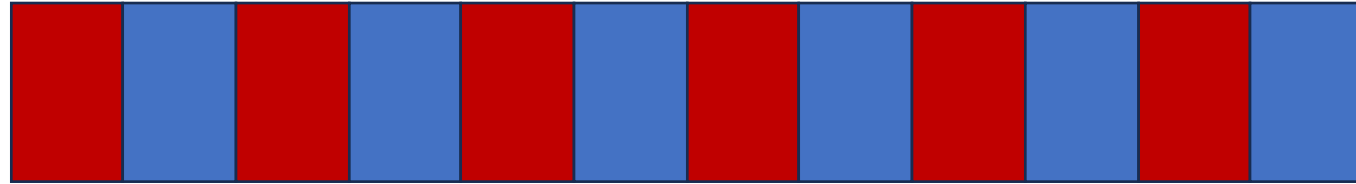
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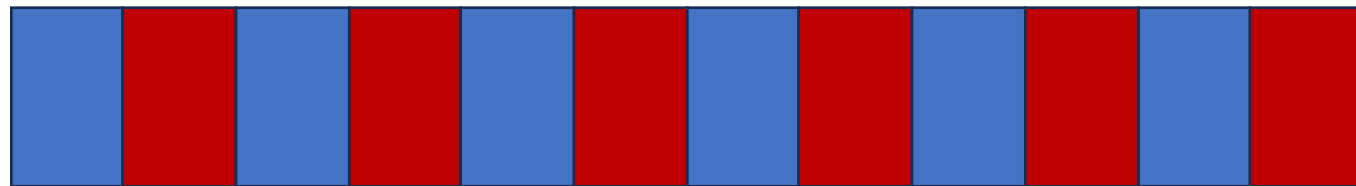
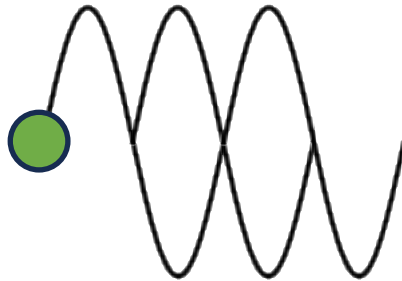
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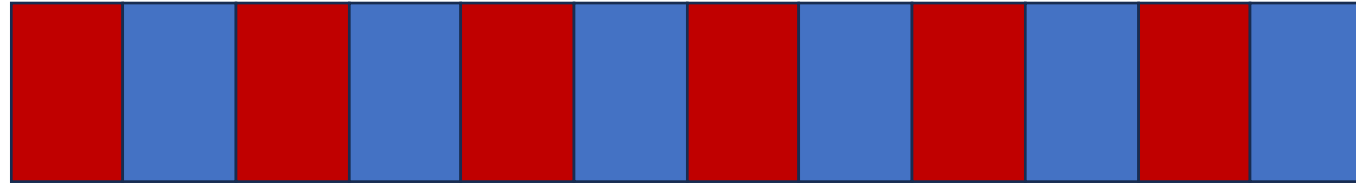
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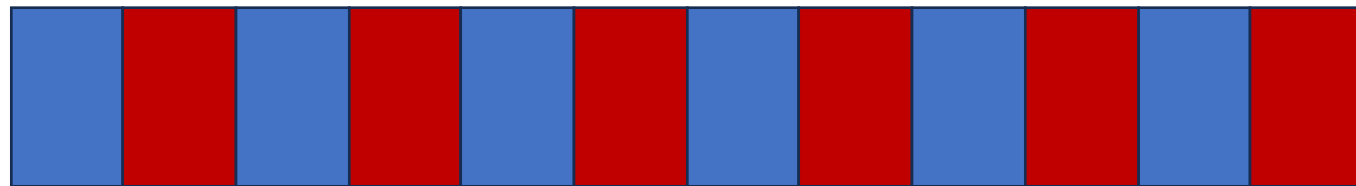
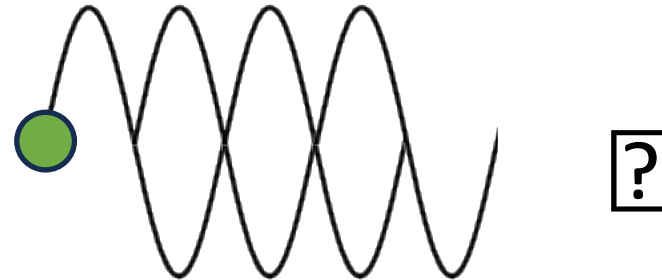
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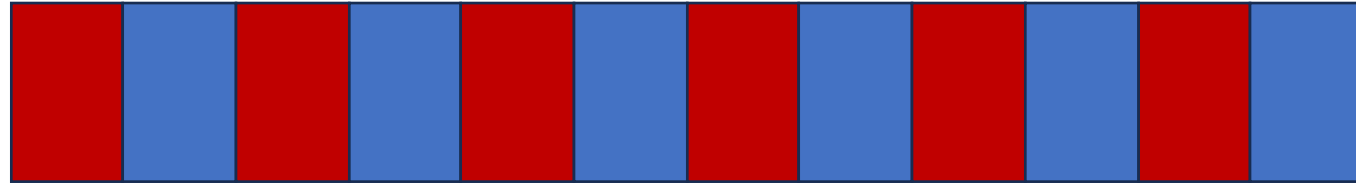
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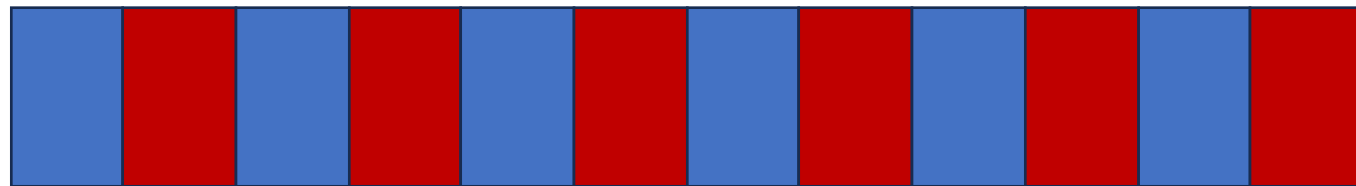
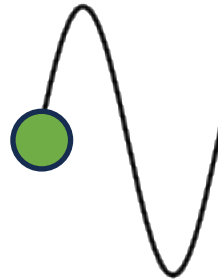
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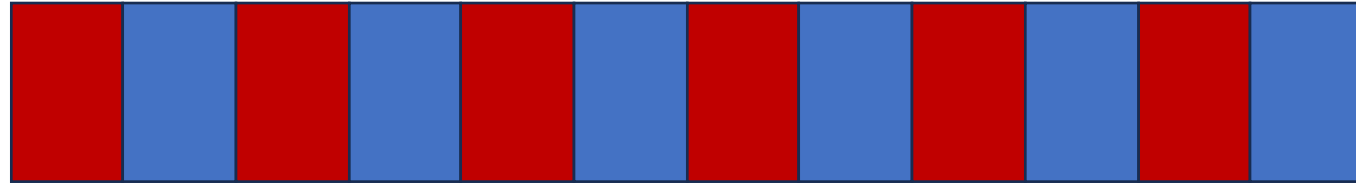
$$\frac{\lambda_u}{2\gamma^2} \left(1 + \frac{K^2}{2}\right) = \frac{\lambda_r}{4}$$



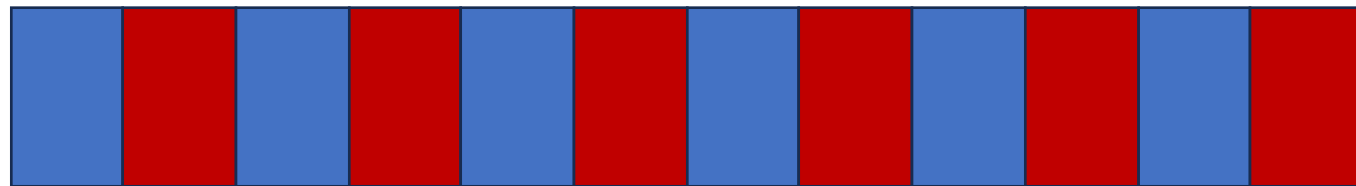
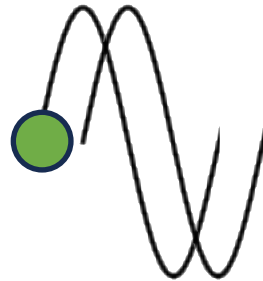
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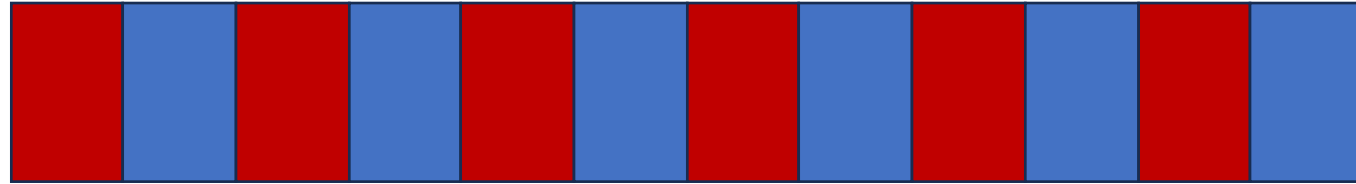
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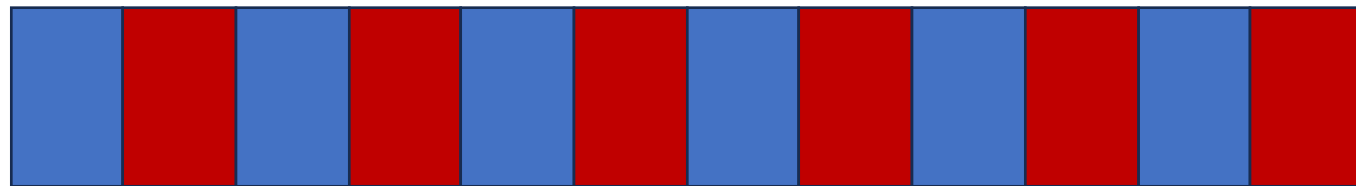
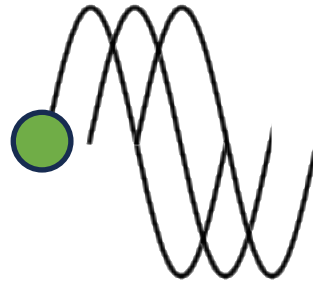
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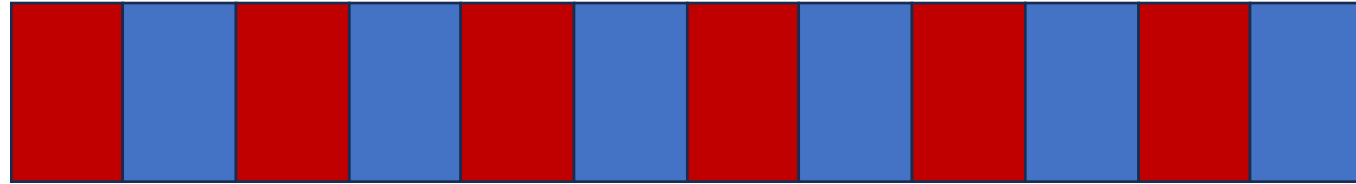
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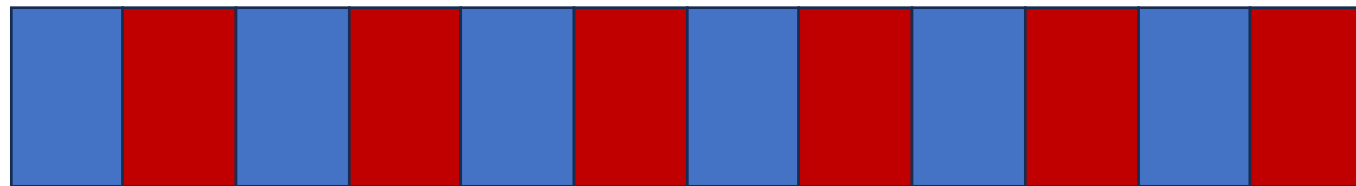
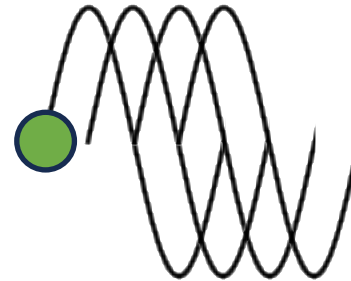
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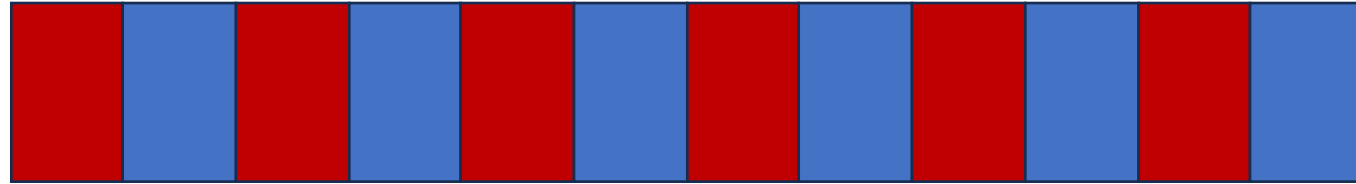
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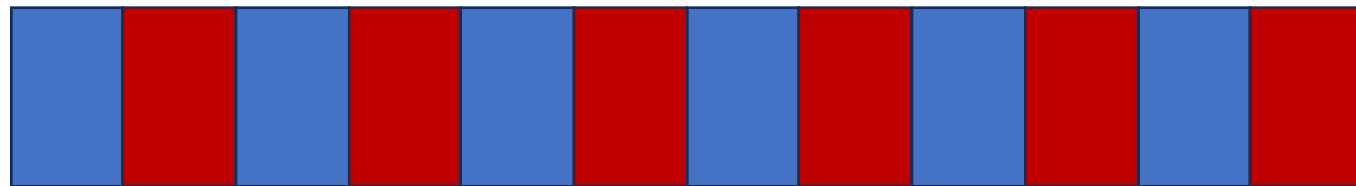
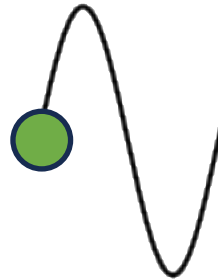
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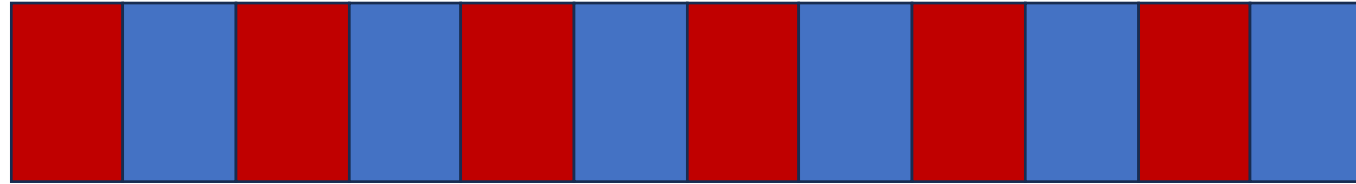
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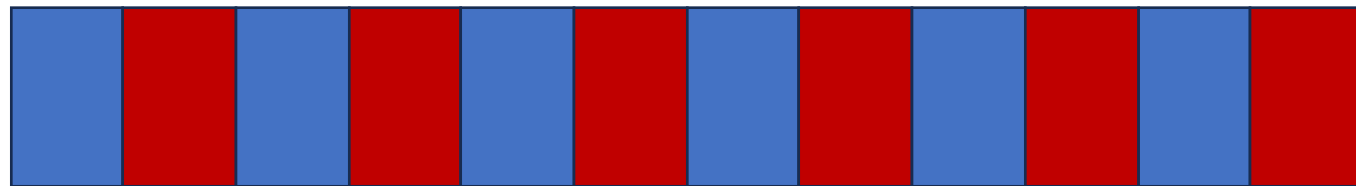
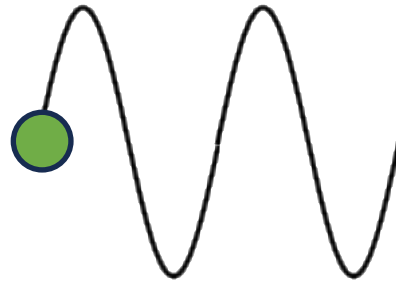
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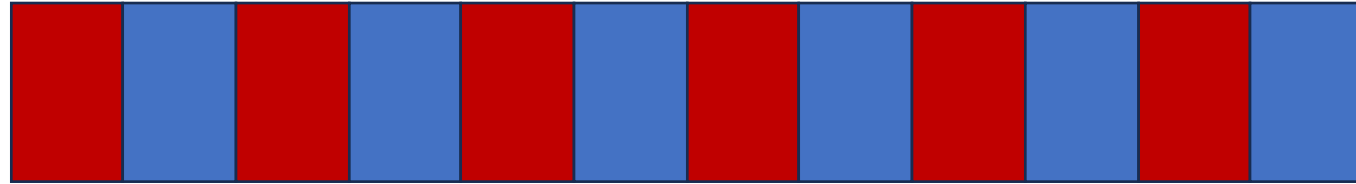
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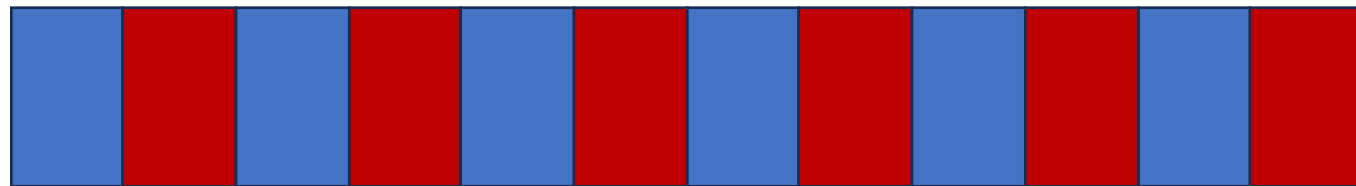
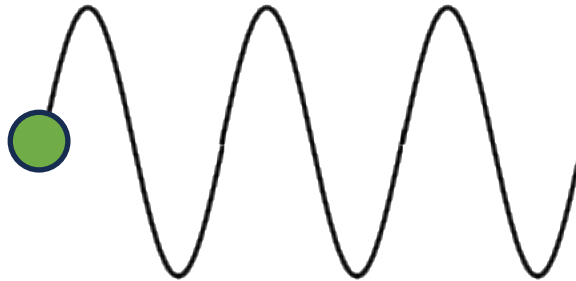
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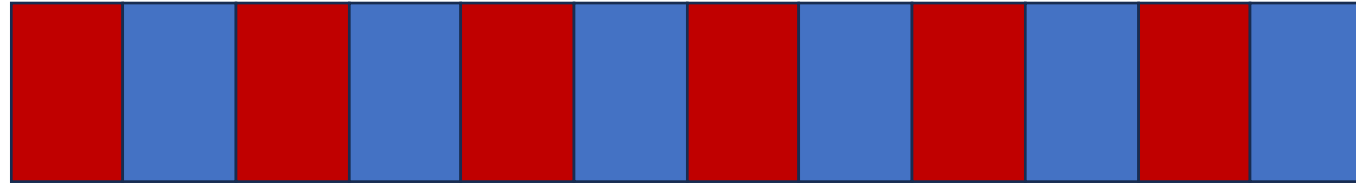
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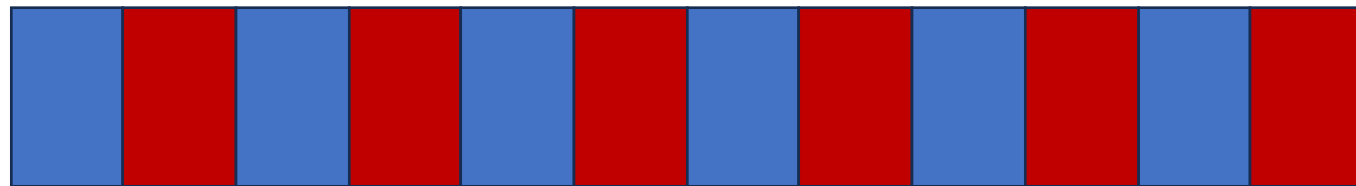
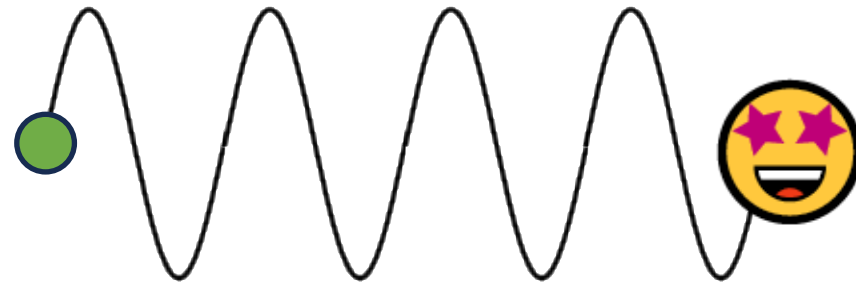
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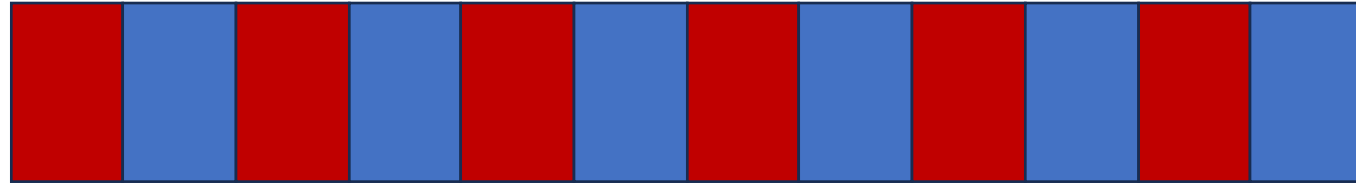
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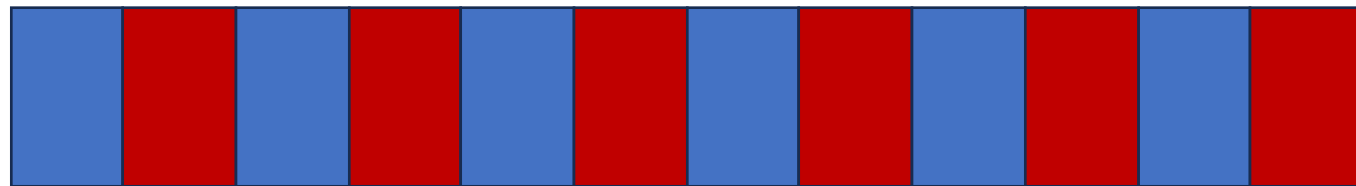
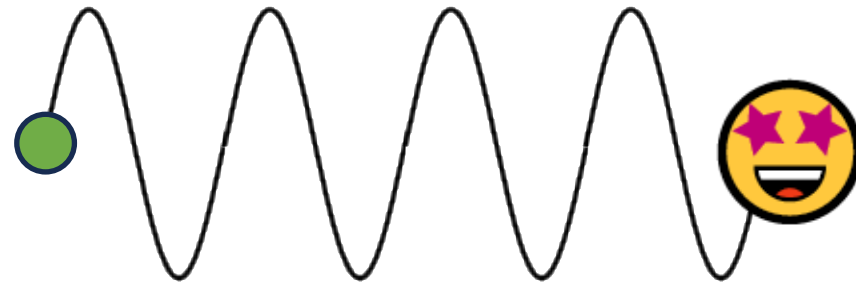
Slippage, phase matching, and the resonance condition

$$\beta_z \simeq 1 - \frac{\left(1 + \frac{K^2}{2}\right)}{2\gamma^2} < 1$$

In the time it takes the electron to travel λ_u , the radiation travels $\frac{\lambda_u}{2\gamma^2} \left(1 + \frac{K^2}{2}\right)$



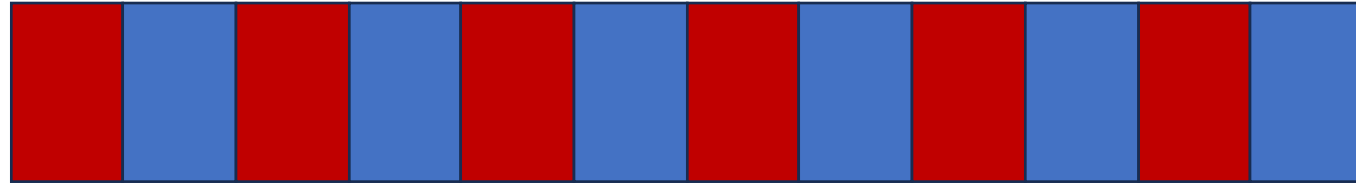
$$\frac{\lambda_u}{2\gamma^2} \left(1 + \frac{K^2}{2}\right) = ?$$



Slippage, phase matching, and the resonance condition

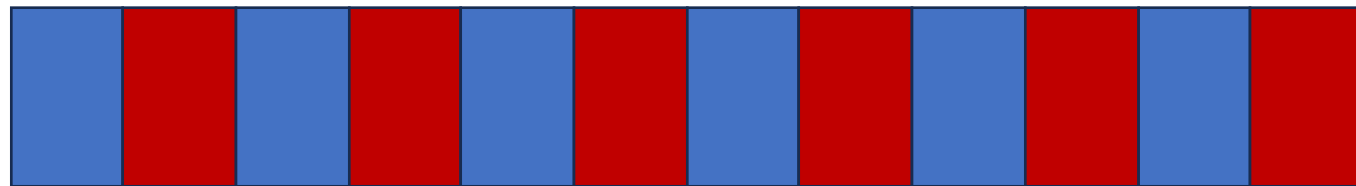
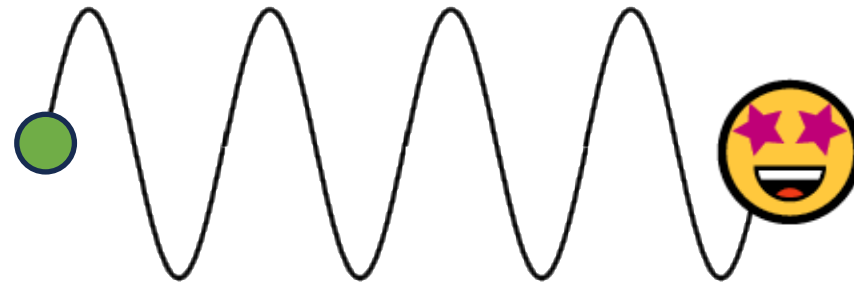
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In the time it takes the electron to travel λ_u , the radiation travels $\frac{\lambda_u}{2\gamma^2} \left(1 + \frac{K^2}{2}\right)$



$$\frac{\lambda_u}{2\gamma^2} \left(1 + \frac{K^2}{2}\right) = h\lambda_r$$

Only for h odd in a planar undulator



Energy exchange equations

$$\vec{E}(z, t) = \hat{x}E_0 \cos(k_r(z - ct) - \phi)$$

$$\begin{aligned} \frac{dU}{dt} &= -e\vec{E} \cdot \vec{v} = -eE_x(z(t), t)v_x(t) \\ &= -\frac{eE_0cK}{\gamma} \cos(k_u z(t)) \cos(k_r(z(t) - ct) - \phi) \\ &\simeq -\frac{eE_0cK}{2\gamma} \left[\underbrace{\cos((k_r + k_u)z(t) - k_r ct - \phi)}_{!} + \underbrace{\cos((k_r - k_u)z(t) + k_r ct + \phi)}_{\text{Hopeless, always } > 0} \right] \end{aligned}$$

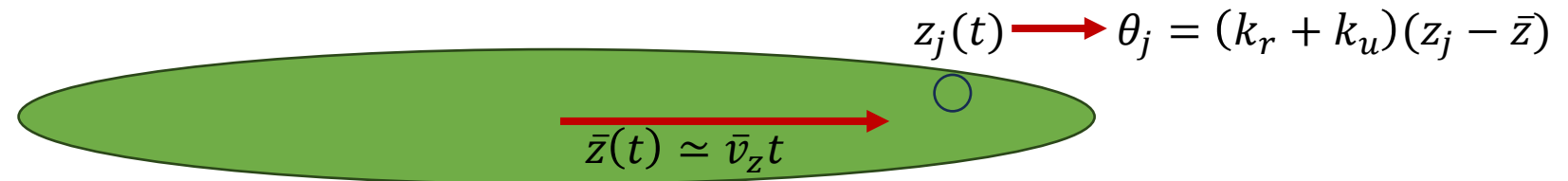
$$\lambda_r = \lambda_u \frac{1 - \bar{\beta}_z}{\bar{\beta}_z} \simeq \frac{\lambda_u}{2\gamma^2} \left(1 + \frac{K^2}{2} \right)$$

Energy exchange equations

$$\frac{dU}{dt} \simeq -\frac{eE_0 c K}{2\gamma} \left[\cos((k_r + k_u)z(t) - k_r ct - \phi) + \cos((k_r - k_u)z(t) + k_r ct + \phi) \right]$$

Introducing the “ponderomotive phase” θ :

$$\theta = (k_u + k_r)z(t) - k_r ct = (k_u + k_r)(z(t) - \bar{v}_z t)$$



Energy exchange equations

$$\frac{dU}{dt} \simeq -\frac{eE_0cK}{2\gamma} [\cos((k_r + k_u)z(t) - k_rct - \phi) + \cos((k_r - k_u)z(t) + k_rct + \phi)]$$

Introducing the “ponderomotive phase” θ :

$$\theta = (k_u + k_r)z(t) - k_rct = (k_u + k_r)(z(t) - \bar{v}_z t)$$

$$\frac{dU}{dt} \simeq -\frac{eE_0cK}{2\gamma} [\cos(\theta - \phi) + \cos(\theta + 2k_rct + \phi)]$$

$$\frac{d\theta_j}{dt} \simeq (k_r + k_u)(v_{z,j} - \bar{v}_z) \simeq 2k_u \frac{(\gamma_j - \bar{\gamma})}{\bar{\gamma}}$$

Equations of motion

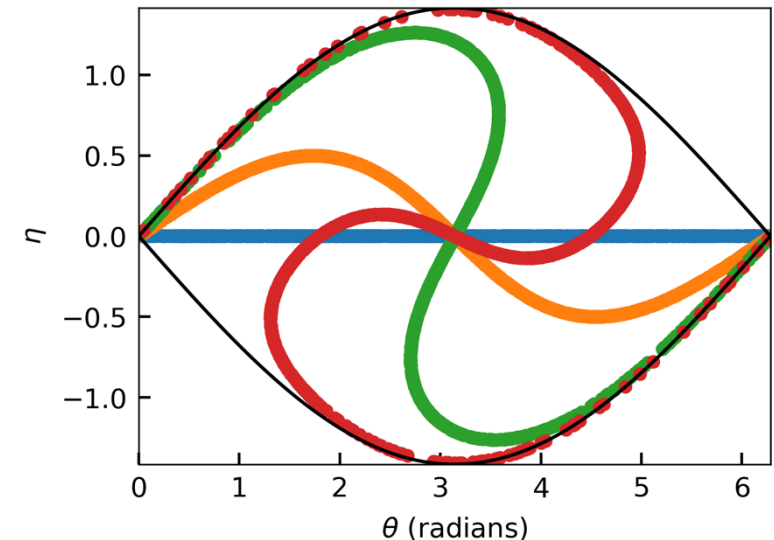
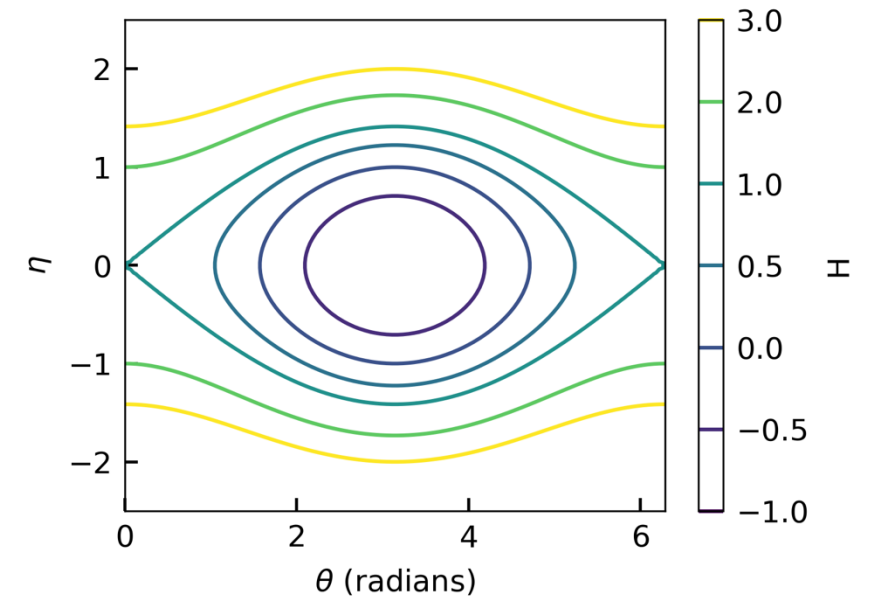
A few housekeeping things:

- Move from t to z as an independent variable
- Average over the fast oscillation in $\bar{\beta}_z$
- Move from total energy U to relative energy $\eta = \frac{U - U_0}{U_0}$

$$\frac{d\eta_j}{dz} = -\frac{eK[JJ]}{2\gamma_0^2 mc^2} E_0 \cos(\theta_j - \phi)$$

$$\frac{d\theta_j}{dz} = 2k_u \eta_j$$

$$[JJ] = J_0 \left(\frac{K^2}{4 + 2K^2} \right) - J_1 \left(\frac{K^2}{4 + 2K^2} \right)$$



The FEL wave equation

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) A_x = -\mu_0 J_x$$

Two approximations

“1D approximation”: treat the system as 1D

Slowly-varying envelope approximation (SVEA):

$$A_x(x, y, z, t) \rightarrow A_x(z, t)$$

$$E_x(z, t) = \bar{E} e^{i(\omega_r t - k_r z)} + c.c.$$

$$J_x(z, t) = -\frac{e}{A_{tr}} \sum_j v_{x,j}(t) \delta(z - z_j(t))$$

$$\frac{\partial^2 \bar{E}_x}{\partial t^2} \ll \omega_r \frac{\partial \bar{E}_x}{\partial t} \quad \frac{\partial^2 \bar{E}_x}{\partial z^2} \ll k_r \frac{\partial \bar{E}_x}{\partial z}$$

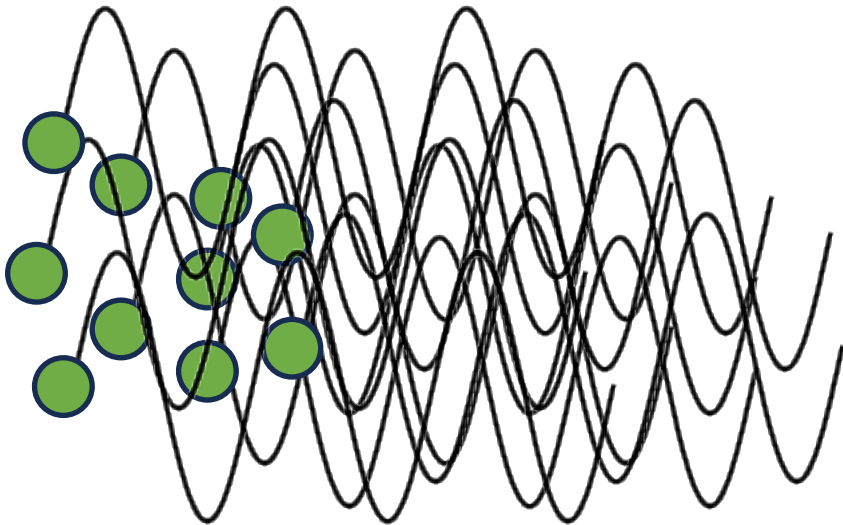
$$\left(\frac{\partial}{\partial z} + \frac{1}{c} \frac{\partial}{\partial t} \right) \bar{E} = \frac{en_e K[JJ]}{4\gamma_0 \epsilon_0} \langle e^{i\theta_j} \rangle$$

The wave equation and bunching

$$\left(\frac{\partial}{\partial z} + \frac{1}{c} \frac{\partial}{\partial t}\right) \bar{E} = \frac{en_e K[JJ]}{4\gamma_0 \epsilon_0} \langle e^{i\theta_j} \rangle$$

n_e : the beam number density

$$b \equiv \langle e^{i\theta_j} \rangle = \frac{1}{N_e} \sum_{j=1}^{N_e} e^{i\theta_j} \quad \text{is the “bunching”}$$

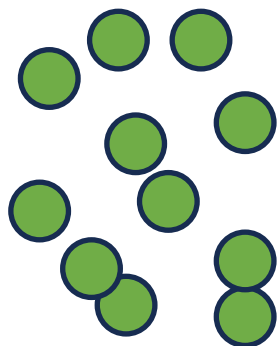


$$E_x(t) = E_0 \sum_j \sin(\omega_X(t - t_j)) \sim N_e E_0 b$$

+ something to do with the complex conjugate

More about bunching

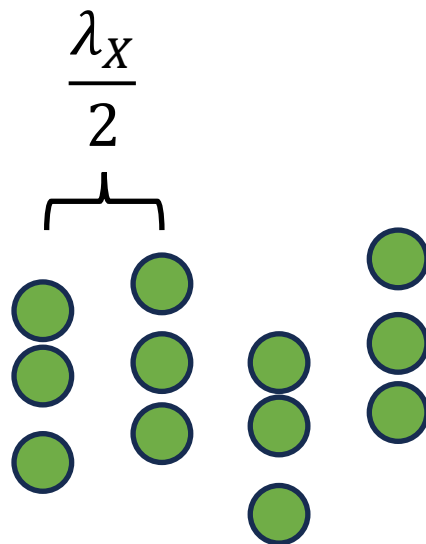
A



$$0 < |b| < 1$$

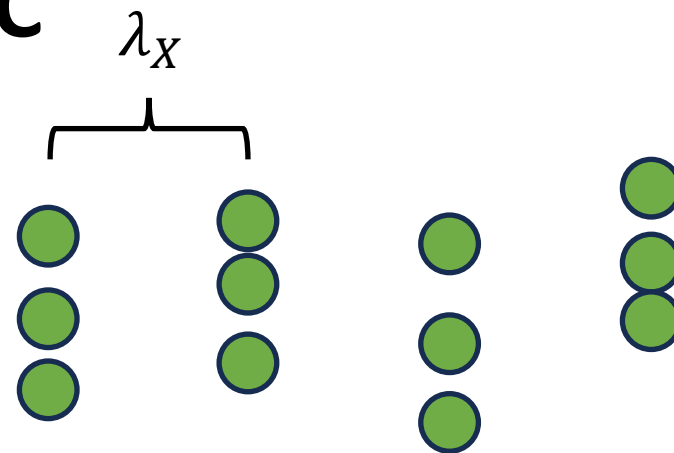
$$\text{Typically } b \sim \frac{1}{\sqrt{N_e}}$$

B



$$b = 0$$

C



$$b = 1$$

Scaling the FEL equations

$$\frac{d\eta_j}{dz} = -\frac{eK[JJ]}{2\gamma_0^2 mc^2} (\bar{E} e^{-i\theta_j} + c.c.)$$

$$\frac{d\theta_j}{dz} = 2k_u \eta_j$$

$$\left(\frac{\partial}{\partial z} + \frac{1}{c} \frac{\partial}{\partial t} \right) \bar{E} = \frac{en_e K[JJ]}{4\gamma_0 \epsilon_0} \langle e^{i\theta_j} \rangle$$

$$\hat{\eta}_j = \frac{\eta_j}{\rho}$$

$$\hat{z} = 2\rho k_u z$$

$$a = \frac{eK[JJ]}{4k_u \rho^2 \gamma_0^2 mc^2} \bar{E}$$

$$\delta = \frac{(\omega_r - \omega_{res})}{\rho \omega_{res}}$$

$$\rho = \left(\frac{e^2 n_e K^2 [JJ]^2}{32 k_u^2 \gamma_0^3 mc^2 \epsilon_0} \right)^{\frac{1}{3}}$$

“Pierce parameter”

Typically $10^{-3} - 10^{-4}$

$$\frac{d\hat{\eta}_j}{d\hat{z}} = -(a e^{-i\theta_j} + c.c.)$$

$$\frac{d\theta_j}{d\hat{z}} = \hat{\eta}_j$$

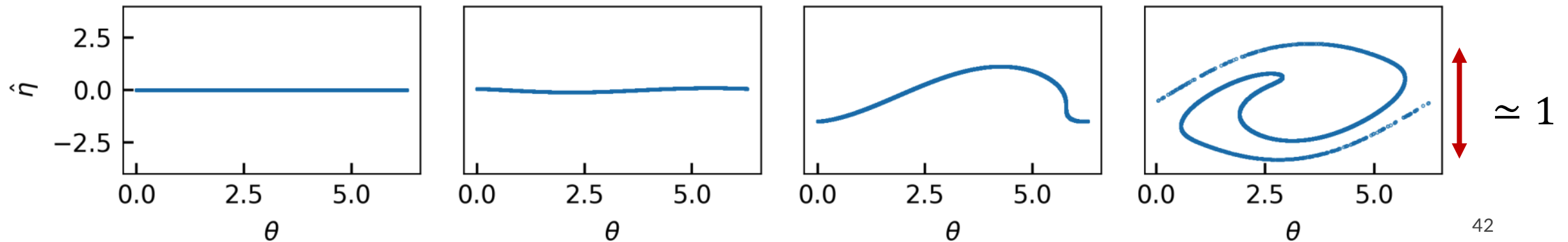
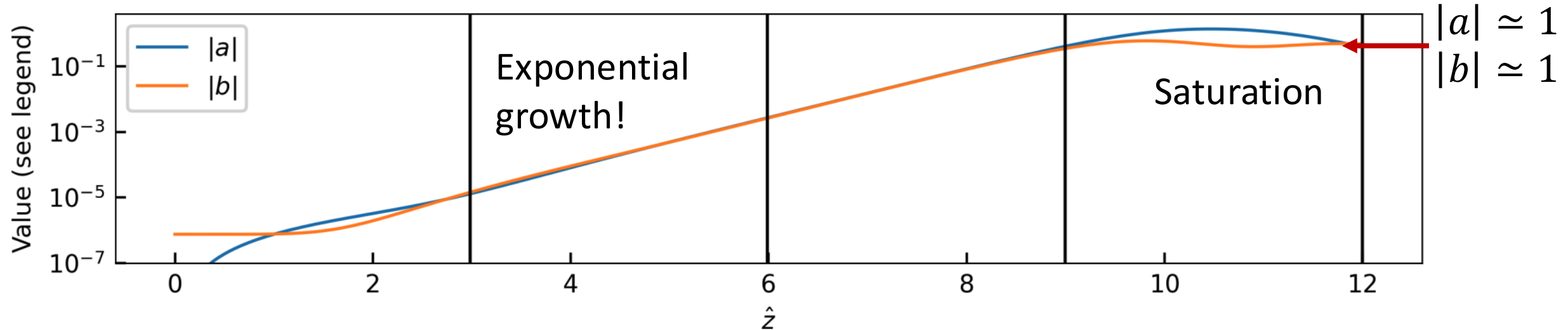
$$\left(\frac{\partial}{\partial \hat{z}} - i\delta \right) a = \langle e^{i\theta_j} \rangle$$

First look at FEL dynamics

$$\frac{d\hat{\eta}_j}{d\hat{z}} = -(ae^{-i\theta_j} + c.c.).$$

$$\frac{d\theta_j}{d\hat{z}} = \hat{\eta}_j$$

$$\left(\frac{\partial}{\partial \hat{z}} - i\delta\right)a = \langle e^{i\theta_j} \rangle$$

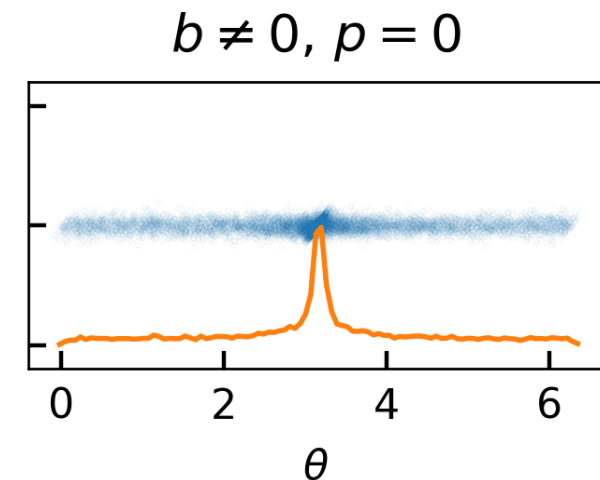
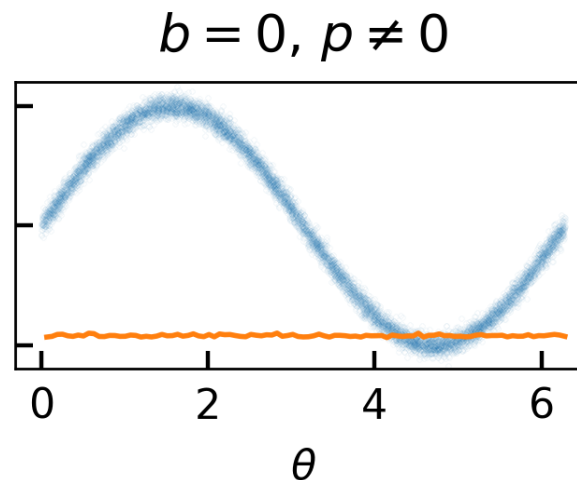
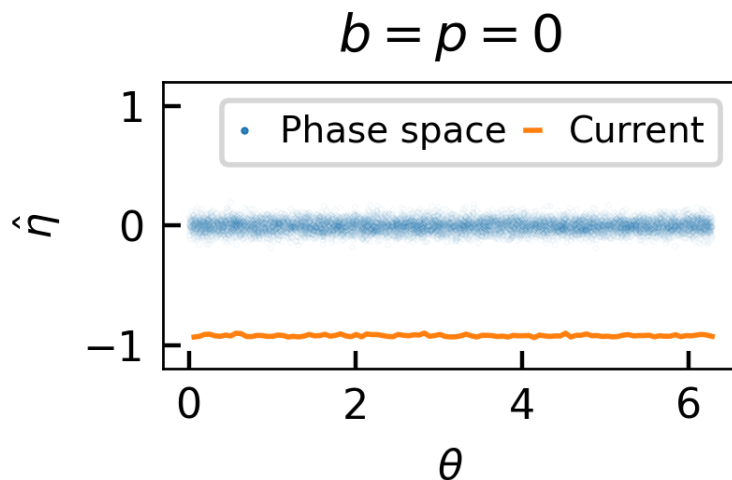


Solving the FEL equations with collective variables

$$\frac{d\hat{\eta}_j}{d\hat{z}} = -(ae^{-i\theta_j} + c.c.). \quad \frac{d\theta_j}{d\hat{z}} = \hat{\eta}_j \quad \left(\frac{\partial}{\partial \hat{z}} - i\delta\right)a = \langle e^{i\theta_j} \rangle$$

a only cares about the net bunching, not about any individual particle

$$\frac{db}{d\hat{z}} = \left\langle i \frac{d\theta_j}{d\hat{z}} e^{i\theta_j} \right\rangle = i \langle \hat{\eta}_j e^{i\theta_j} \rangle \equiv ip$$



Solving the FEL equations with collective variables

$$\frac{d\hat{\eta}_j}{d\hat{z}} = -(ae^{-i\theta_j} + c.c.). \quad \frac{d\theta_j}{d\hat{z}} = \hat{\eta}_j \quad \left(\frac{\partial}{\partial \hat{z}} - i\delta\right)a = \langle e^{i\theta_j} \rangle$$

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$$\frac{db}{d\hat{z}} = \left\langle i \frac{d\theta_j}{d\hat{z}} e^{i\theta_j} \right\rangle = i \langle \hat{\eta}_j e^{i\theta_j} \rangle \equiv ip$$

$$\frac{dp}{dz} = \left\langle \frac{d\hat{\eta}_j}{d\hat{z}} e^{i\theta_j} + i\hat{\eta}_j \frac{d\theta_j}{d\hat{z}} e^{i\theta_j} \right\rangle = [-a - \cancel{a^* \langle e^{2i\theta_j} \rangle} + \cancel{i \langle \hat{\eta}_j^2 e^{i\theta_j} \rangle}] \simeq -a$$

Nonlinear terms, important for saturation

Collective variables and exponential gain

$$\frac{db}{d\hat{z}} = ip \quad \frac{dp}{d\hat{z}} = -a \quad \left(\frac{\partial}{\partial \hat{z}} - i\delta \right) a = b$$

$$a, b, p \propto e^{i\mu\hat{z}}$$

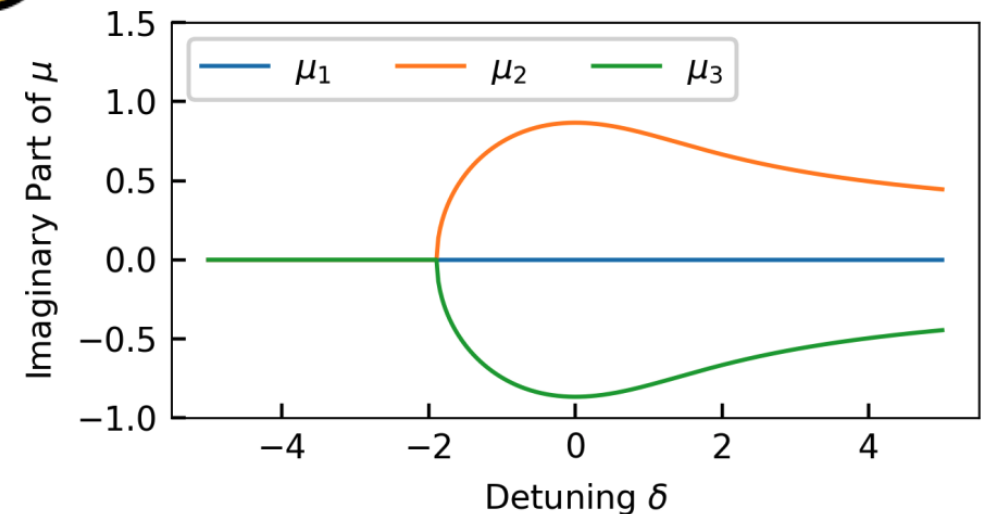
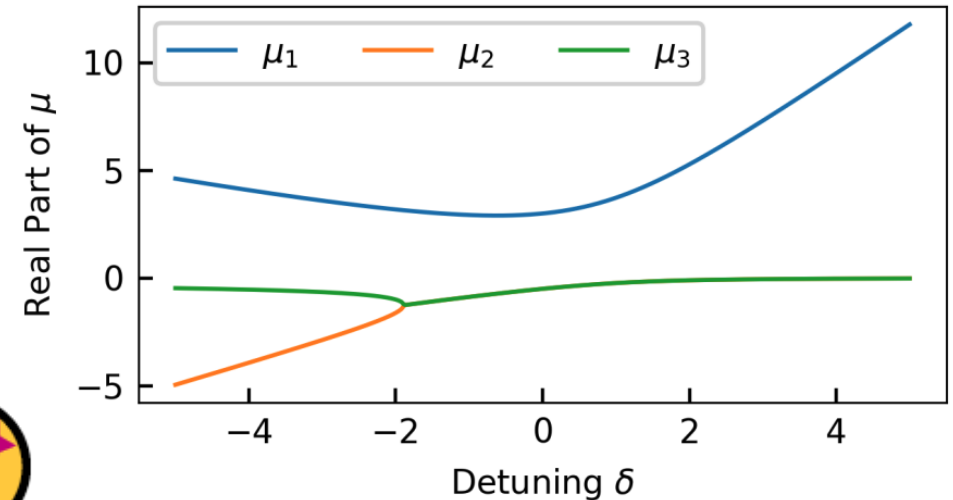
$$\mu^2(\mu - \delta) - 1 = 0$$

$$\mu_1 = 1 \quad \mu_2 = \frac{i\sqrt{3}-1}{2} \quad \mu_3 = -\frac{i\sqrt{3}+1}{2}$$



$$|a|^2 \propto e^{-2\text{Im}[\mu_3]\hat{z}} \\ = e^{z/L_g}$$

$$L_g = \frac{\lambda_u}{4\pi\sqrt{3}\rho}$$



FEL theory: some key results

For typical XFELs:

Gain length:

$$L_g = \frac{\lambda_u}{4\pi\sqrt{3}\rho}$$

A few meters

Saturation length $\simeq 20$ gain lengths

Relative bandwidth at saturation:

$$\frac{\sigma_\omega}{\omega} \simeq \rho$$

0.01%—0.1% (or, a few eV)

Peak power at saturation:

$$P_{sat} \simeq \rho P_{beam}$$

GW-scale

Energy change at saturation:

$$\Delta\eta_{sat} \simeq -\rho$$

A few MeV

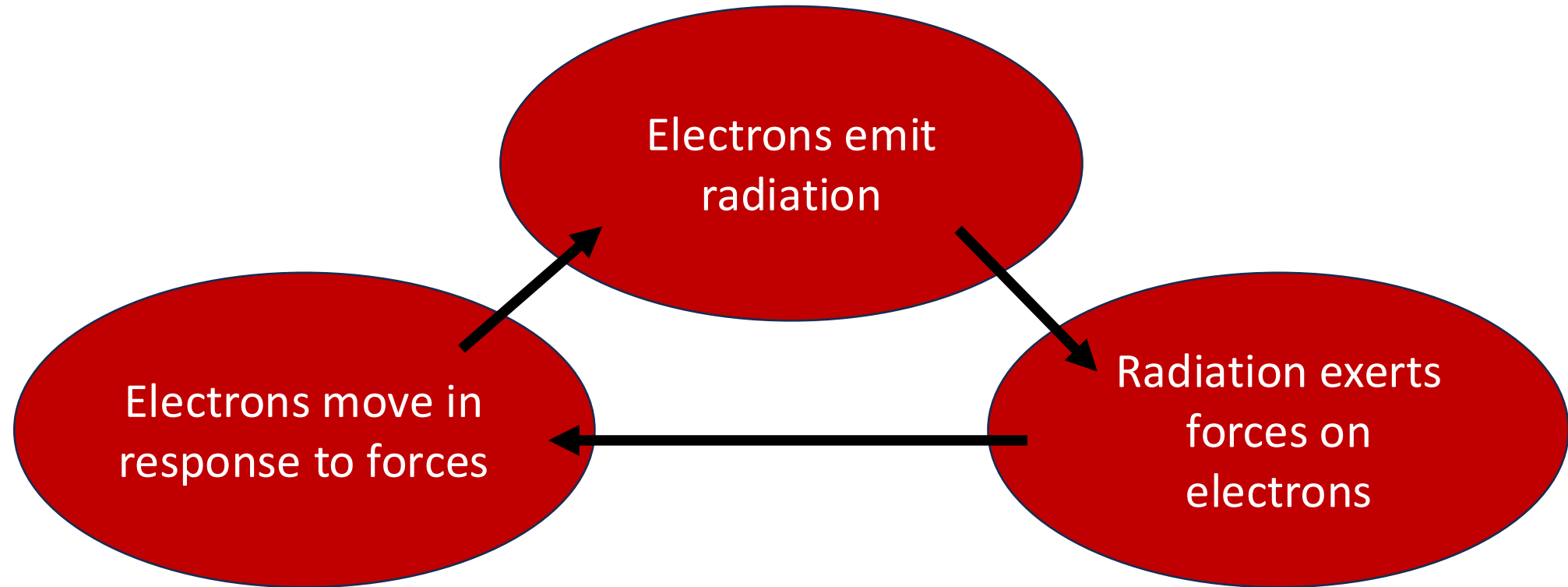
Energy spread at saturation:

$$\sigma_{\eta,sat} \simeq \rho$$

A few MeV

The FEL instability

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \vec{A} = -\mu_0 \vec{J}$$

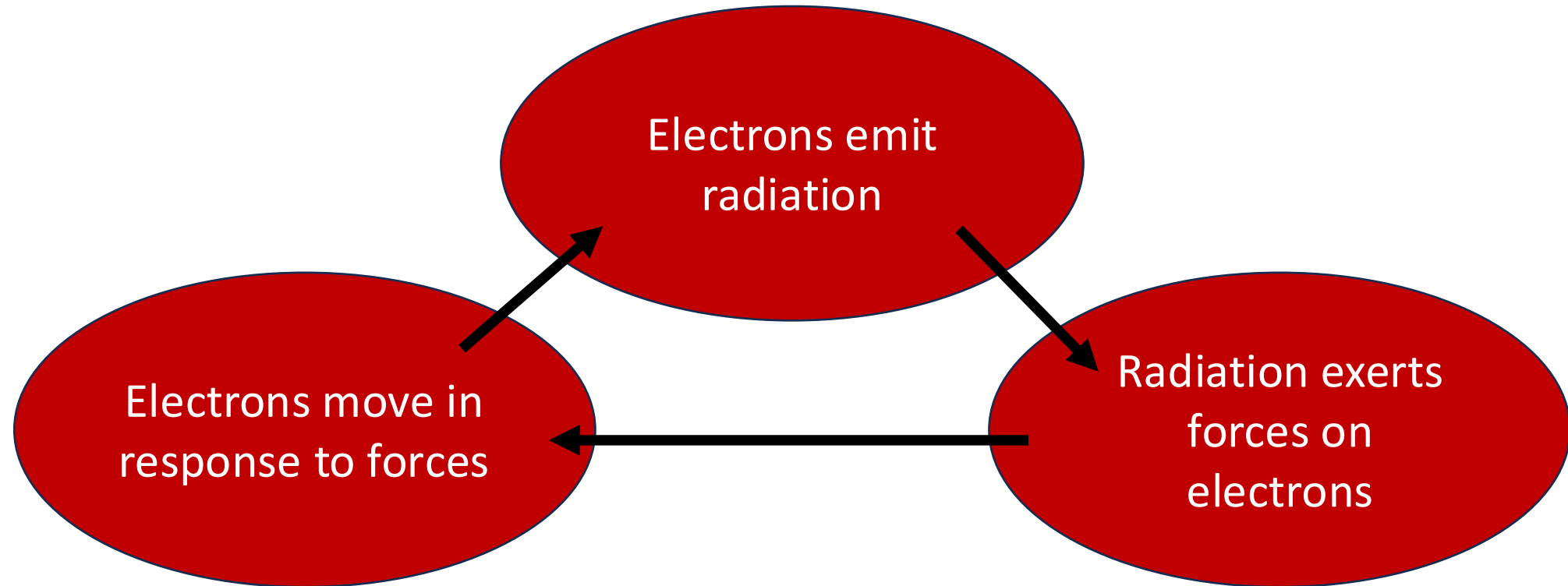


$$\frac{d\vec{x}}{dt} = \frac{\vec{p}}{\gamma m}$$

$$\frac{d\vec{p}}{dt} = \vec{F}$$

The FEL instability

$$\left(\frac{\partial}{\partial \hat{z}} - i\delta \right) a = b$$

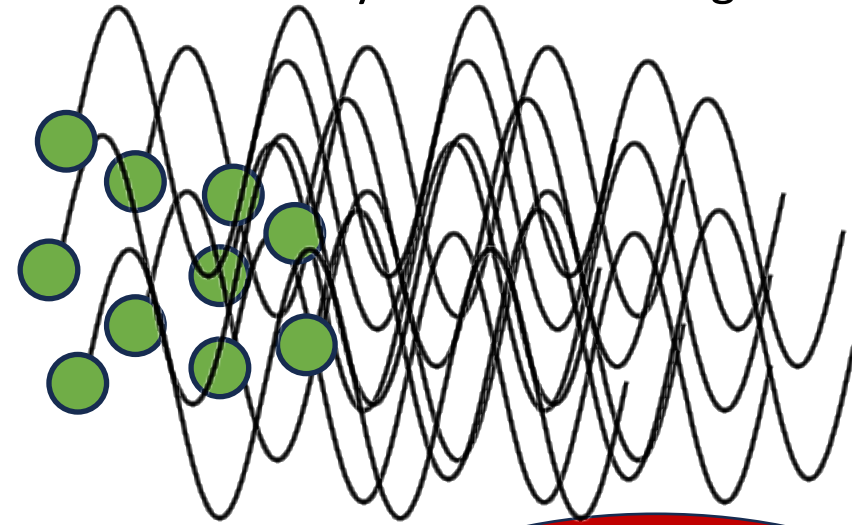


$$\frac{db}{d\hat{z}} = ip$$

$$\frac{dp}{d\hat{z}} = -a$$

How do we enter the loop?

There's always *some* bunching!



$$\left(\frac{\partial}{\partial \hat{z}} - i\delta \right) a = b$$

Electrons emit
radiation

Direct seeding, useful up to around
UV, but impractical so far for X-rays

Electrons move in
response to forces

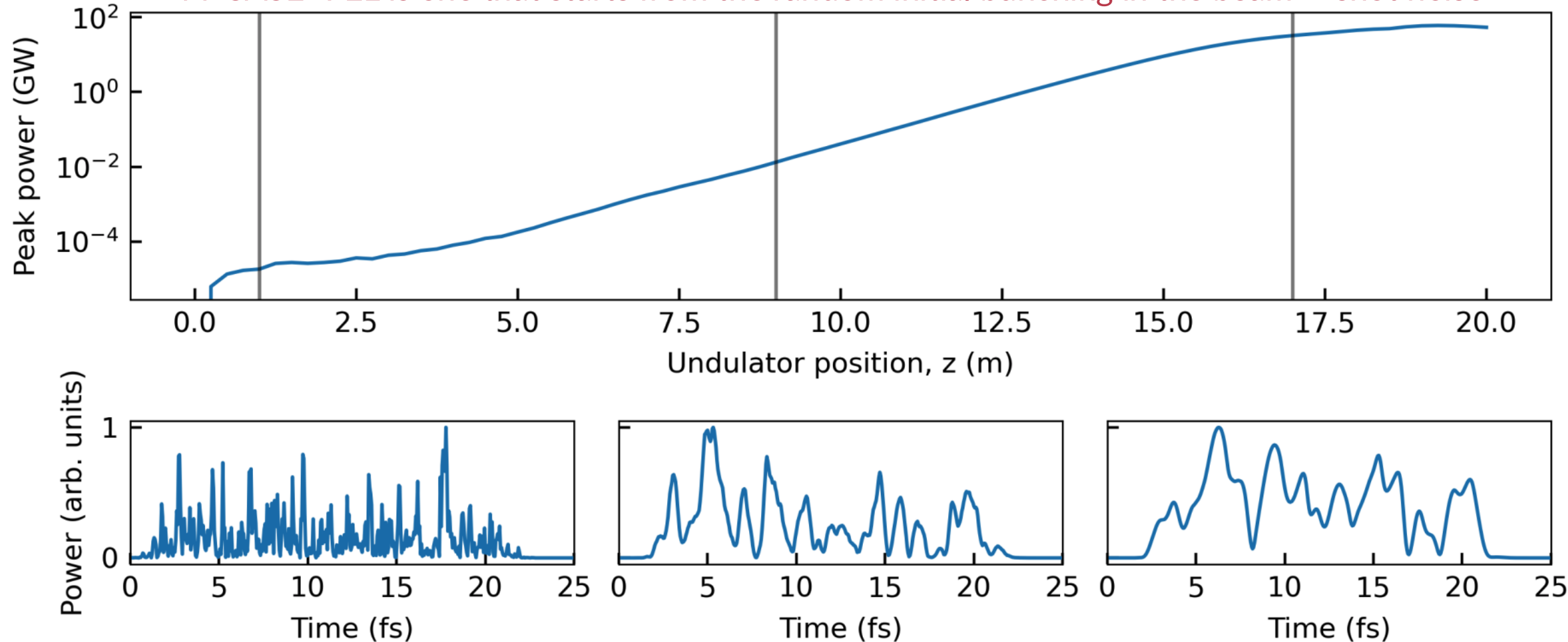
Radiation exerts
forces on
electrons

$$\frac{db}{d\hat{z}} = ip$$

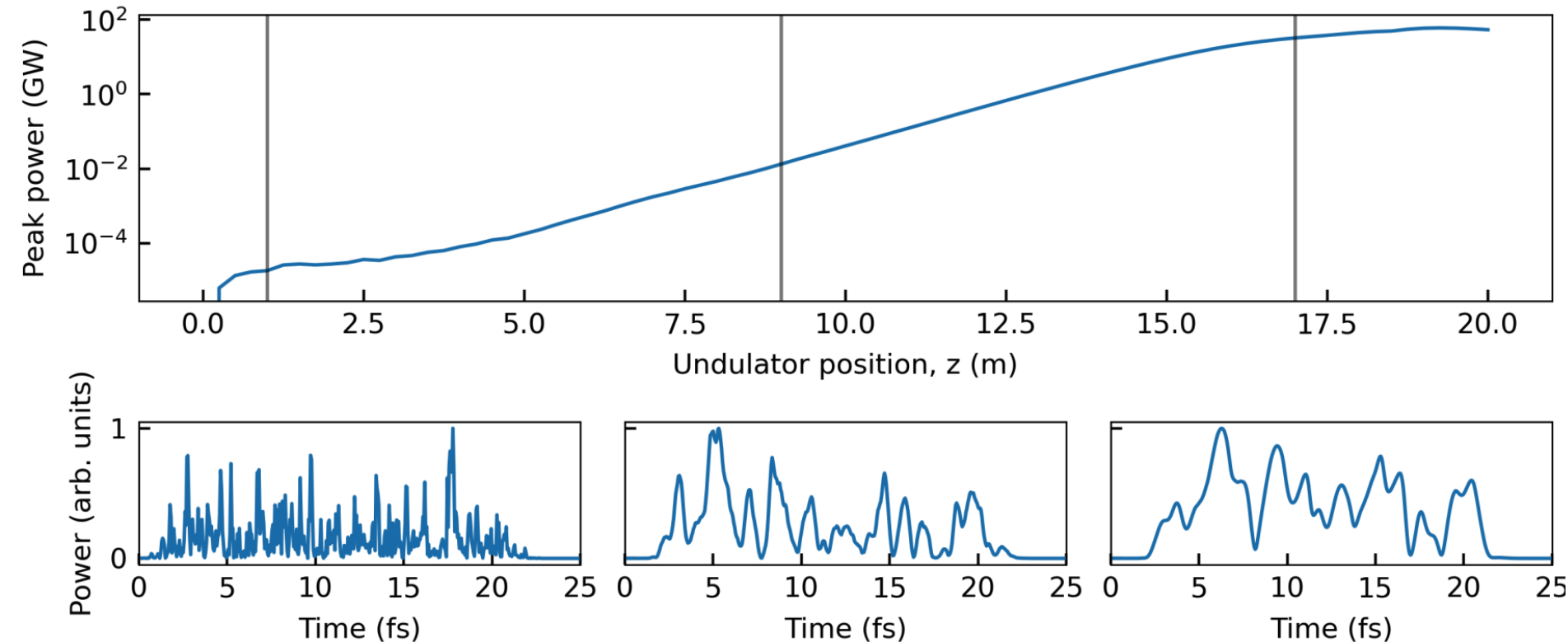
$$\frac{dp}{d\hat{z}} = -a$$

Self-Amplified Spontaneous Emission (SASE)

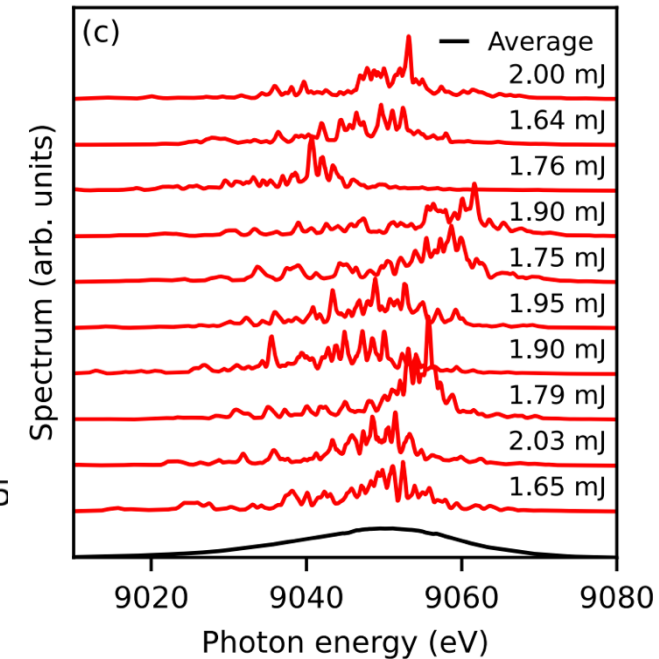
A “SASE” FEL is one that starts from the random initial bunching in the beam -- shot noise



Self-Amplified Spontaneous Emission (SASE)



Spikes in time
=
Spikes in spectrum



Determined by:

Pulse length: **Bunch length**

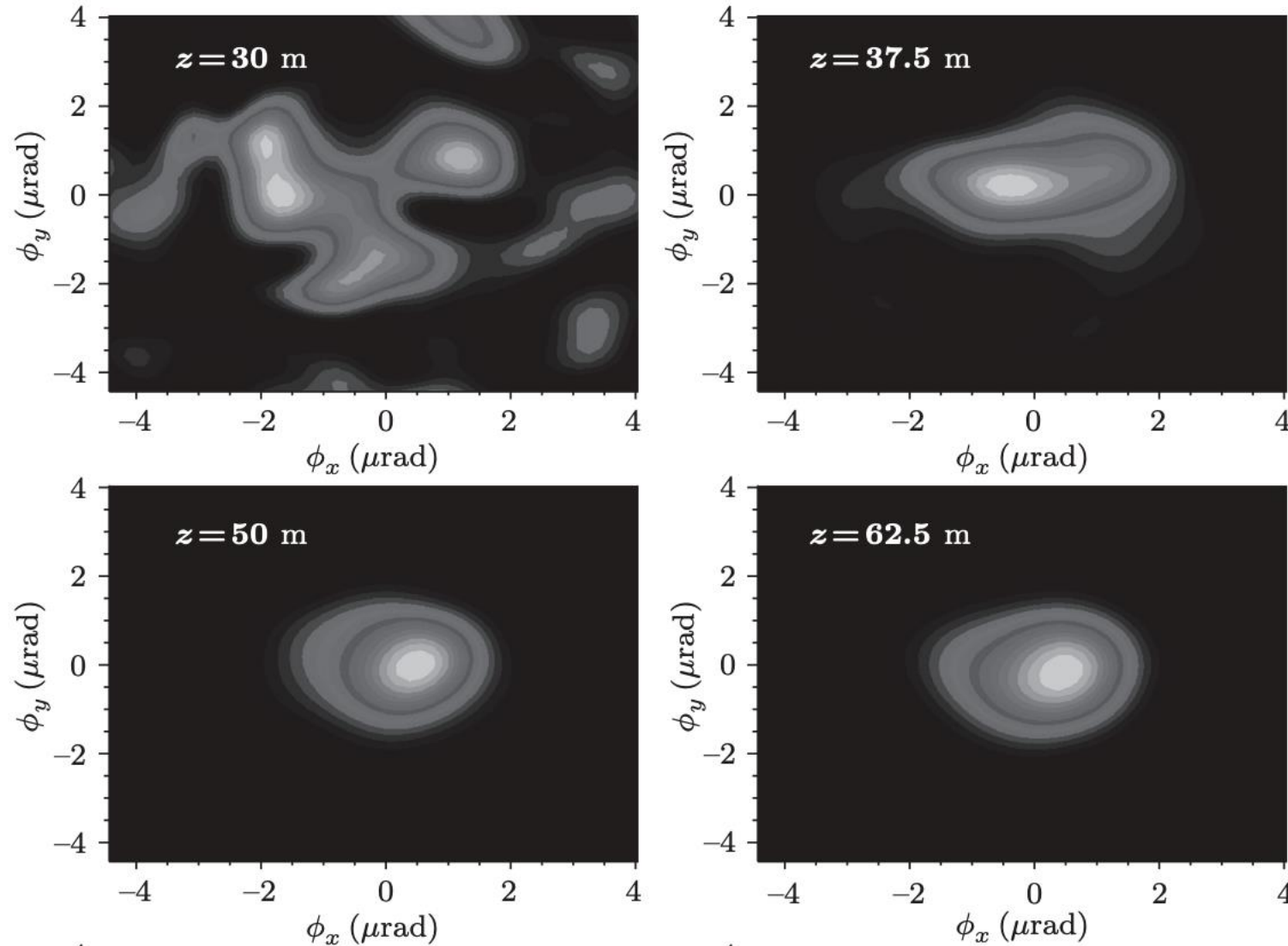
Bandwidth: **Gain length (ρ)**

Determined by:

Spike length: **Gain length (ρ)**

Spike width: **Bunch length**

Transverse characteristics of SASE FELs: gain guiding



Overcoming saturation: tapering

Why does the FEL saturate? Partly because we lose the resonance condition

$$\lambda_r = \frac{\lambda_u}{2\gamma^2} \left(1 + \frac{K^2}{2} \right)$$

At saturation, field grows roughly linearly:

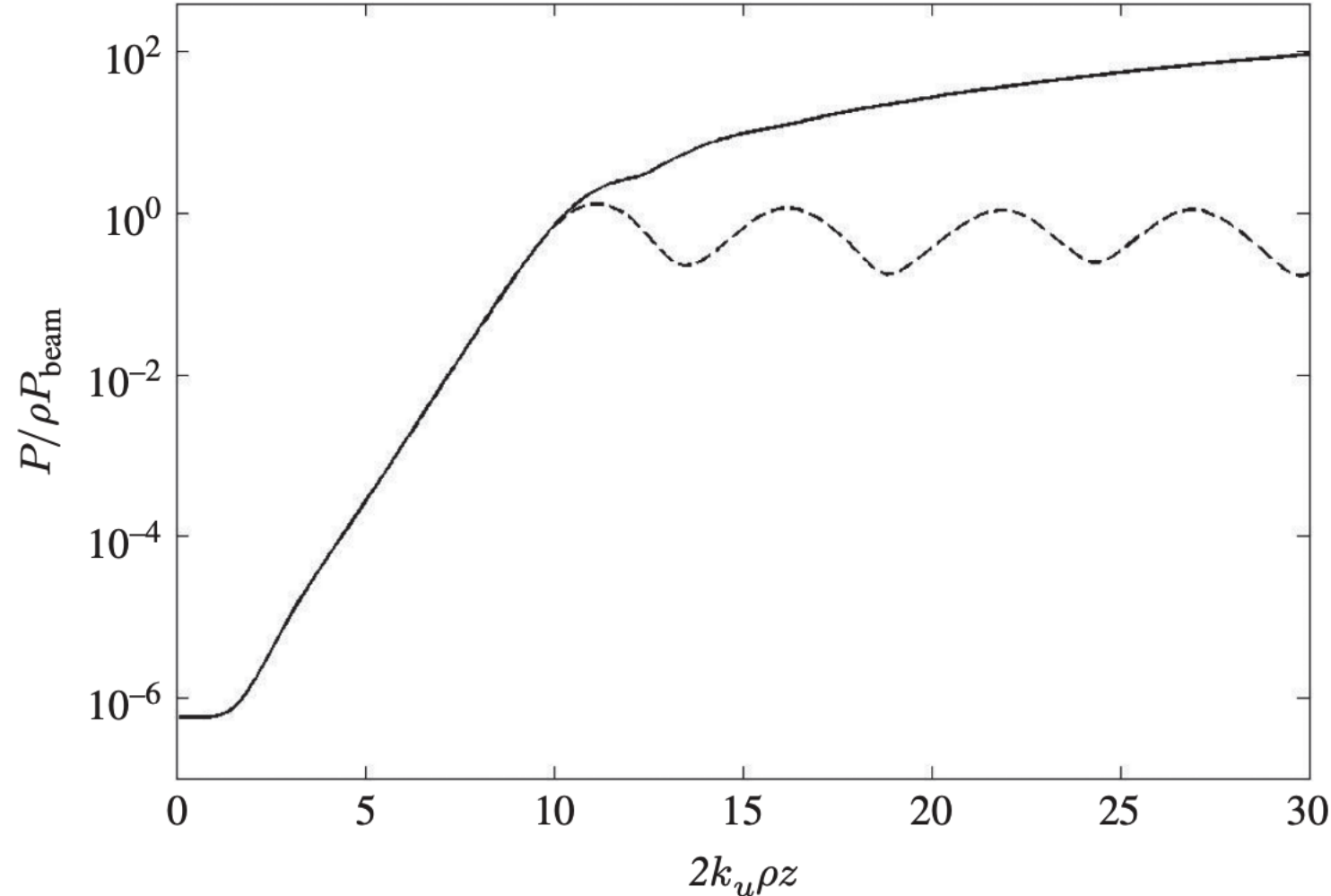
$$\frac{da}{d\hat{z}} = b \rightarrow a(\hat{z}) \simeq b\hat{z}$$

As a result, energy loss is roughly quadratic with z

$$\frac{d\hat{\eta}}{d\hat{z}} = -(ae^{i\theta_j} + c.c.) \rightarrow \hat{\eta} \propto \hat{z}^2$$

To keep lasing, we need to maintain the resonance condition:

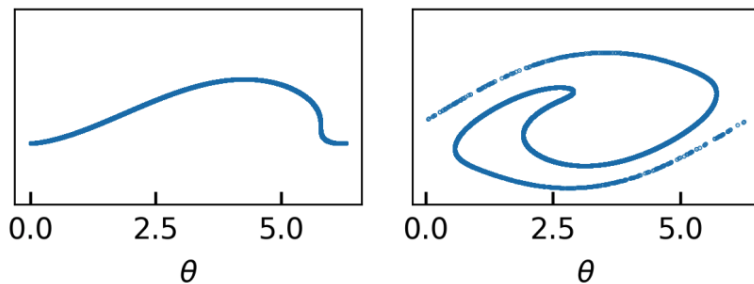
$$K \propto -\hat{z}^2$$



Non-ideal effects: energy spread, emittance, diffraction, Ming Xie

Energy spread

Recall:



Energy spread at saturation $\simeq \rho$

$$\sigma_{\eta, \text{initial}} \ll \rho$$

Emittance

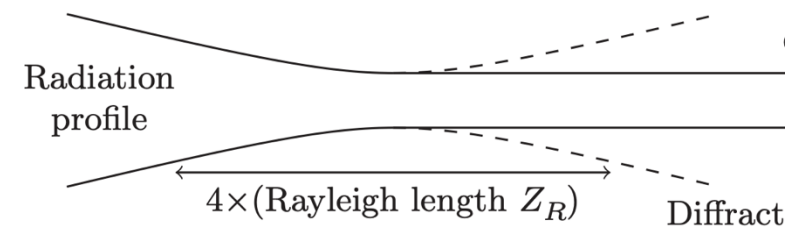
$$\lambda_r = \frac{\lambda_u}{2\gamma^2} \left(1 + \frac{K^2}{2} \right)$$

$$\lambda_r = \frac{\lambda_u}{2\gamma^2} \left(1 + \frac{K^2}{2} + \gamma^2 \theta^2 \right)$$

Need detuning for typical angle $\theta \simeq \frac{\epsilon_x}{\beta_x}$
to be small relative to ρ

$$\frac{\epsilon_x}{\beta_x} k_r L_G \ll 1$$

Diffraction



Diffraction too fast, and you can't gain fast enough to keep the mode confined

$$L_g \ll Z_R \simeq \frac{k_r \sigma_x^2}{2}$$

$$L_{g,3D} \simeq L_{g,1D} (1 + \Lambda)$$

$$\eta_d = \frac{L_{g,1D}}{2k_r \sigma_x^2}$$

$$\eta_\epsilon = 2 \frac{\epsilon_x}{\beta_x} k_r L_{g,1D}$$

$$\eta_\gamma = \frac{\sigma_\eta}{\sqrt{3}\rho}$$

$$P_{\text{sat},3D} \simeq \frac{1.6}{(1 + \Lambda)^2} P_{\text{sat},1D}$$

$$\Lambda = a_1 \eta_d^{a_2} + a_3 \eta_\epsilon^{a_4} + a_5 \eta_\gamma^{a_6} + a_7 \eta_\epsilon^{a_8} \eta_\gamma^{a_9}$$

$$+ a_{10} \eta_d^{a_{11}} \eta_\gamma^{a_{12}} + a_{13} \eta_d^{a_{14}} \eta_\epsilon^{a_{15}} + a_{16} \eta_d^{a_{17}} \eta_\epsilon^{a_{18}} \eta_\gamma^{a_{19}}$$

Non-ideal effects: energy spread, emittance, diffraction, Ming Xie

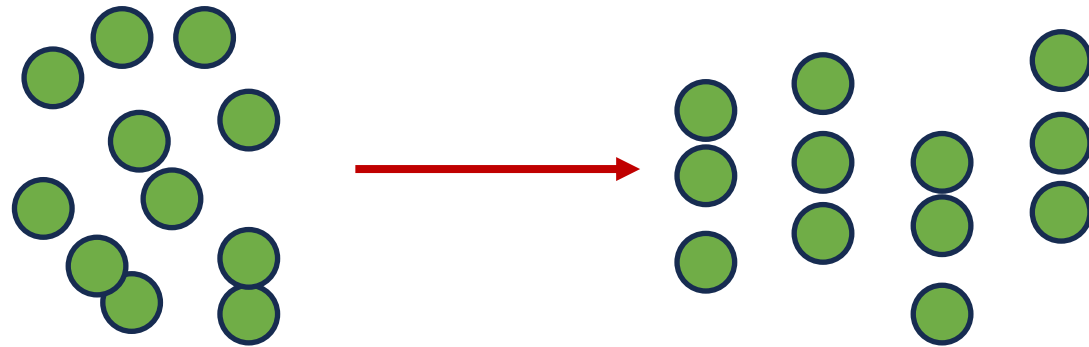
Where, obviously....

$$\begin{array}{lllll} a_1 = 0.45, & a_2 = 0.57, & a_3 = 0.55, & a_4 = 1.6, & a_5 = 3, \\ a_6 = 2, & a_7 = 0.35, & a_8 = 2.9, & a_9 = 2.4, & a_{10} = 51, \\ a_{11} = 0.95, & a_{12} = 3, & a_{13} = 5.4, & a_{14} = 0.7, & a_{15} = 1.9, \\ a_{16} = 1140, & a_{17} = 2.2 & a_{18} = 2.9, & a_{19} = 3.2. \end{array}$$

$$\begin{aligned} L_{g,3D} &\simeq L_{g,1D}(1 + \Lambda) & \eta_d &= \frac{L_{g,1D}}{2k_r\sigma_x^2} & \eta_\epsilon &= 2\frac{\epsilon_x}{\beta_x}k_rL_{g,1D} & \eta_\gamma &= \frac{\sigma_\eta}{\sqrt{3}\rho} \\ P_{sat,3D} &\simeq \frac{1.6}{(1 + \Lambda)^2}P_{sat,1D} & \Lambda &= a_1\eta_d^{a_2} + a_3\eta_\epsilon^{a_4} + a_5\eta_\gamma^{a_6} + a_7\eta_\epsilon^{a_8}\eta_\gamma^{a_9} \\ & & & & & + a_{10}\eta_d^{a_{11}}\eta_\gamma^{a_{12}} + a_{13}\eta_d^{a_{14}}\eta_\epsilon^{a_{15}} + a_{16}\eta_d^{a_{17}}\eta_\epsilon^{a_{18}}\eta_\gamma^{a_{19}} \end{aligned}$$

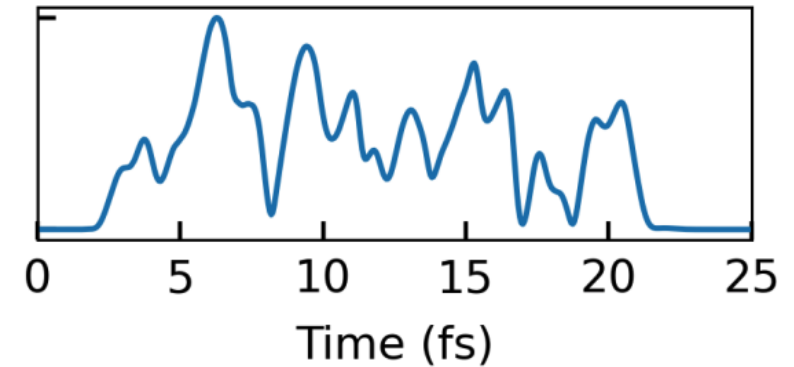
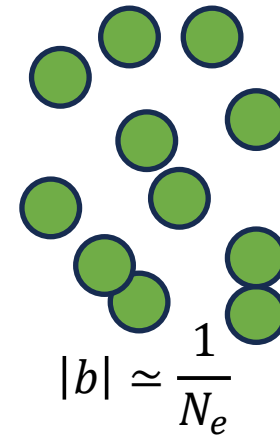
Key takeaways

The magic of the FEL is all about bunching



Factor of N_e enhancement in radiation power!

XFELs generally start from random bunching: SASE



Most features of the FEL are characterized by ρ

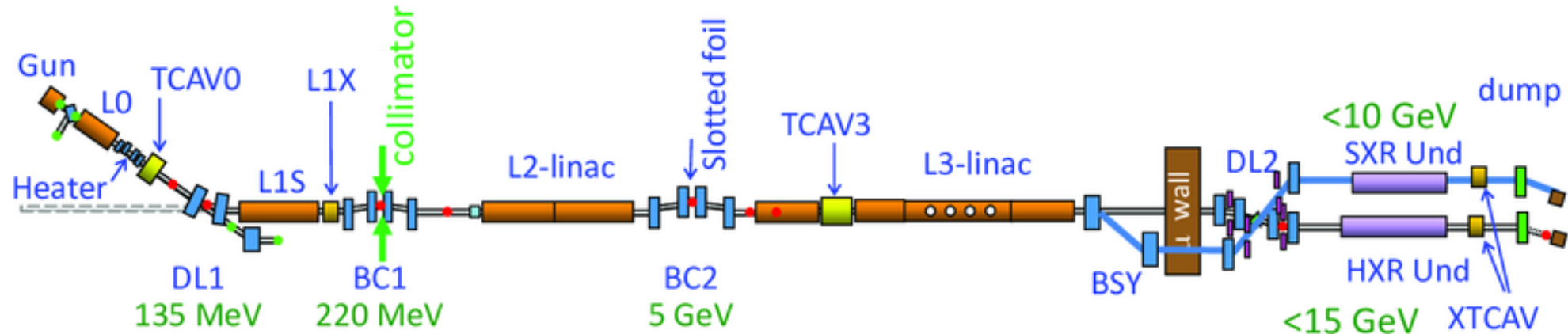
$$\rho = \left(\frac{e^2 n_e K^2 [JJ]^2}{32 k_u^2 \gamma_0^3 m c^2 \epsilon_0} \right)^{\frac{1}{3}}$$

“Pierce parameter”
Typically $10^{-3} - 10^{-4}$

Gain length, saturation
power, spike length,
bandwidth, energy
efficiency, electron
energy loss and spread, ...

FELs in Practice

Typical layout of an XFEL



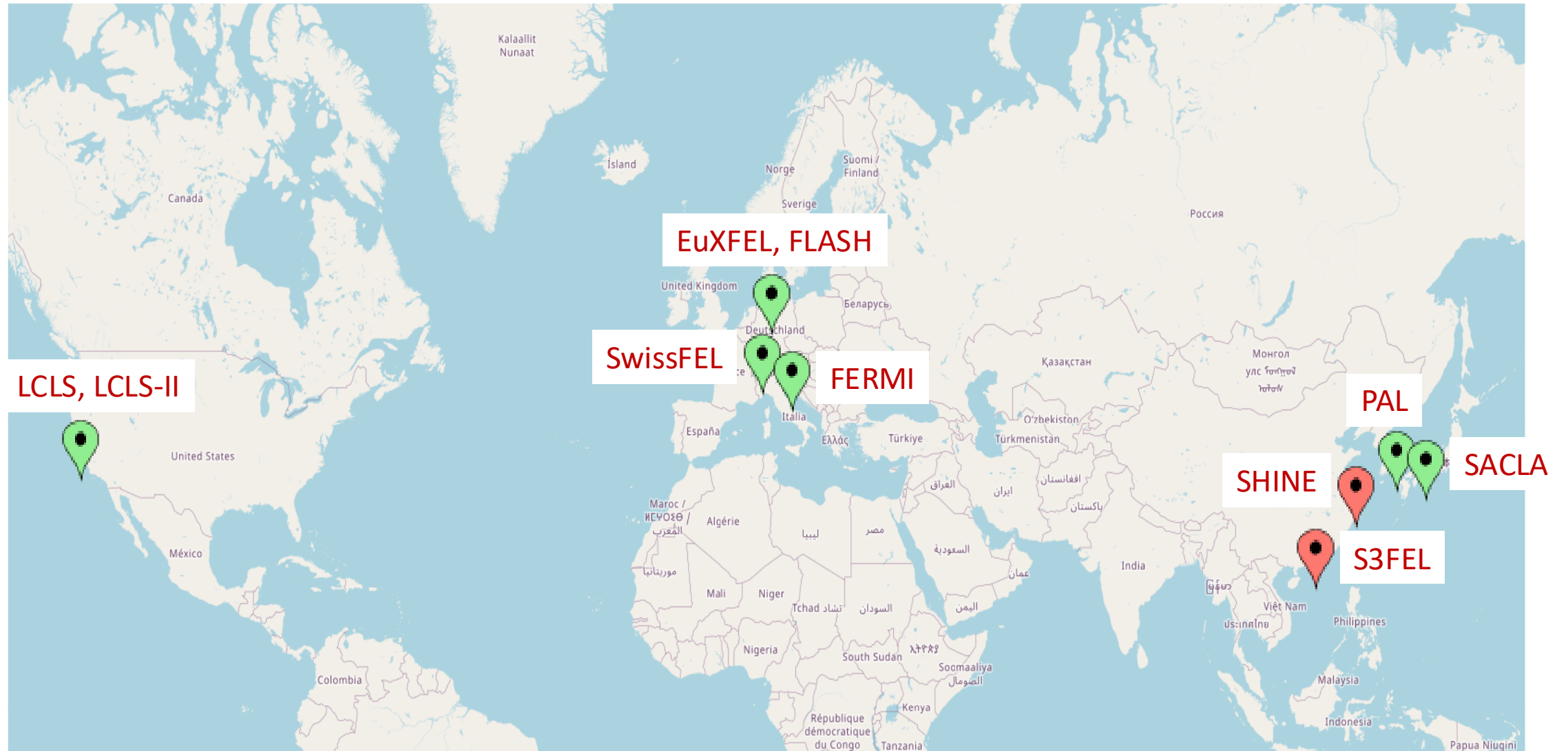
Typical beam requirements of an FEL:

- Charge: few 100 pC
- Peak current: few kA
- Bunch length: ≈ 50 fs/15 μ m
- Beam energy: 1 – 20 GeV
- Normalized emittance: < 1 mm-mrad
- Rel. energy spread: $\leq 0.1\%$

Typical features of an FEL:

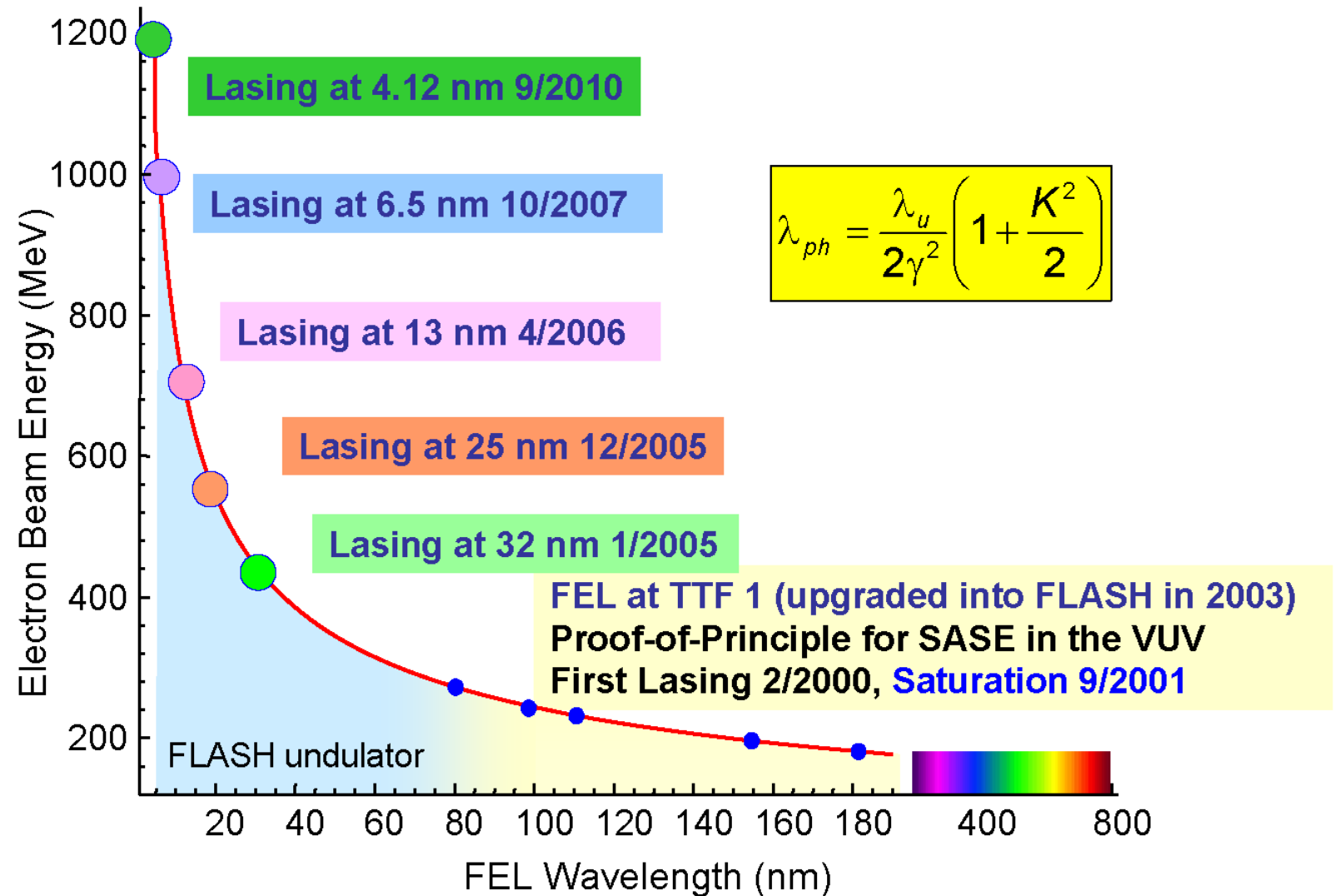
- High-gradient photoinjector electron source
- Multiple stages of bunch compression with careful optics control
- Instability mitigation strategies (e.g., laser heater)
- Active phase space linearization with harmonic cavities

XFELs around the world



FLASH FEL

World's first XUV/soft X-ray FEL
First lasing in 2005

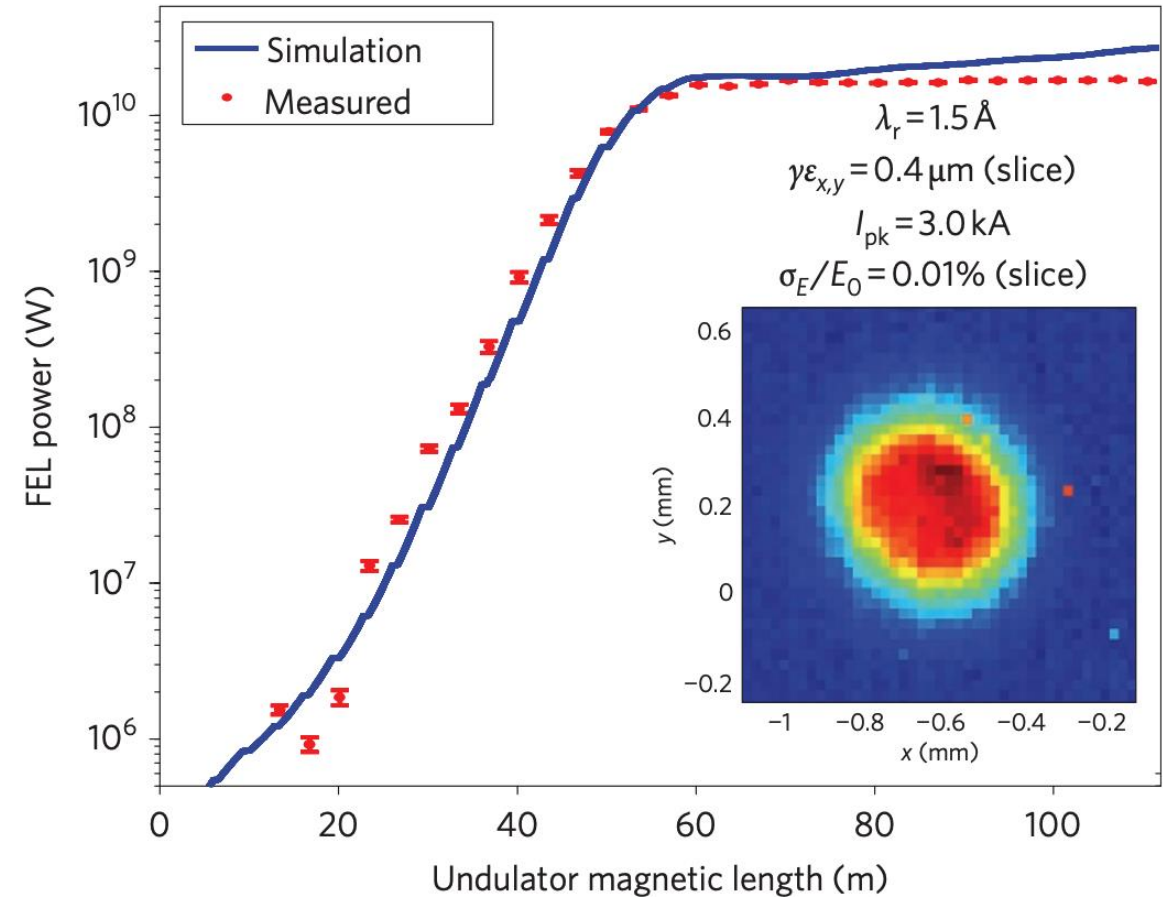


The Linac Coherent Light Source (LCLS)

World's first hard X-ray FEL, first user operations in 2009

4 – 15 GeV beams from a 120 Hz copper linac

Lasing soft X-rays ($\simeq 200$ eV) to hard ($\simeq 25$ keV)



The Linac Coherent Light Source (LCLS)

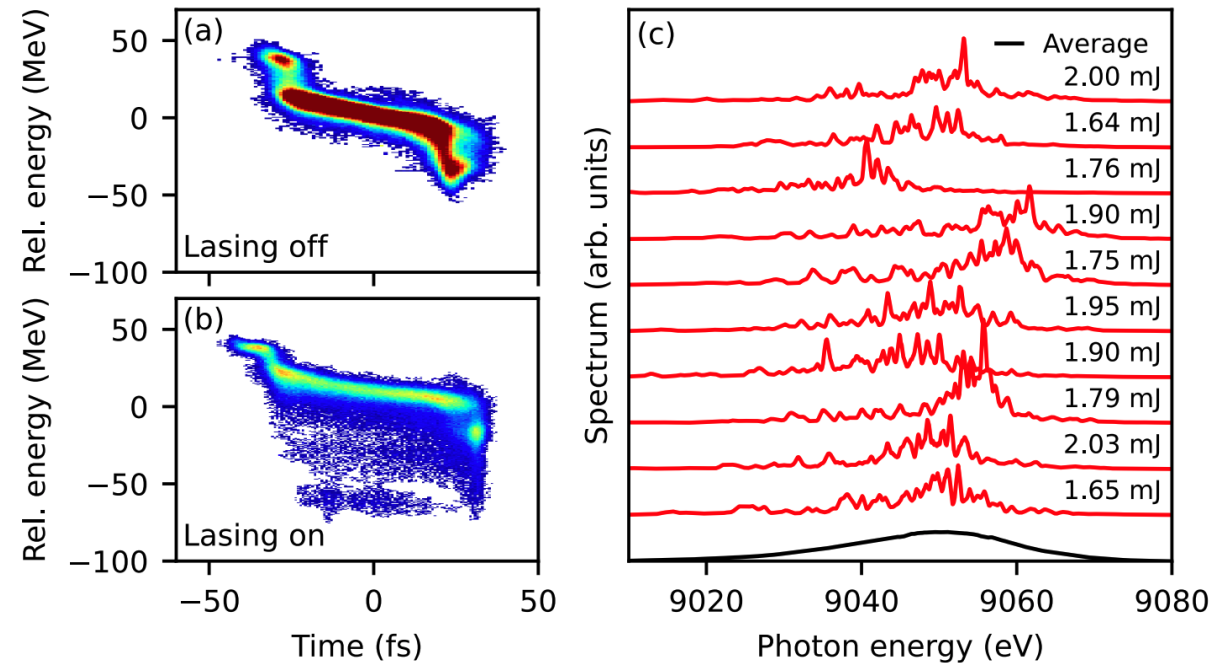
World's first hard X-ray FEL, first user operations in 2009

4 – 15 GeV beams from a 120 Hz copper linac

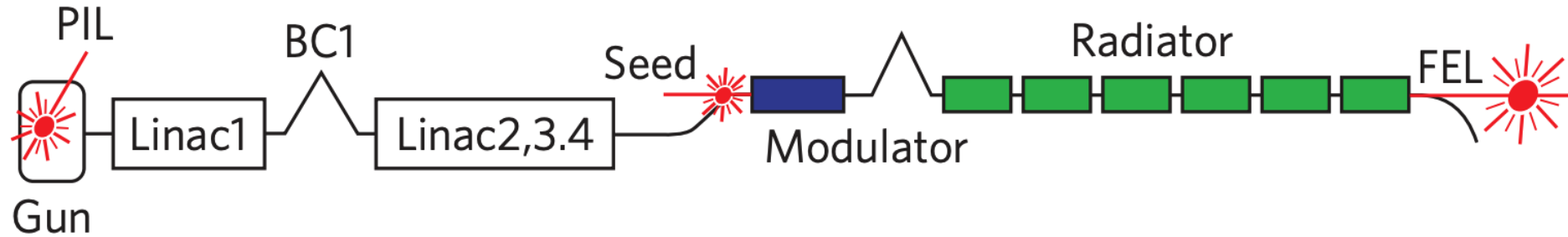
Lasing soft X-rays ($\simeq 200$ eV) to hard ($\simeq 25$ keV)



Joined later by other SASE XFELs: SACLA, EuXFEL, PAL, SwissFEL, LCLS-II, (soon) SHINE

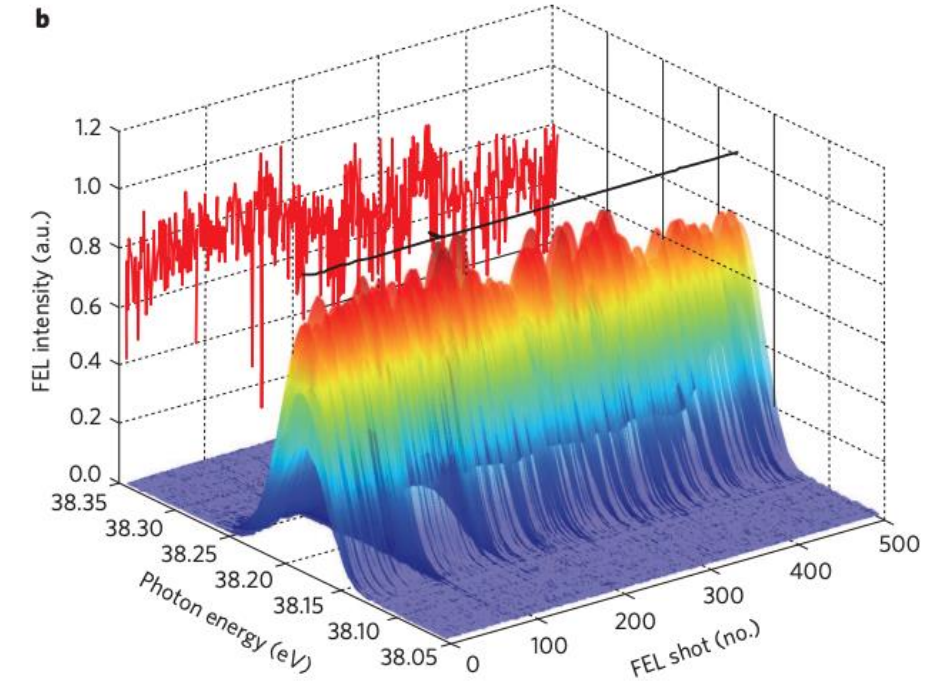
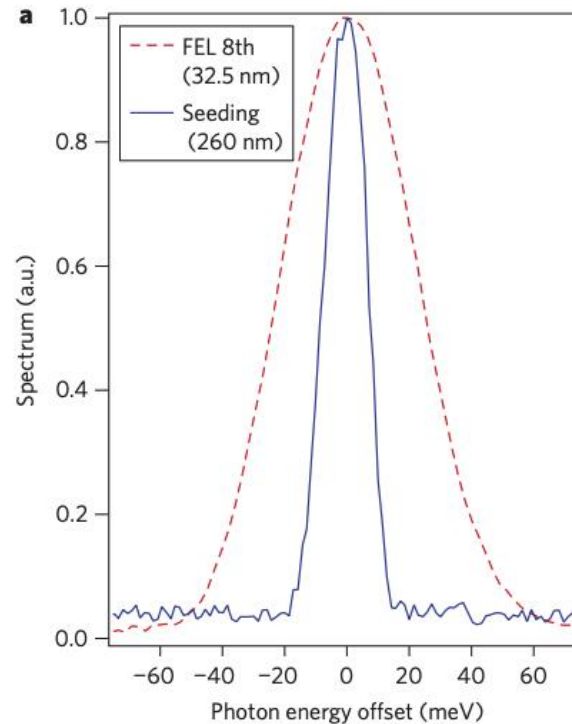


Fermi FEL

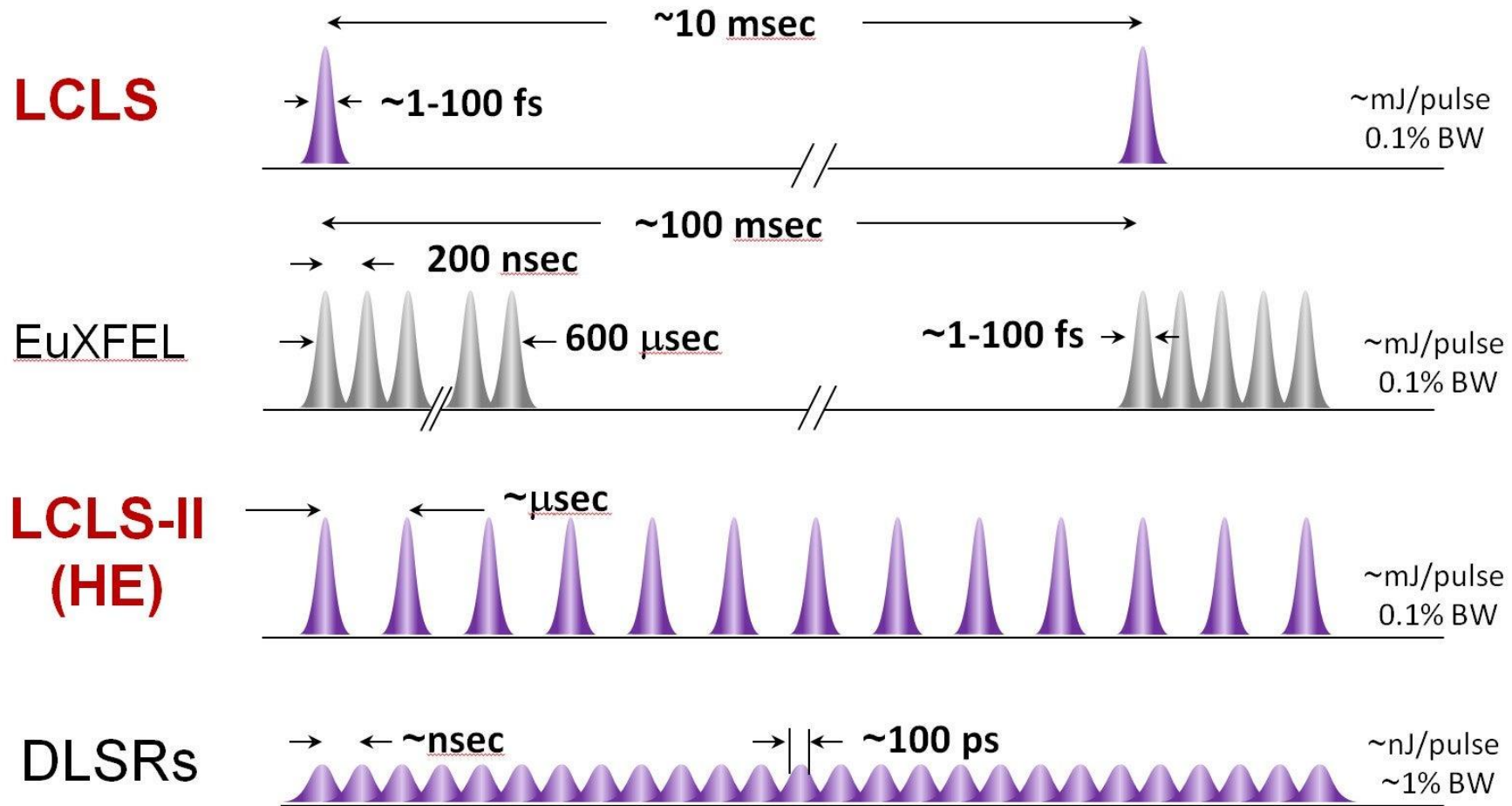


Externally seeded, start up not from noise but from a coherent input laser

Lasing up to the start of the soft X-ray regime, but not much further due to falloff of seeding efficiency



Towards higher repetition rates



Modern XFEL R&D

Better temporal coherence

How do we get clean up the ugly structure of SASE?

Flexible pulse structure

How do we tailor pulse properties (chirp, polarization, shape) for different experiments?

Higher power/pulse energy

How do we surpass the usual saturation limits?

Shorter pulses

How do we make femtosecond and even attosecond pulse durations?

Harder X-rays

How do we go beyond ~25 keV?

*NOT a comprehensive review – look more into specific topics if you find them interesting, or you can ask me for further references

*What I show is mostly what I am most familiar with, namely from SLAC

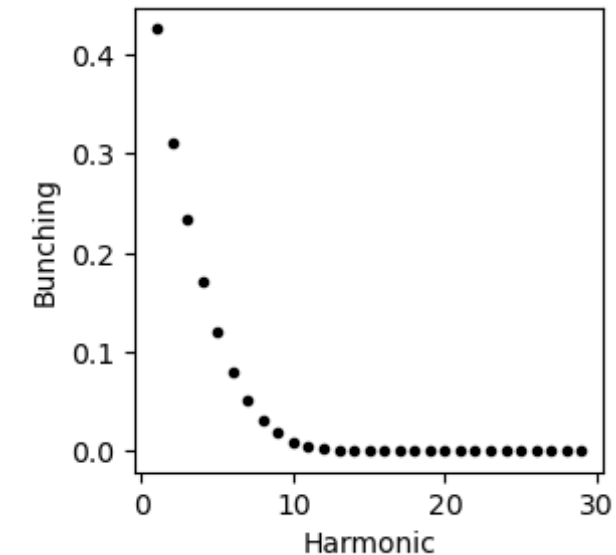
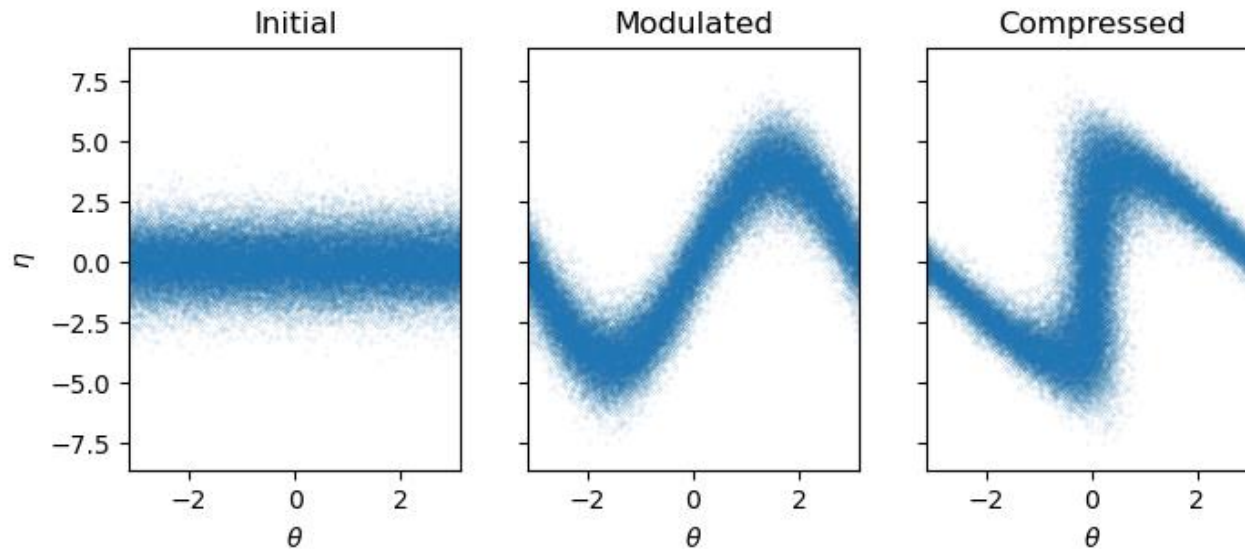
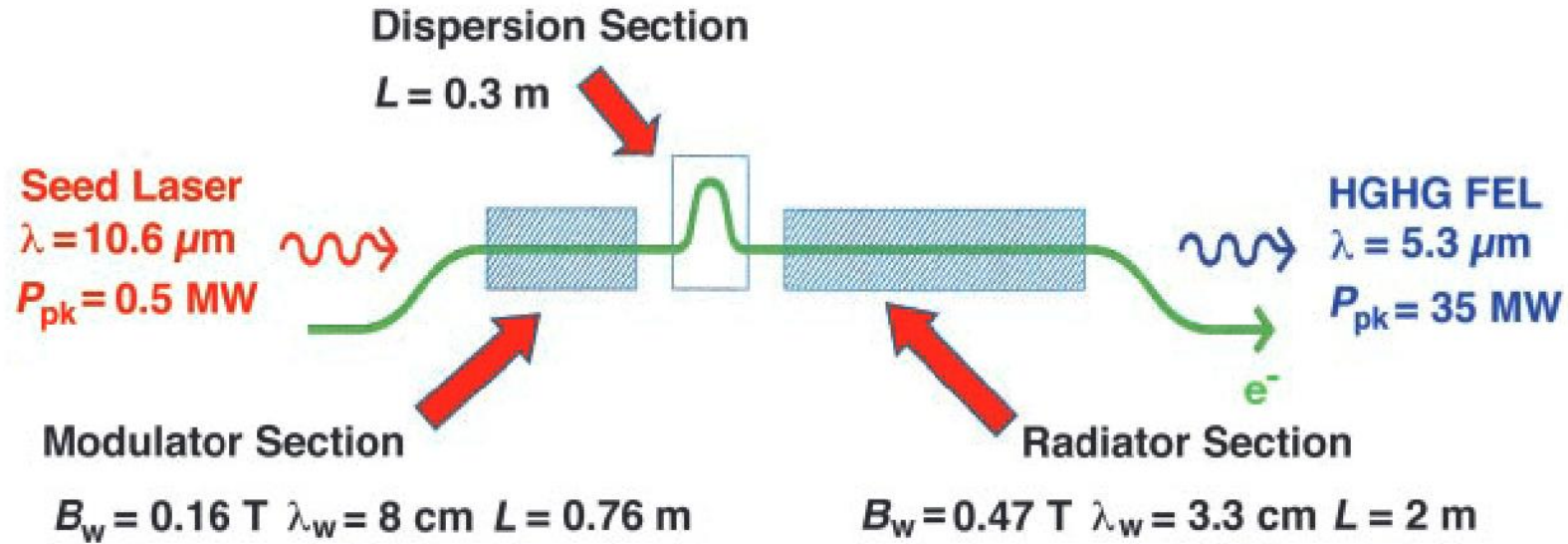
Modern XFEL R&D

Better temporal coherence

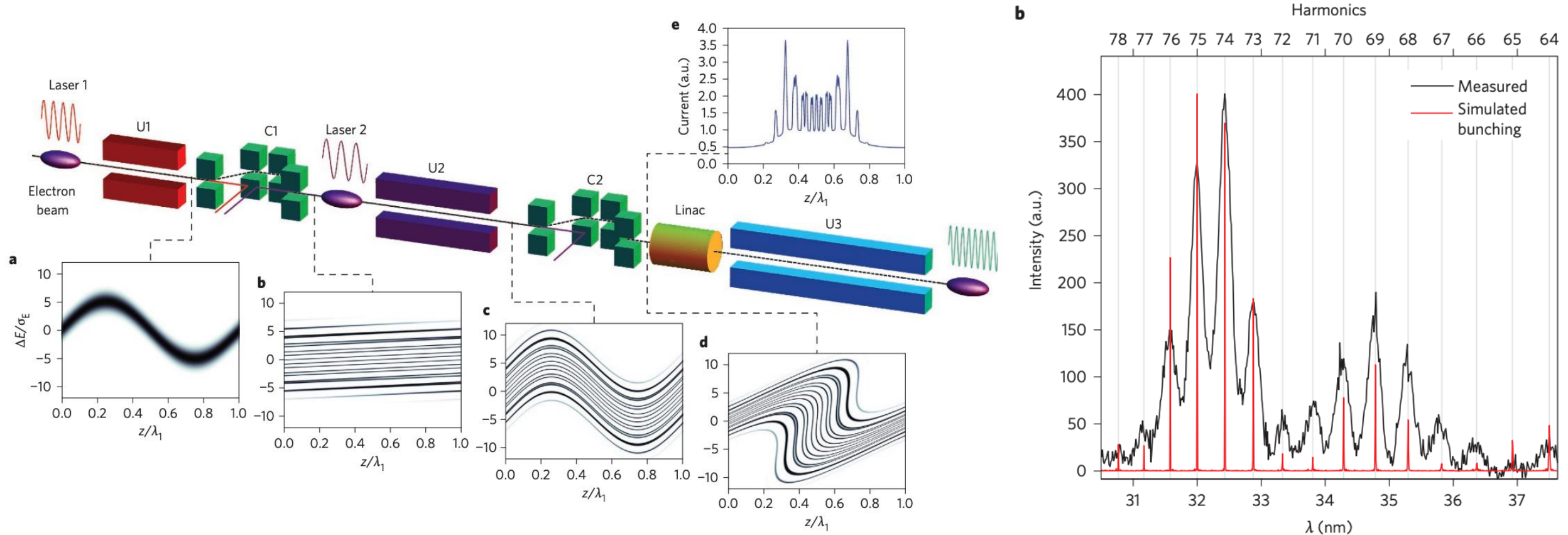
How do we get clean up the
ugly structure of SASE?

High-Gain Harmonic Generation

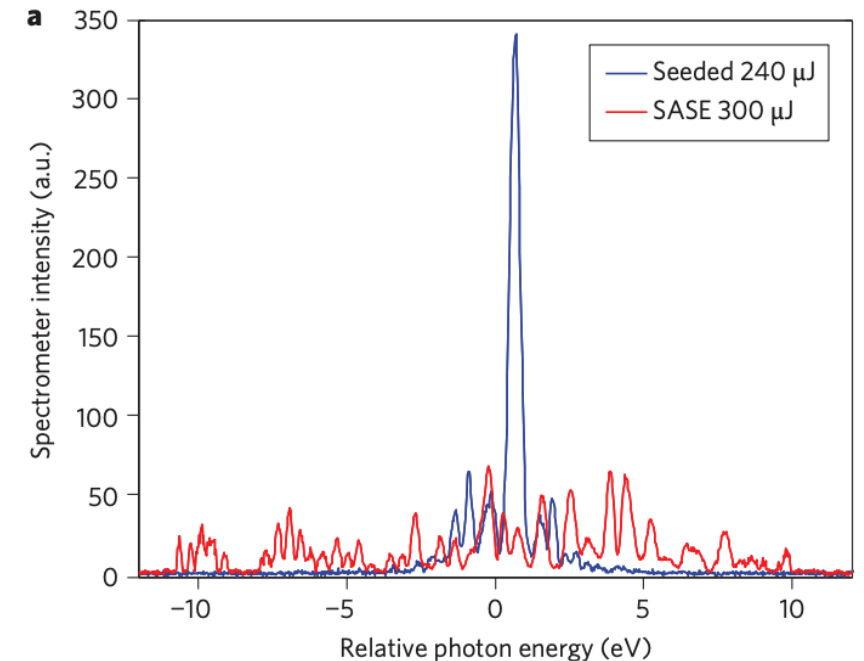
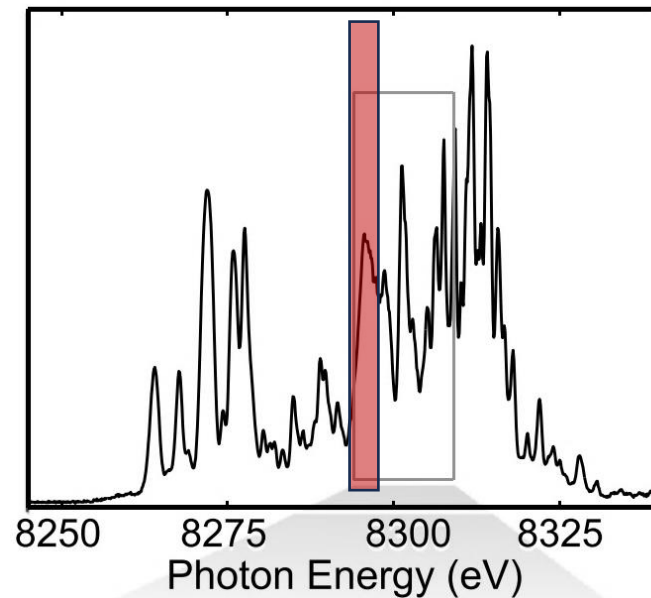
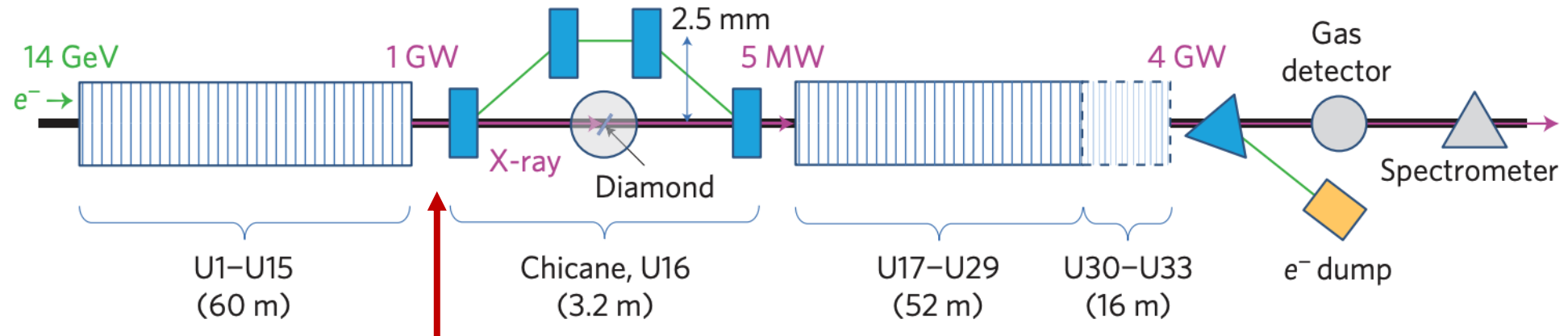
Yu, L-H., et al. *Science* 289.5481 (2000): 932-934.



Echo-Enabled Harmonic Generation

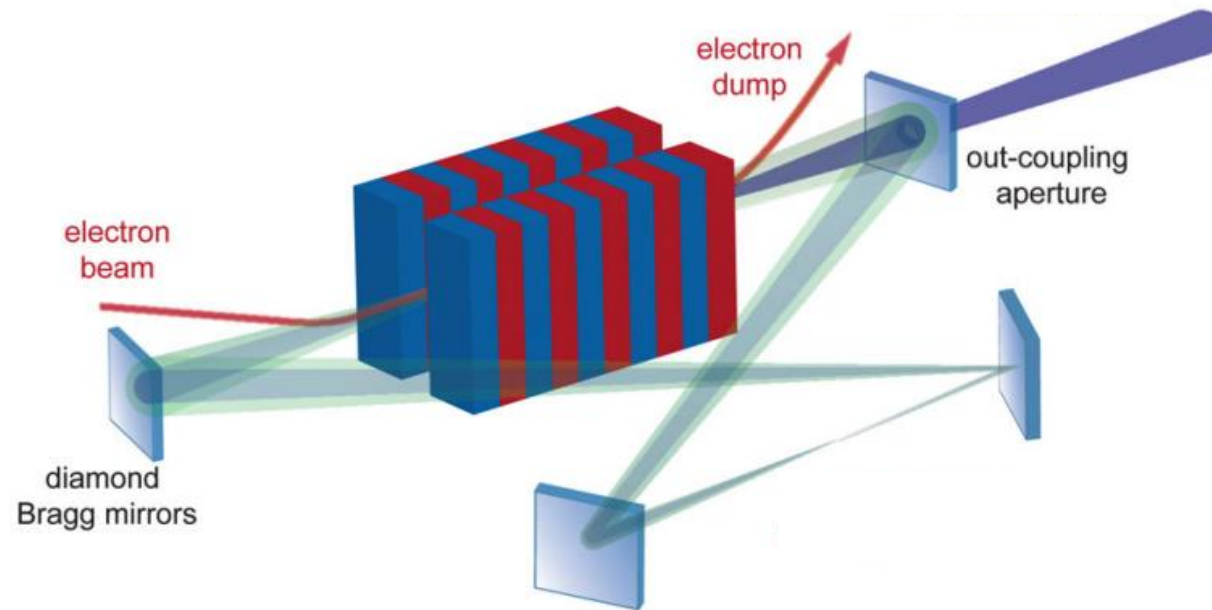


Self-Seeding



Cavity-based XFELs

Marcus, Gabriel, et al. *Physical Review Letters* 125.25 (2020): 254801.



Requirements:

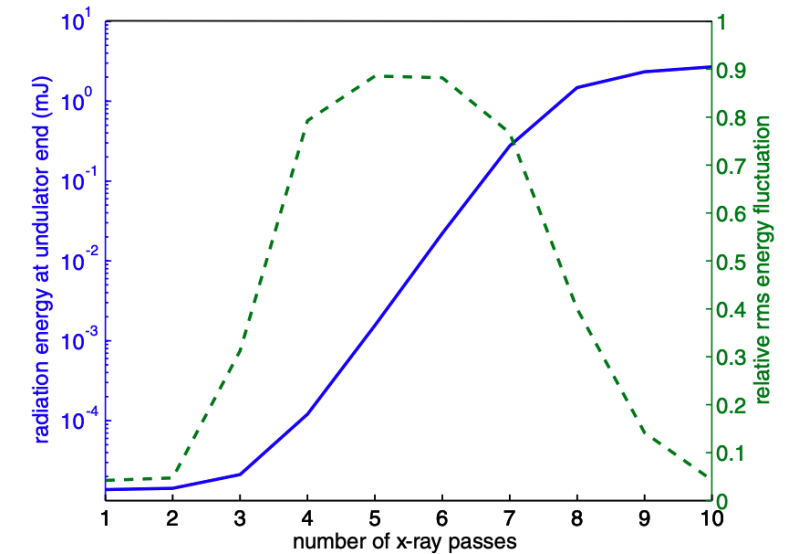
- High-reflectivity X-ray mirrors with large reflection angles
- High-repetition rate accelerator (rep. rate sets necessary cavity length)
- Arrival time stability to maintain spatiotemporal overlap of electrons with X-rays pass by pass

SLAC

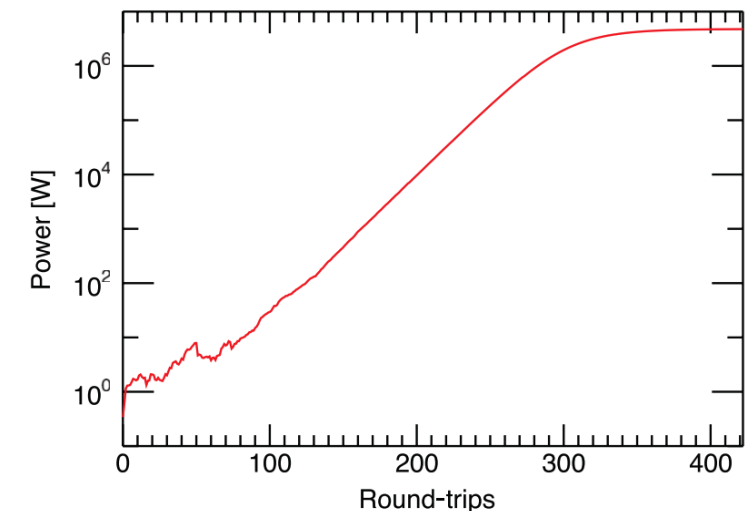
Huang, Zhirong, and Ronald D. Ruth. *Physical review letters* 96.14 (2006): 144801.

Kim, Kwang-Je, Yuri Shvyd'ko, and Sven Reiche. *Physical review letters* 100.24 (2008): 244802

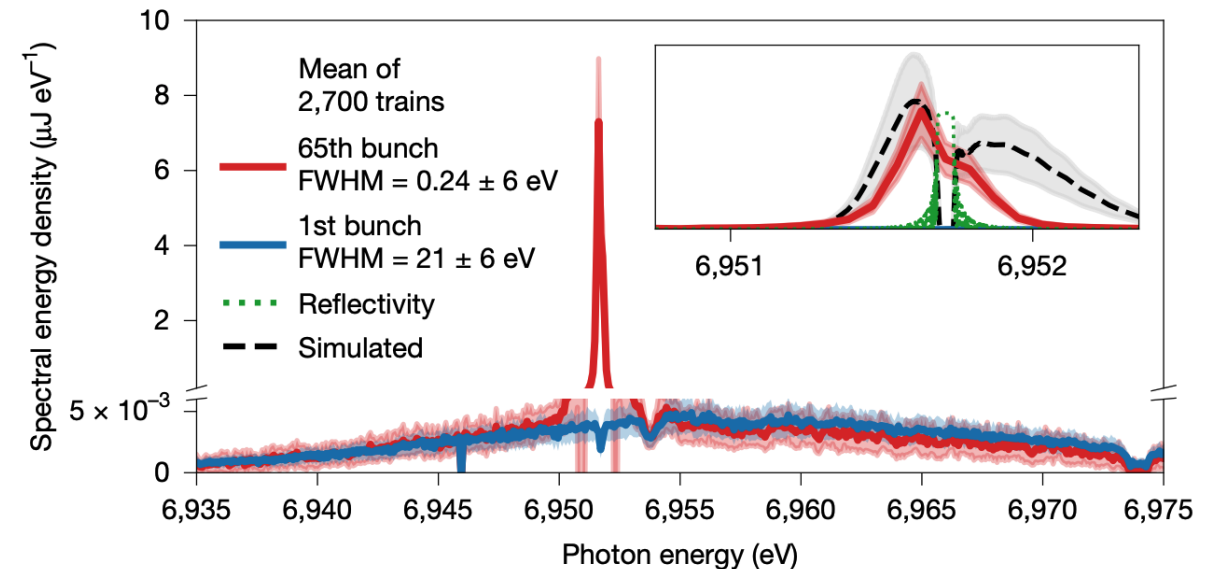
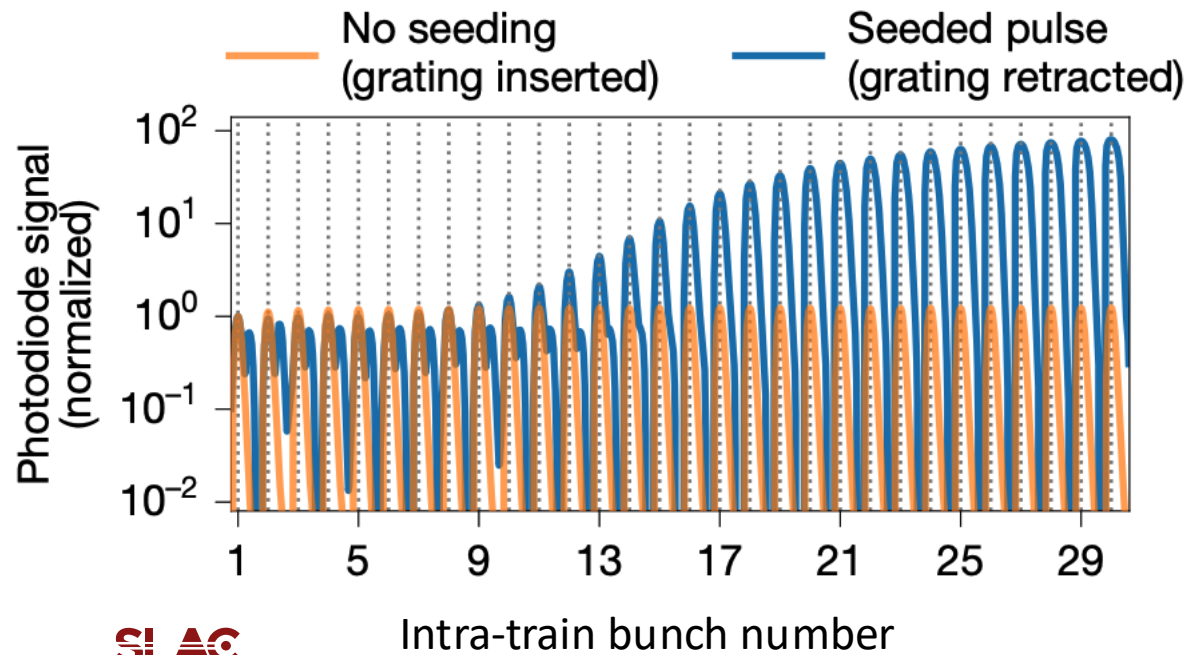
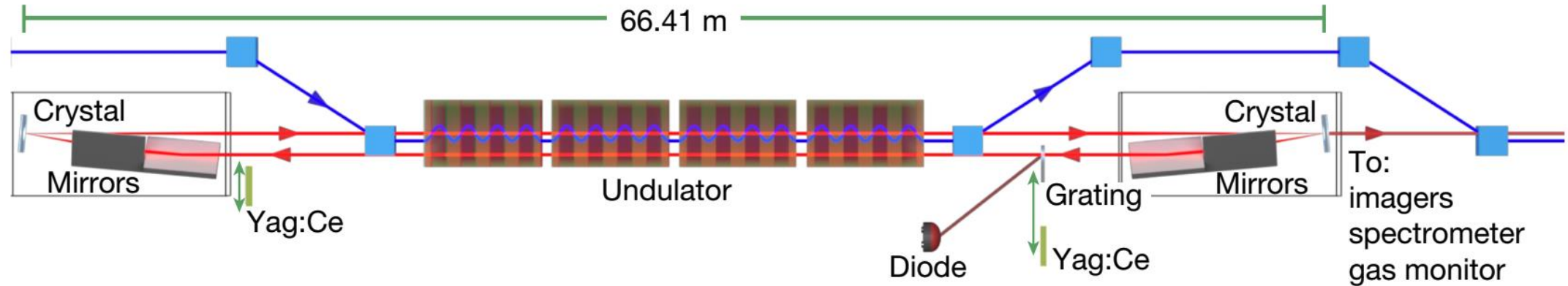
Regenerative Amplifier (RAFEL)



FEL Oscillator (FELO)



Cavity-based XFELs in reality



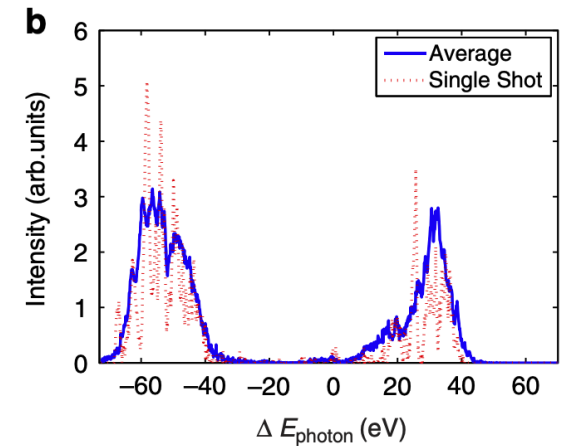
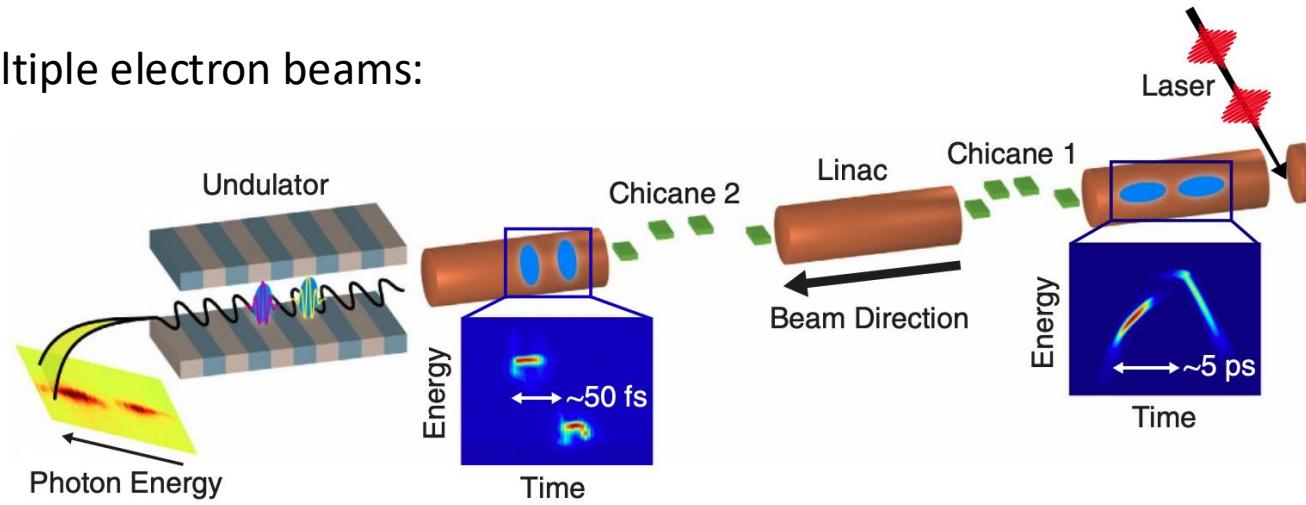
Modern XFEL R&D

Flexible pulse structure

How do we tailor pulse properties (chirp, polarization, shape) for different experiments?

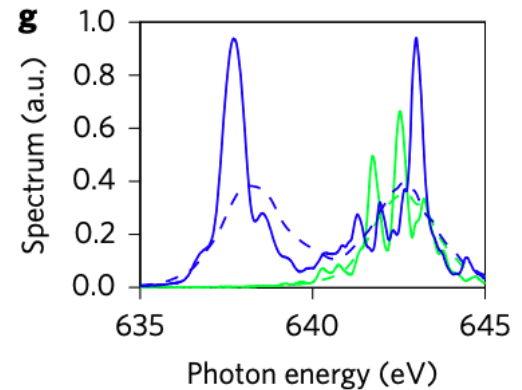
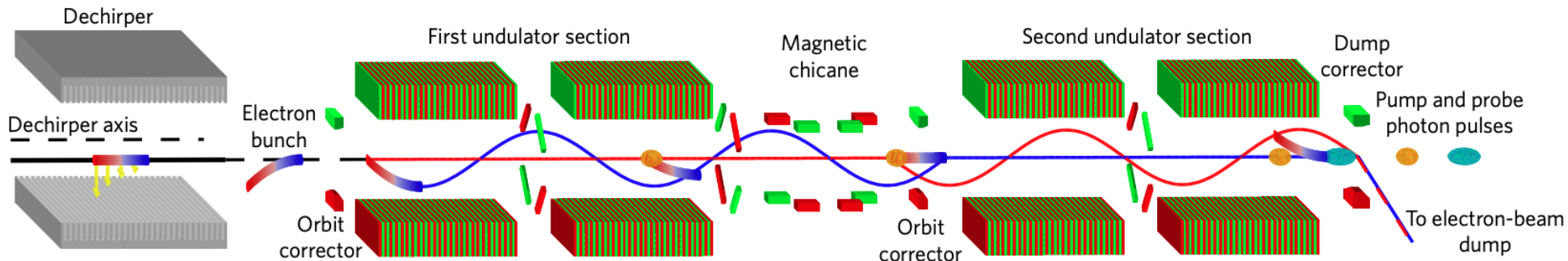
Multiple pulses

By making multiple electron beams:



Marinelli, A., et al. *Nature communications* 6.1 (2015): 6369.

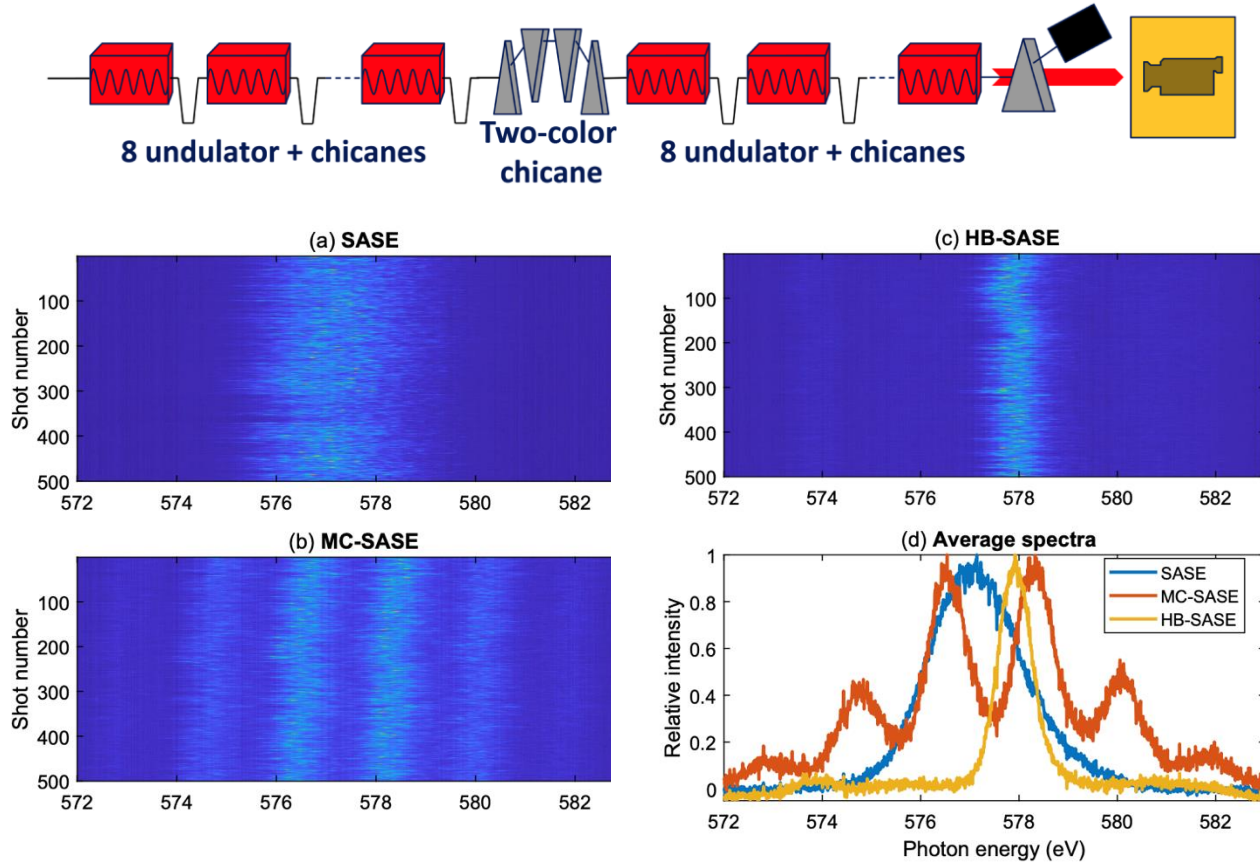
By making the same beam lase multiple times:



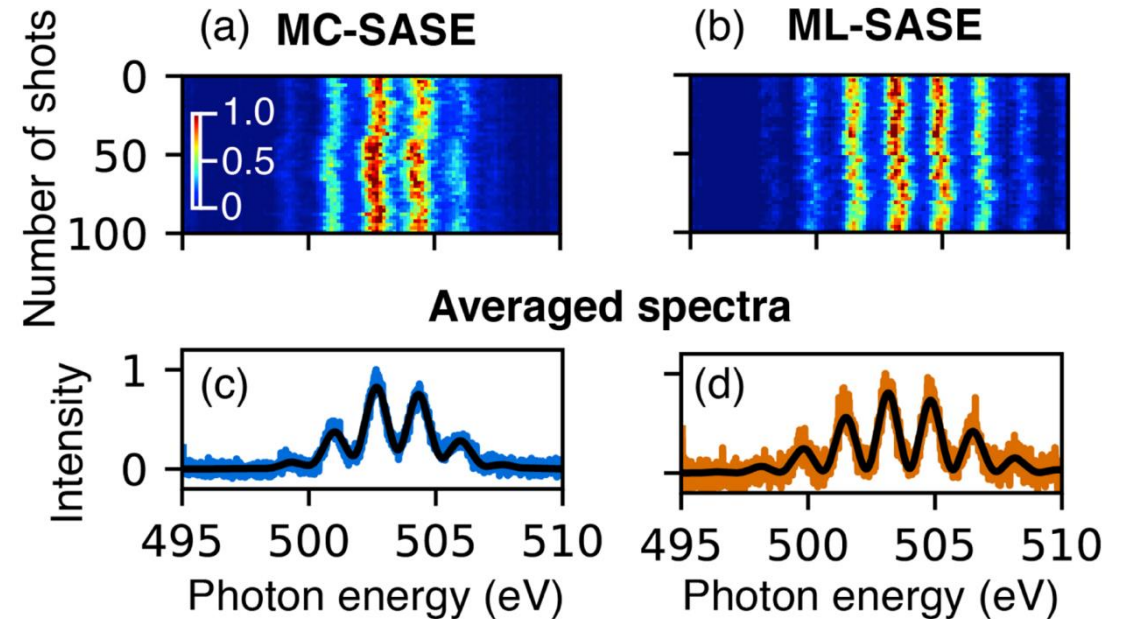
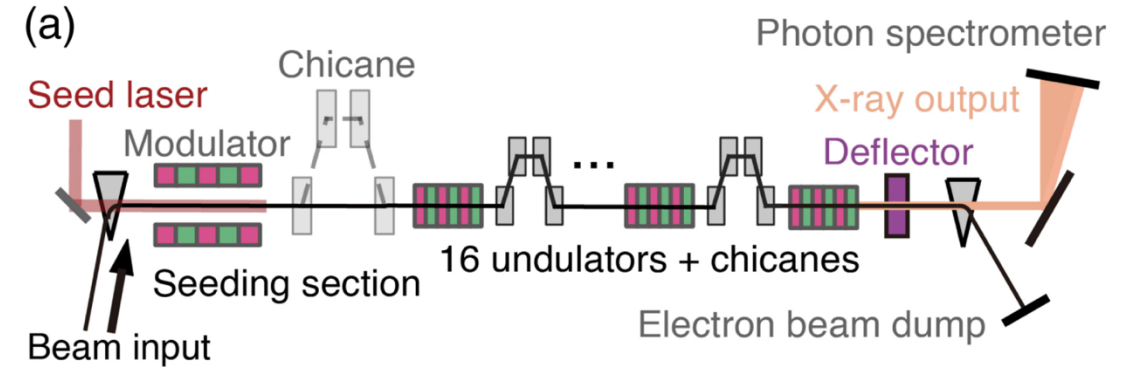
SLAC

Lutman, Alberto A., et al. *Nature Photonics* 10.11 (2016): 745-750.

Tailoring the spectrum of SASE



Prat, Eduard, et al. *Physical Review Letters* 133.20 (2024): 205001.



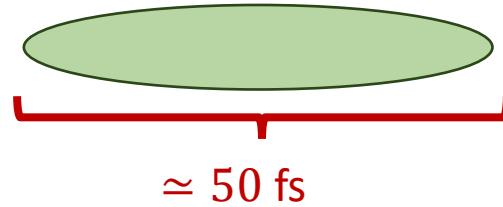
Hu, Wenxiang, et al. *Physical Review Letters* 135.26 (2025): 265001.

Modern XFEL R&D

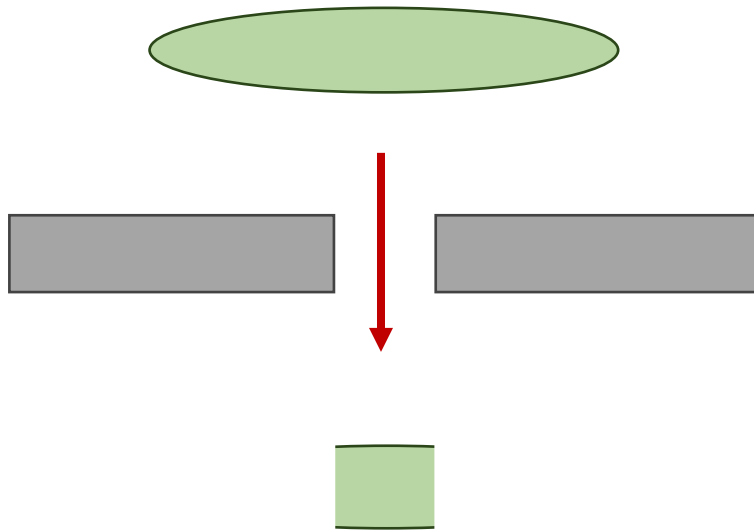
Shorter pulses

How do we make femtosecond
and even attosecond pulse
durations?

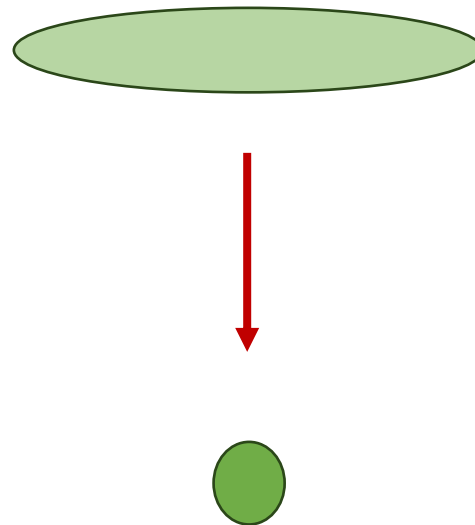
Shorter pulses – schools of thought



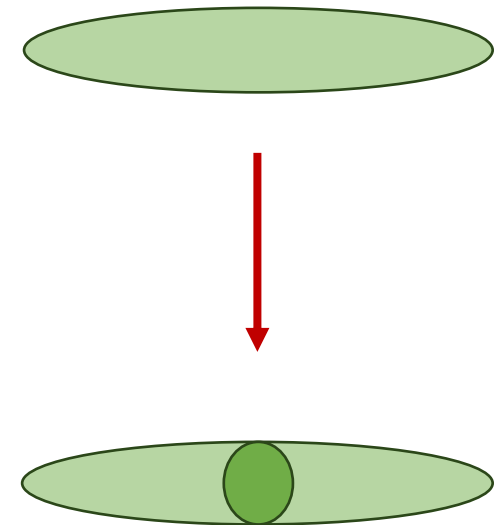
Spoil most of the beam



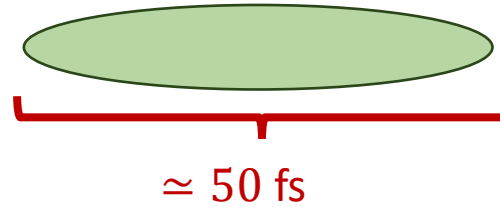
Compress the whole beam



Compress part of the beam

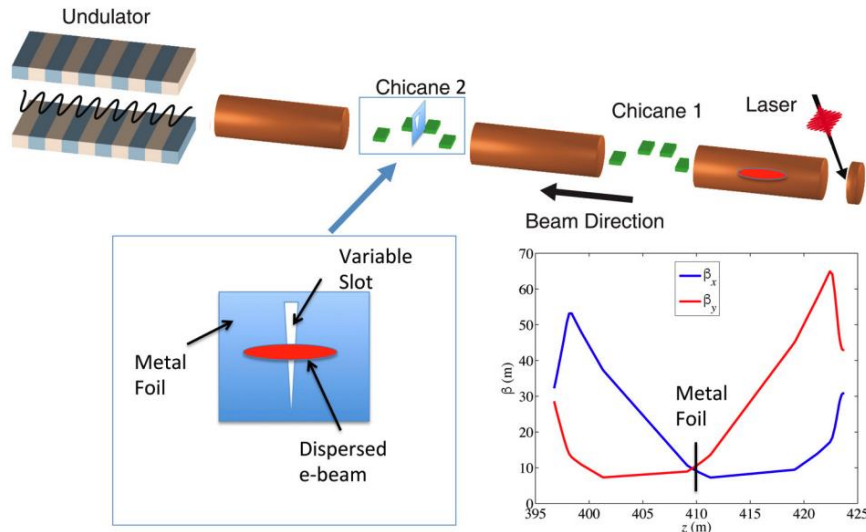


Shorter pulses – schools of thought



Spoil most of the beam

E.g. emittance spoiling, orbit control



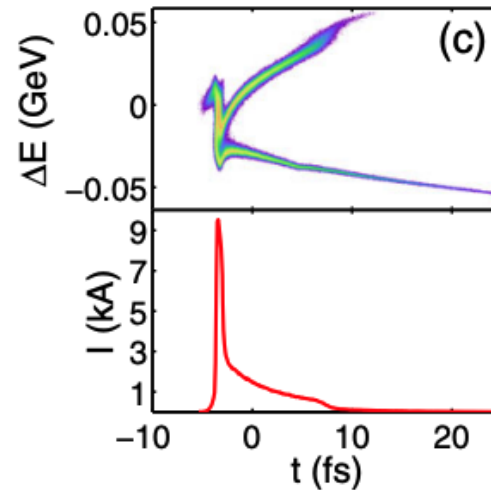
Marinelli, A., et al. *Applied Physics Letters* 111.15 (2017).

Good for <10 fs pulses

SLAC

Compress the whole beam

E.g. nonlinear compression



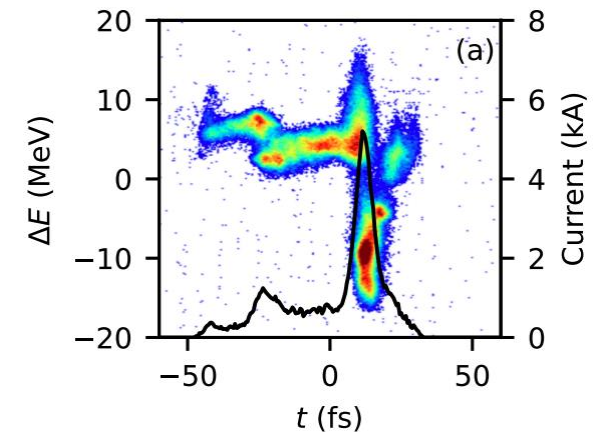
Huang, S., et al. *Physical review letters* 119.15 (2017): 154801.

Good for attosecond pulses

Most common at EuXFEL

Compress part of the beam

E.g. cathode/laser heater shaping



Robles, R. PhD Dissertation (2026).

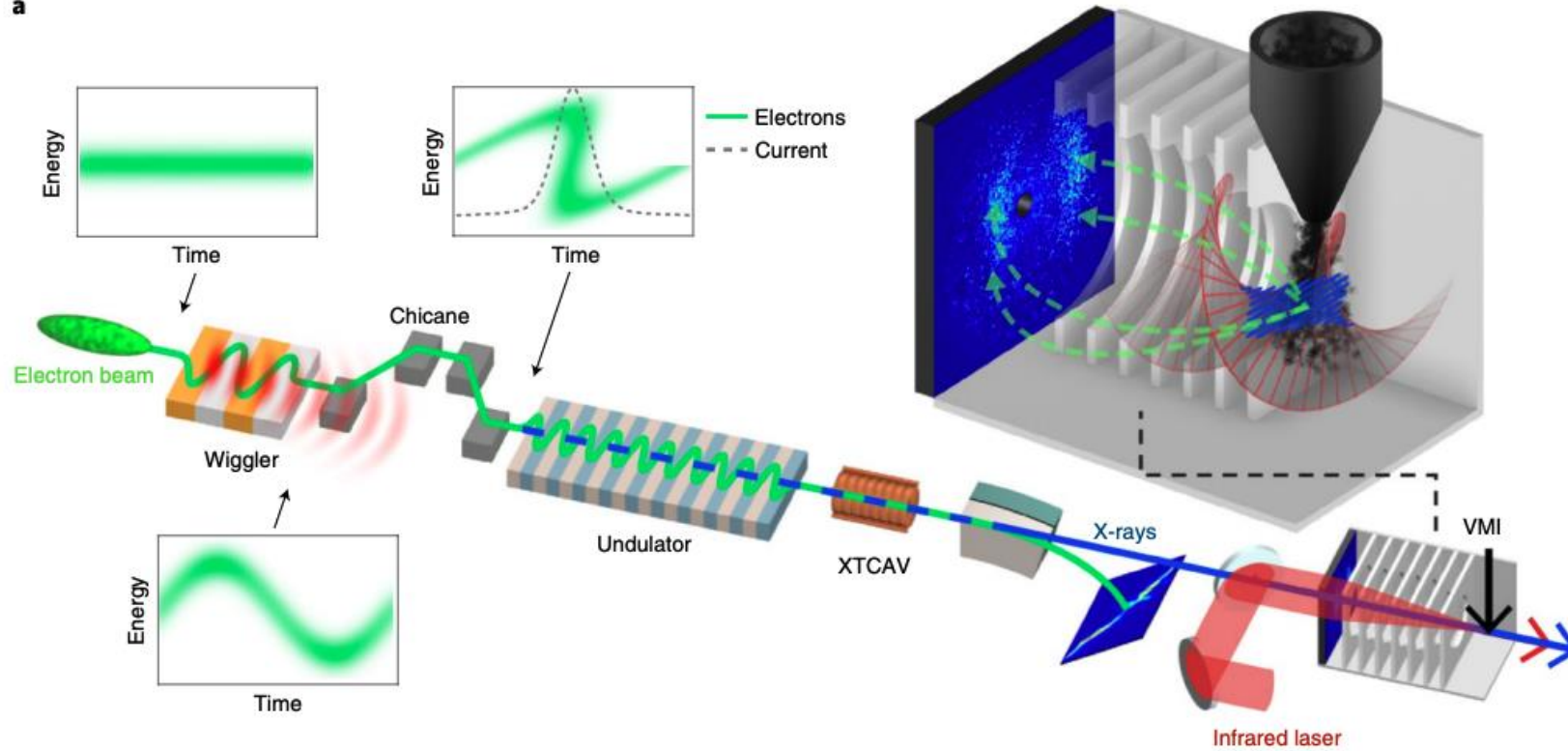
Good for attosecond pulses

Most common at SLAC

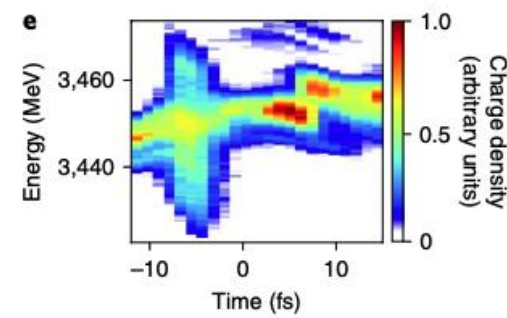
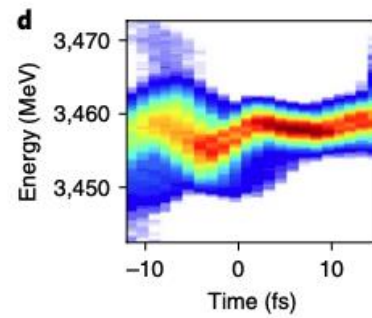
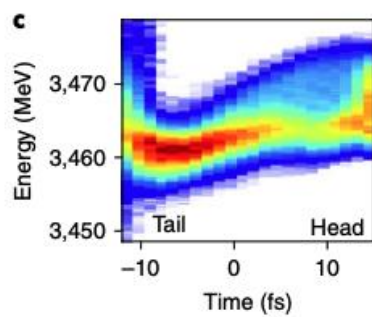
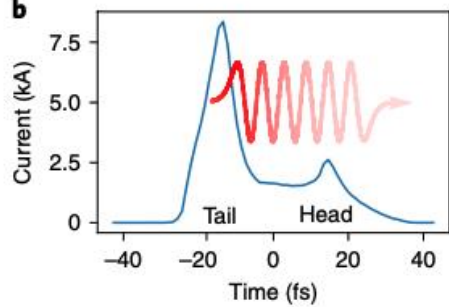
SwissFEL still doing a mix

Attosecond pulses

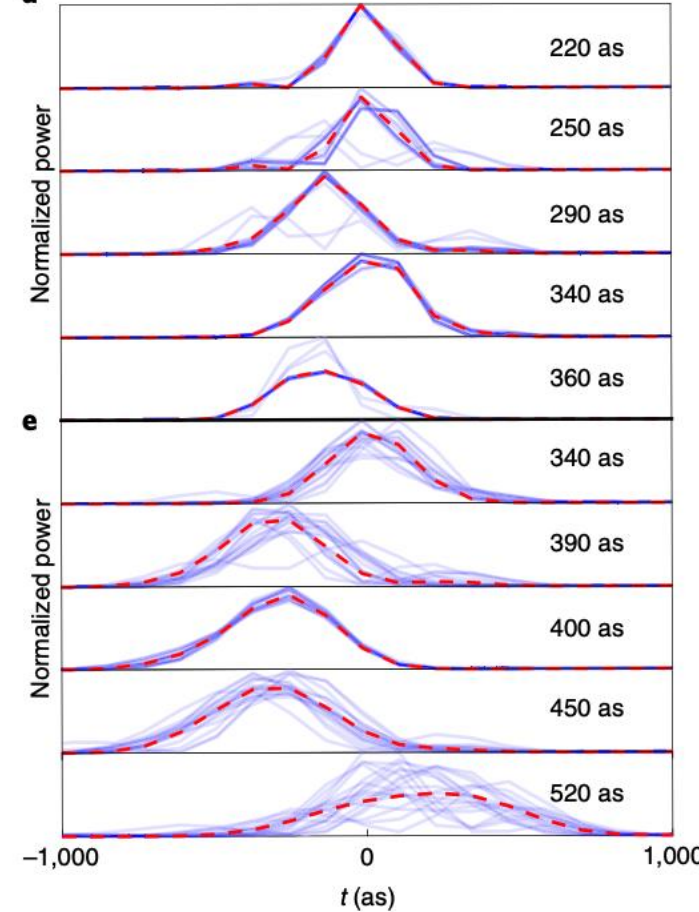
a



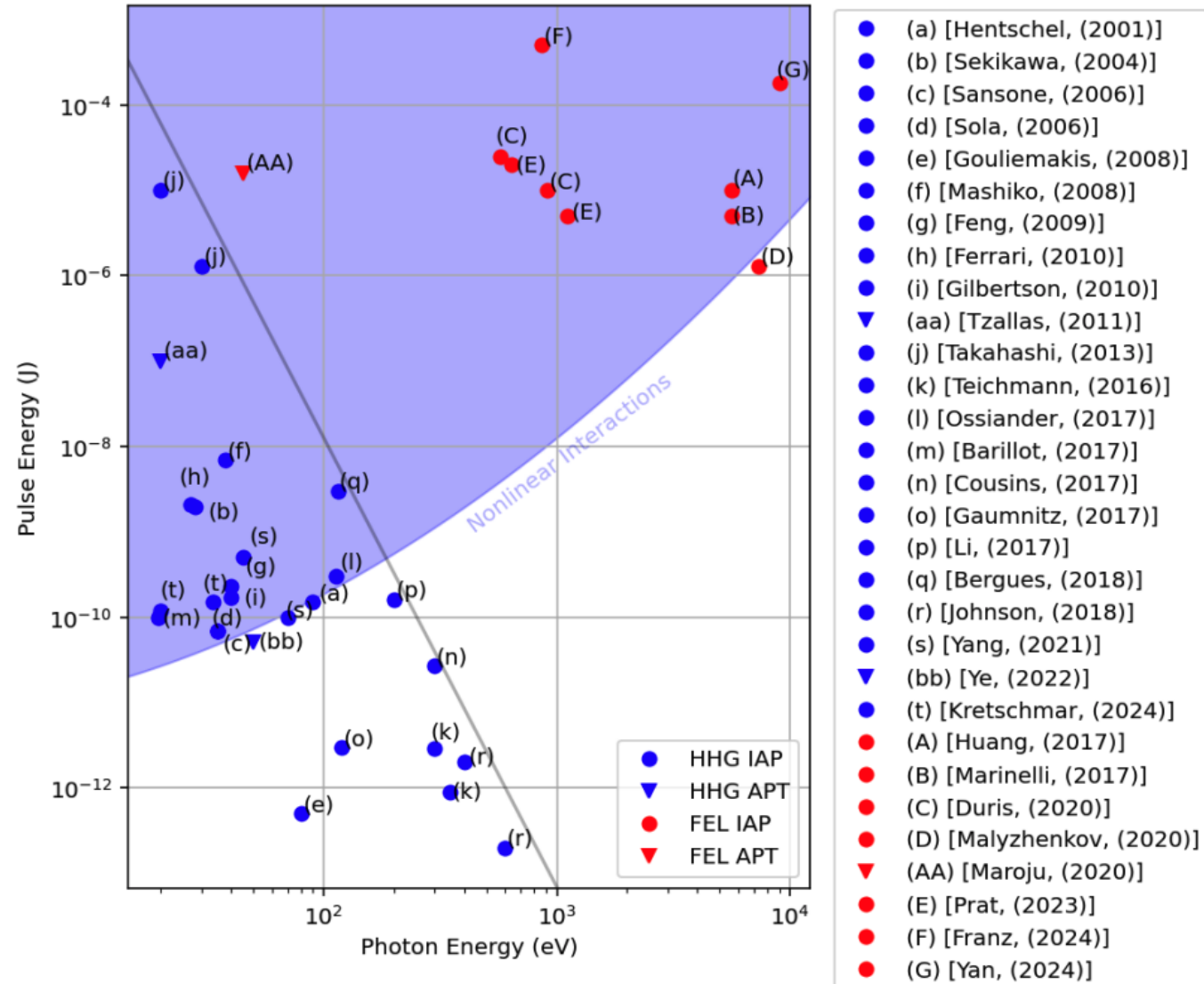
b



d

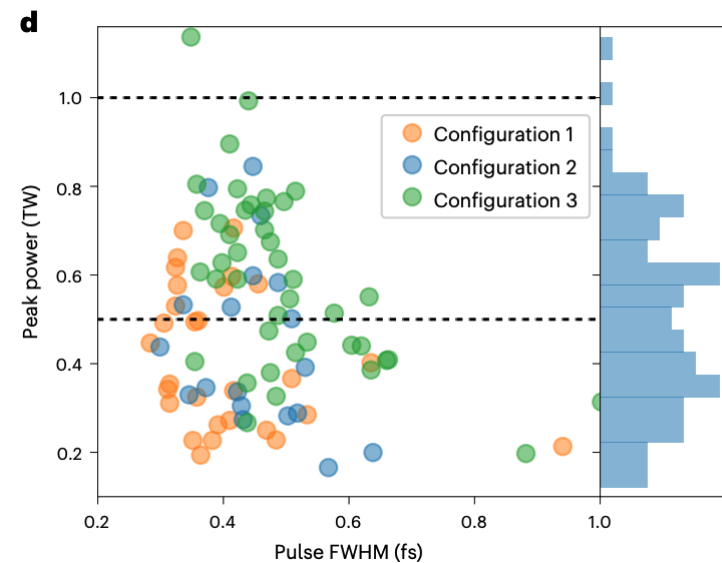


Attosecond pulses



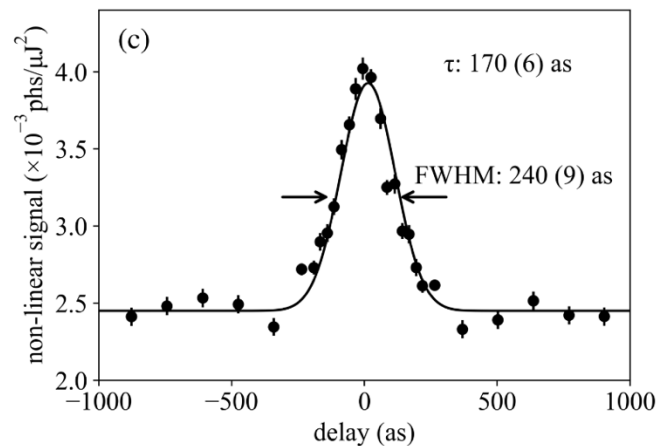
Attosecond pulses: recent results from SLAC

Terawatt-attosecond pulses



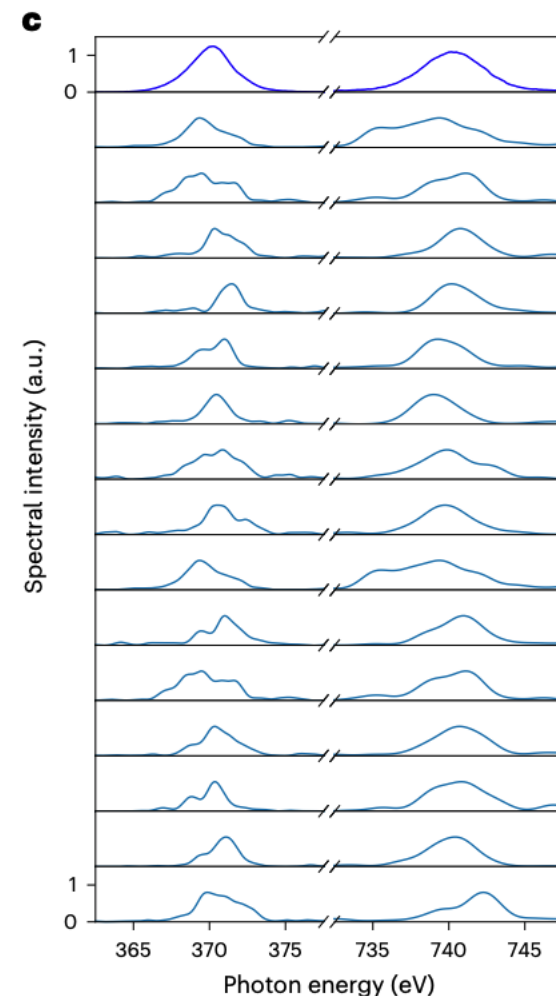
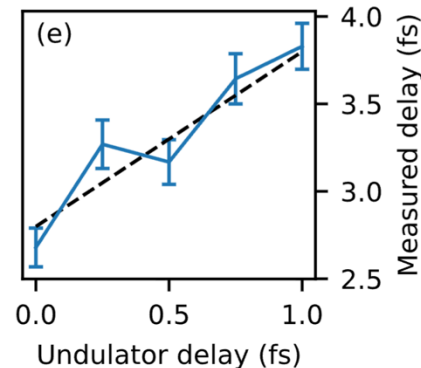
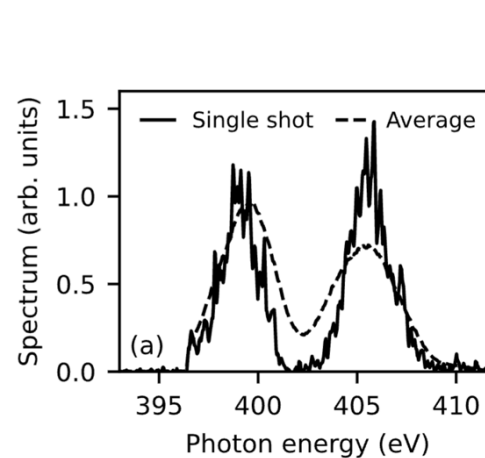
Franz, Paris, et al. *Nature Photonics* 18.7 (2024): 698-703.
Yan, Jiawei, et al. *Nature Photonics* 18.12 (2024): 1293-1298.

Hard X-ray attosecond pulses



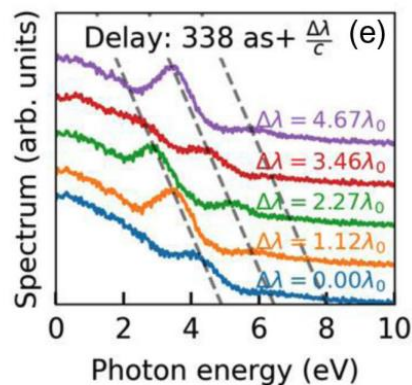
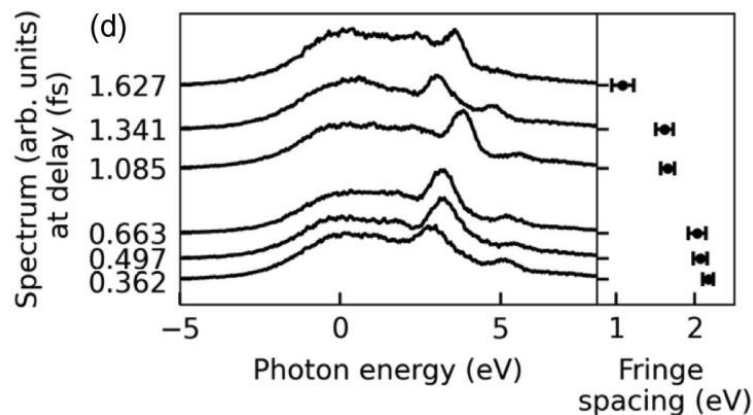
Sun, Y, et al. "Hard X-ray attosecond metrology". Submitted.
Robles, River, et al. *arXiv preprint arXiv:2604.09969* (2026).
Inoue, Ichiro, et al. *arXiv preprint arXiv:2506.07968* (2025).

Attosecond pulse pairs



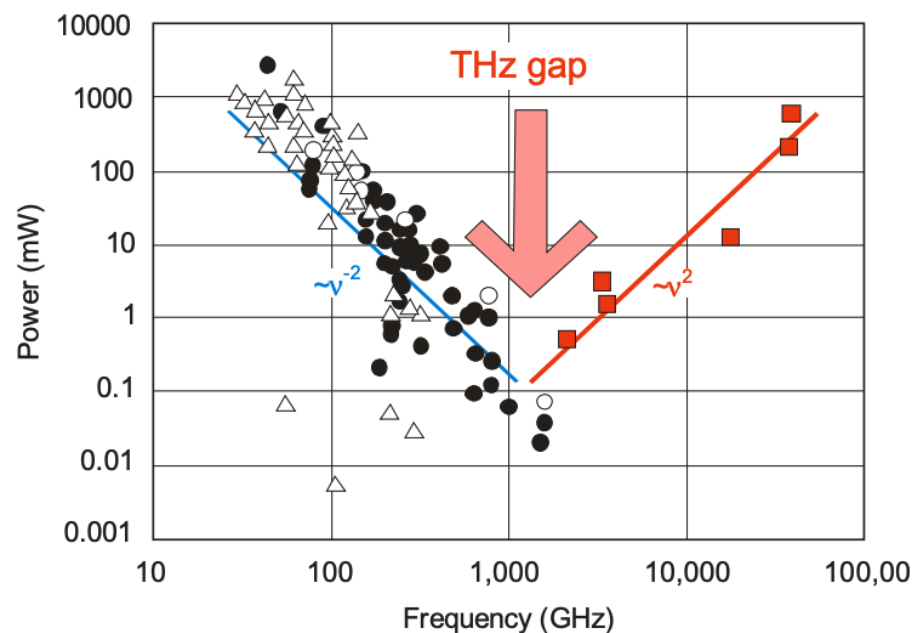
Guo, Zhaoheng, et al. *Nature photonics* 18.7 (2024): 691-697.
Robles, R. et al. In preparation.

Attosecond pulse shaping



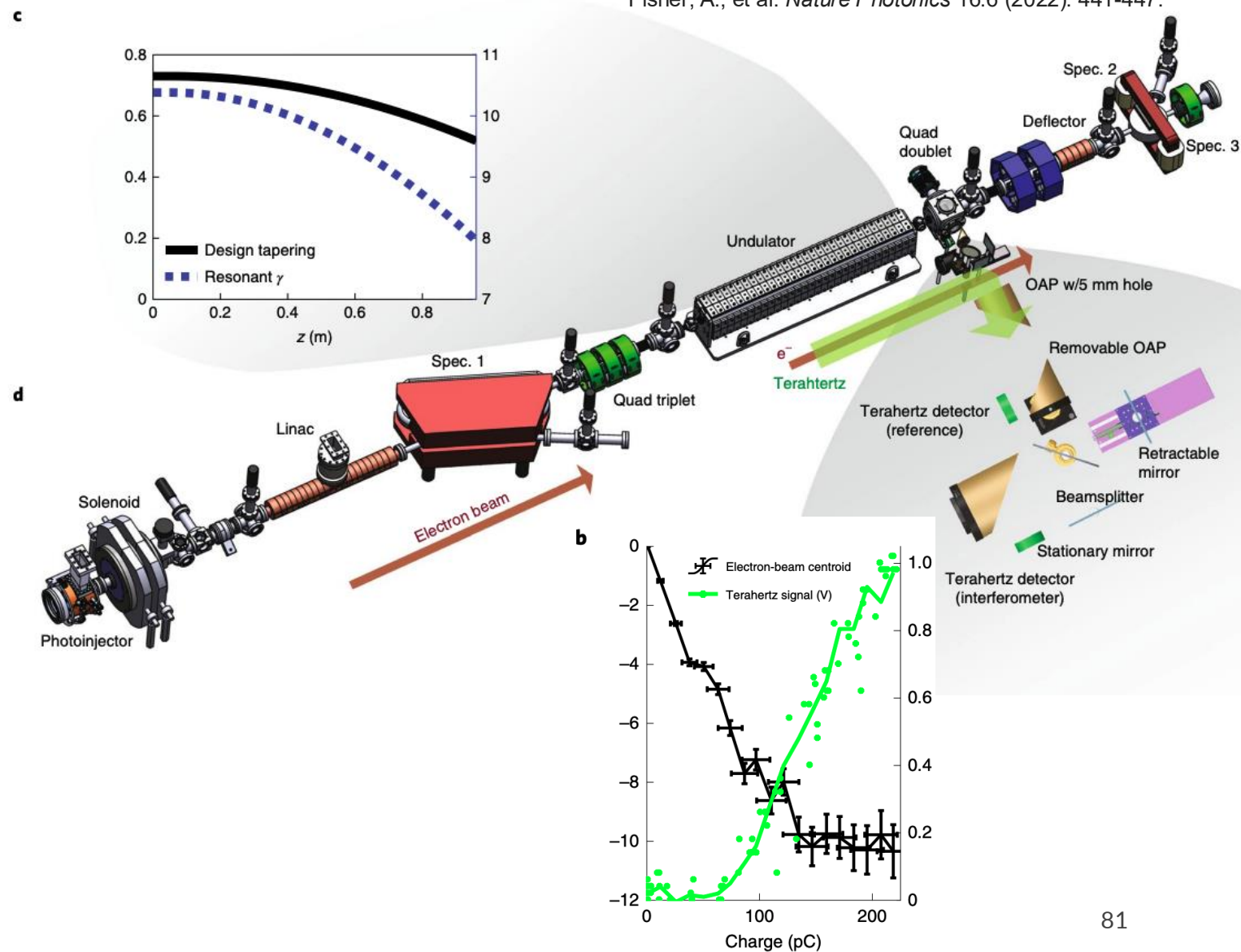
Robles, River R., et al. *Physical Review Letters* 134.11 (2025): 115001.

University-based FEL research: THz FELs



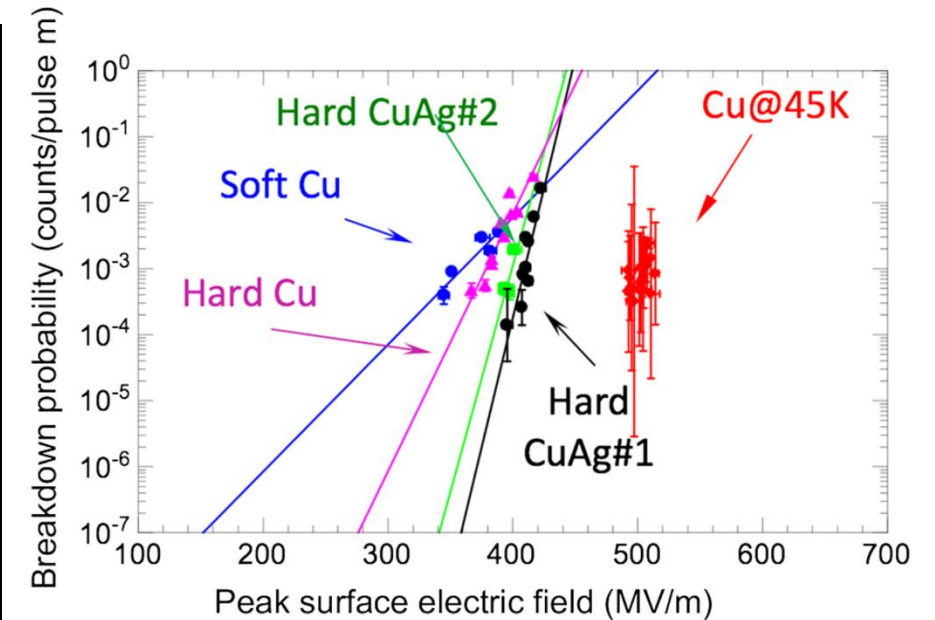
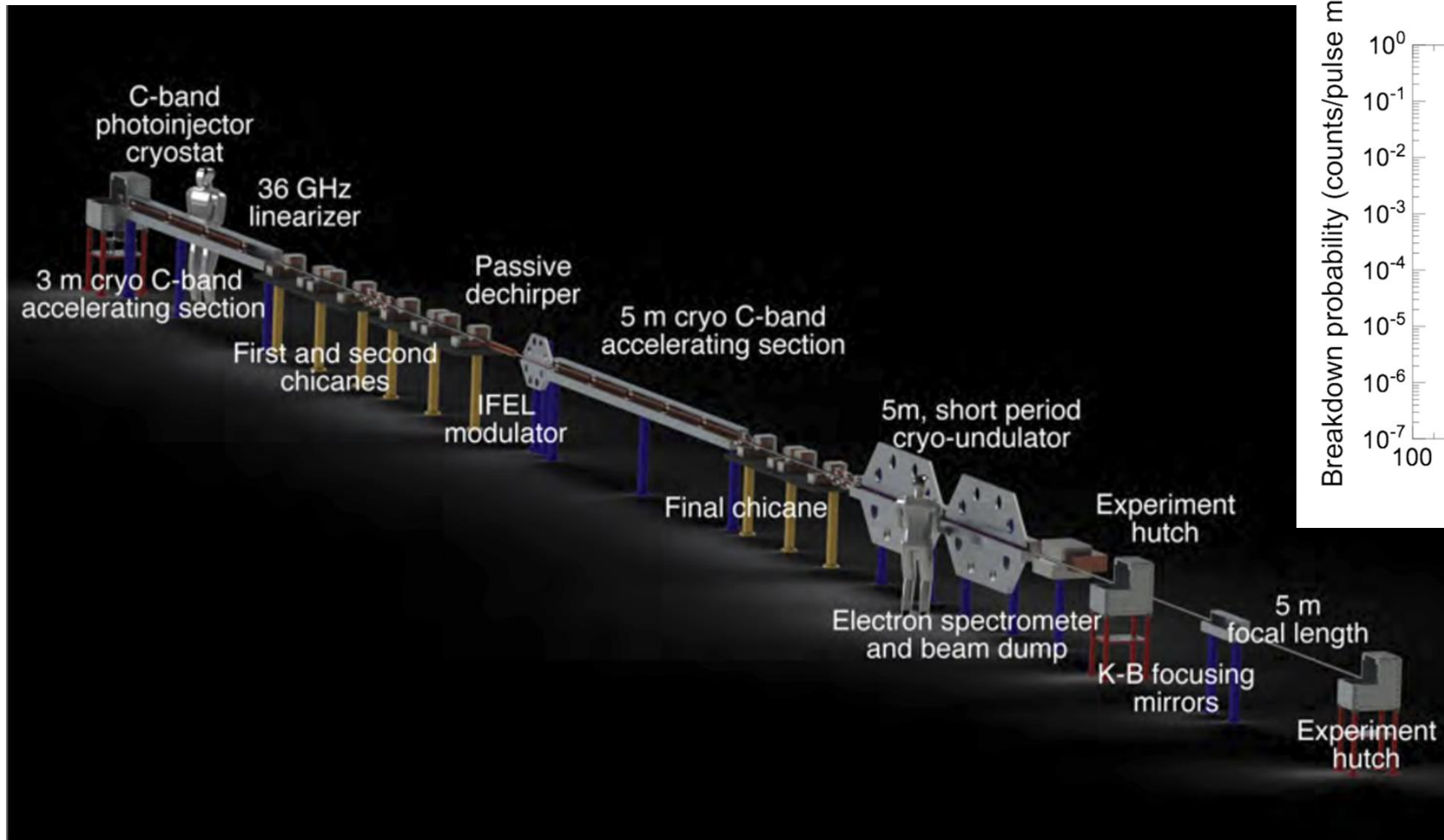
Sizov, F., and A. Rogalski. *Progress in quantum electronics* 34.5 (2010): 278-347.

Fisher, A., et al. *Nature Photonics* 16.6 (2022): 441-447.



University-based FEL research: compact FELs

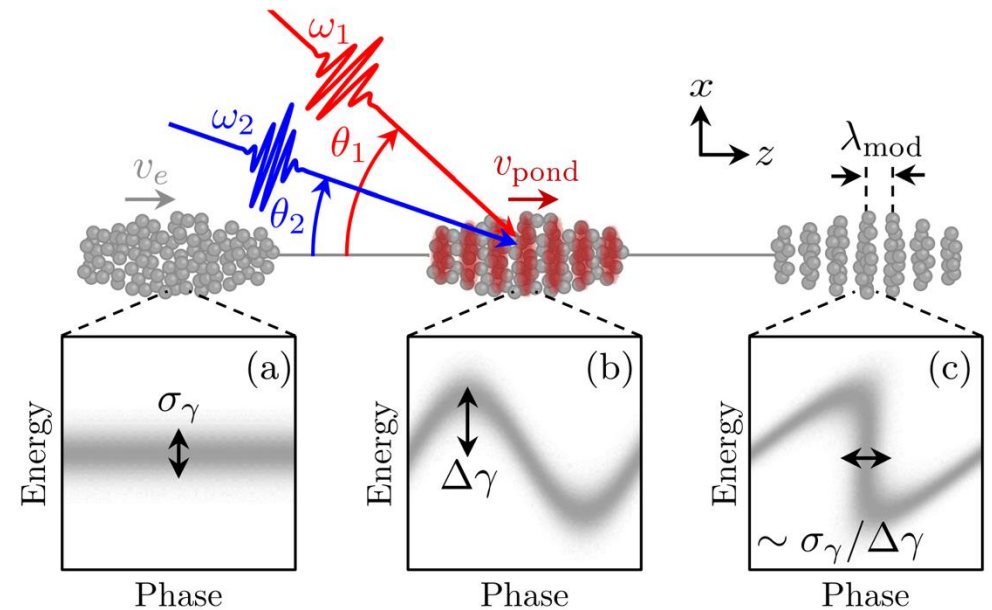
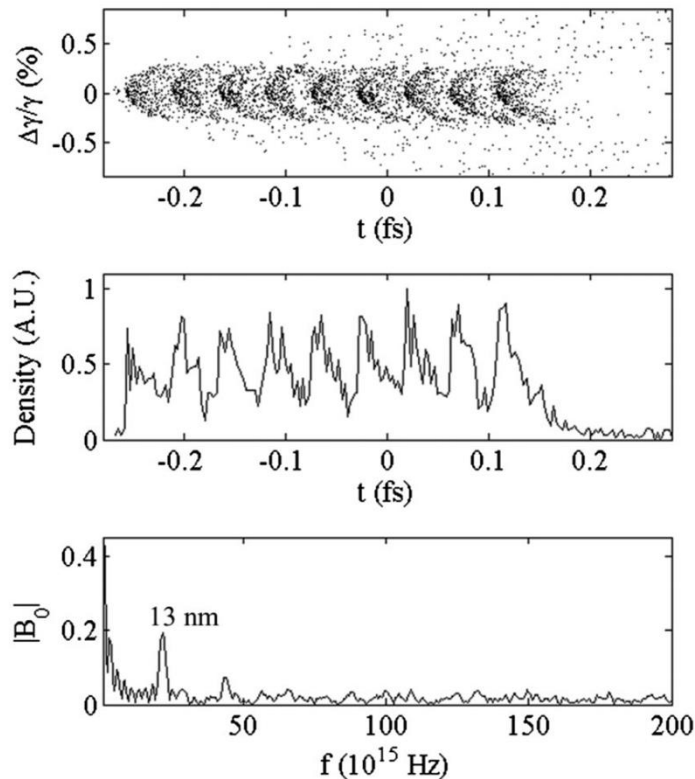
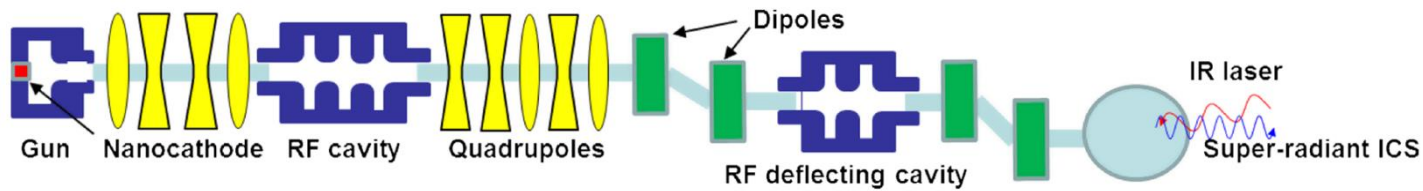
Strategy 1: leverage new technologies and make better beams



Rosenzweig, J. B., et al. *Physical Review Accelerators and Beams* 22.2 (2019): 023403.

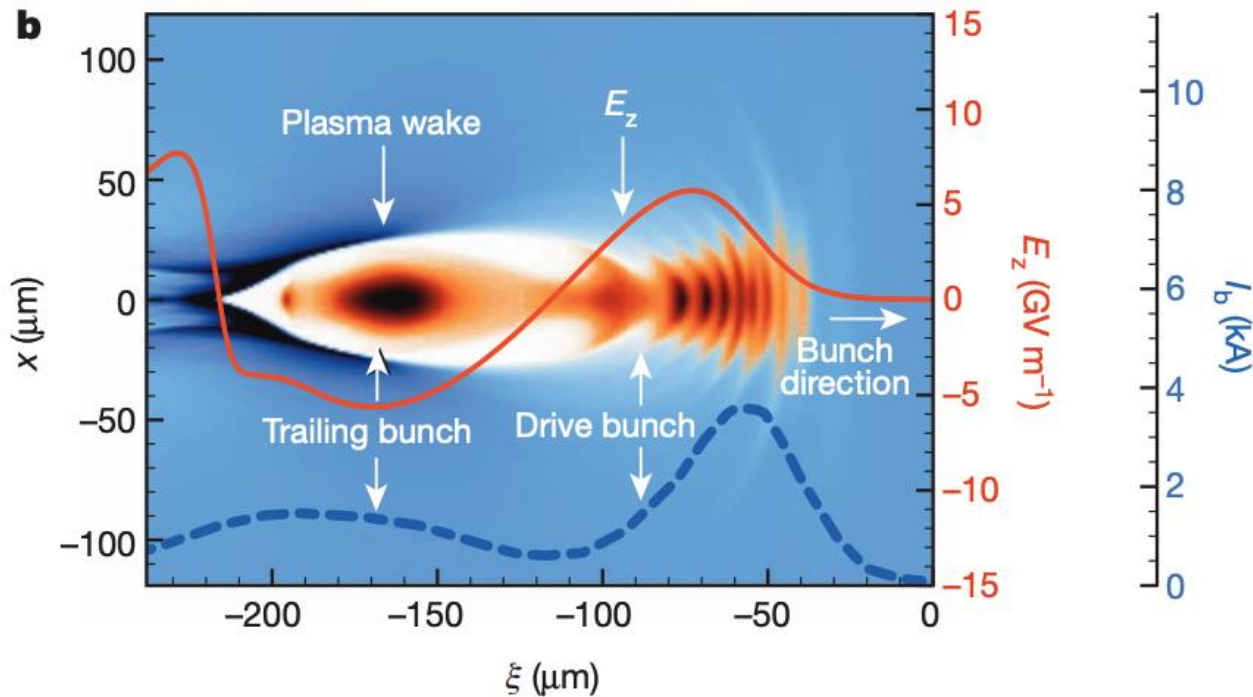
University-based FEL research: compact FELs

Strategy 2: pre-bunch your beam



Schaap, Brian H., et al. *Physical Review Accelerators and Beams* 26.7 (2023): 074401.

Plasma wakefield accelerators (for FELs)



Litos, Michael, et al. *Nature* 515.7525 (2014): 92-95.

Pros:

- 10 GeV beams could be possible in just a few meters
- Self-injected beams can be extremely bright

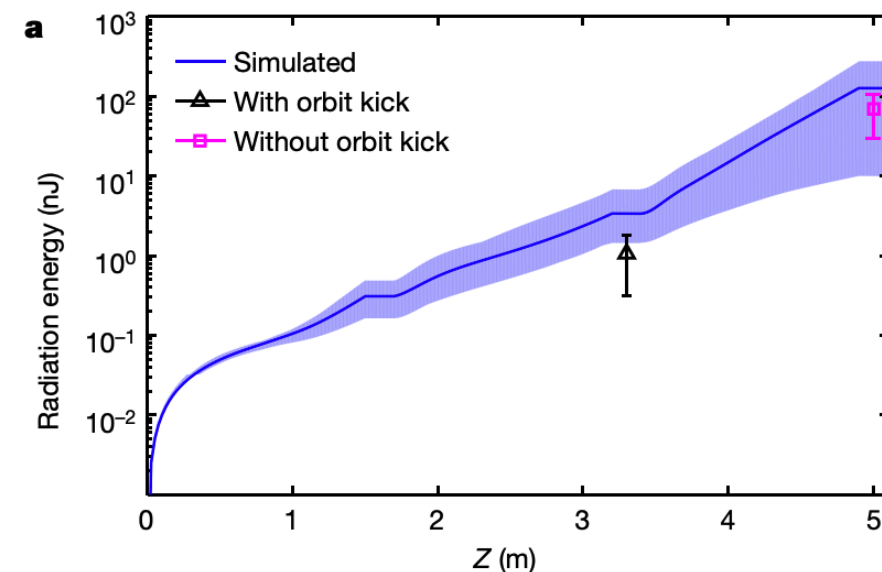
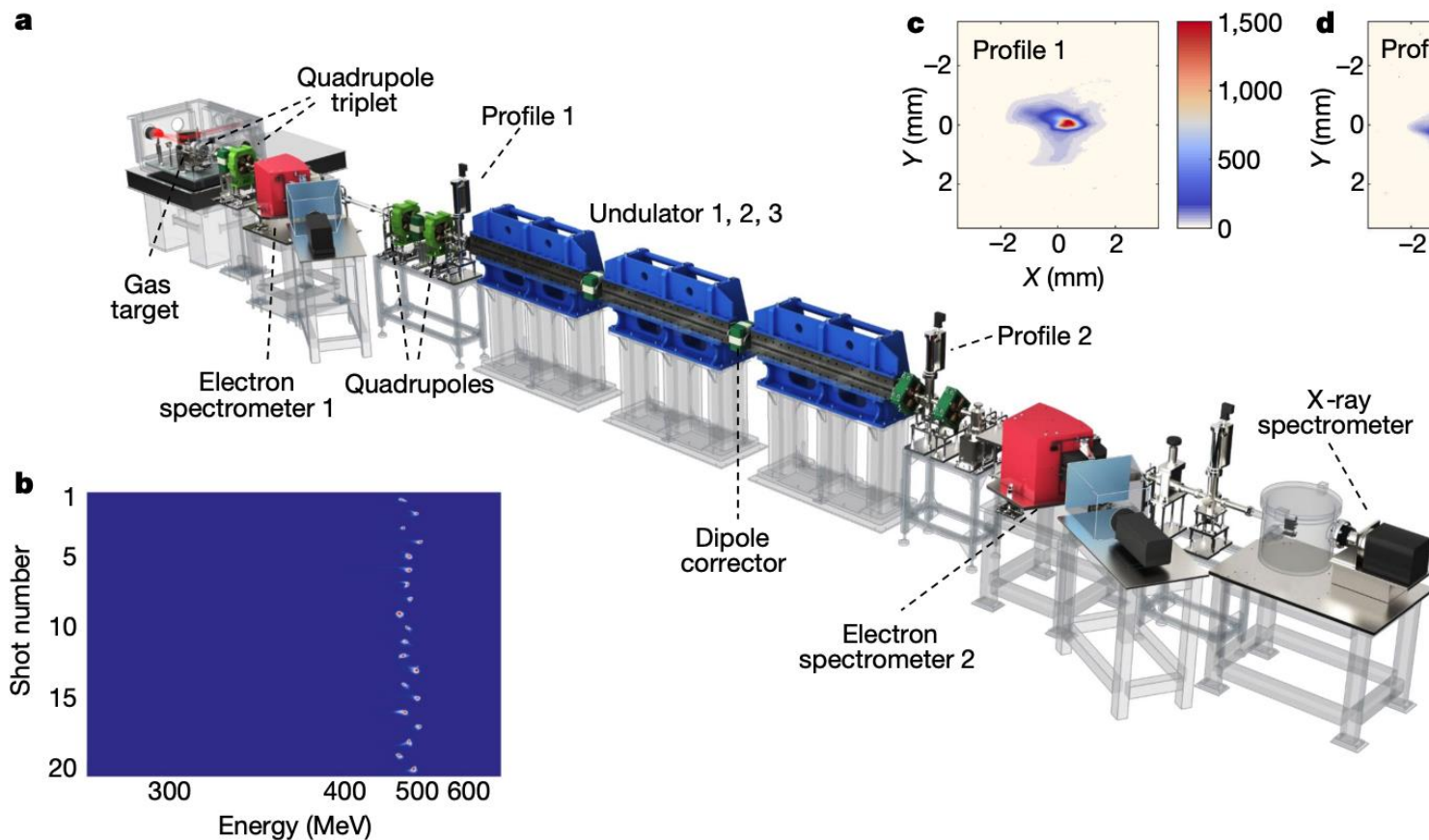
Challenges:

- Large (>% level) energy spread
- Significant shot-to-shot jitter
- Difficult to maintain good emittance
- Hard thing (FEL lasing) + hard thing (PWFA) = very hard thing

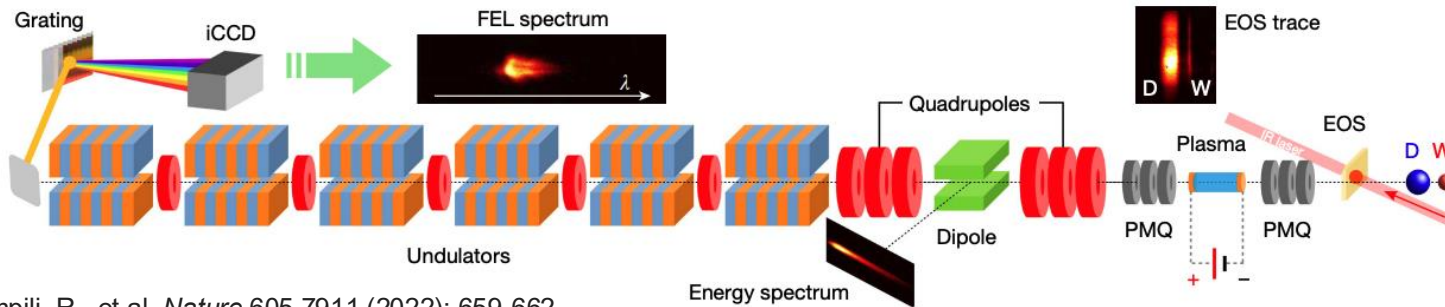


First FEL lasing from a plasma wakefield accelerator

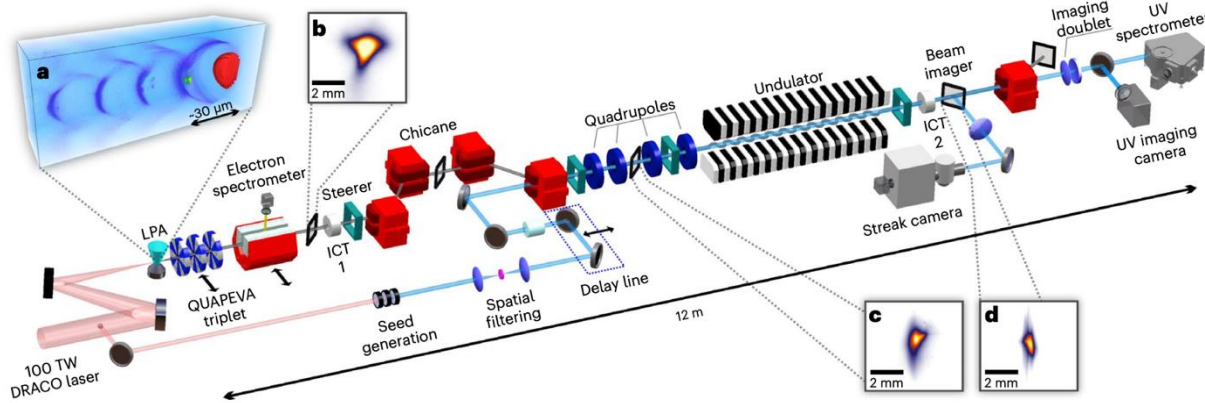
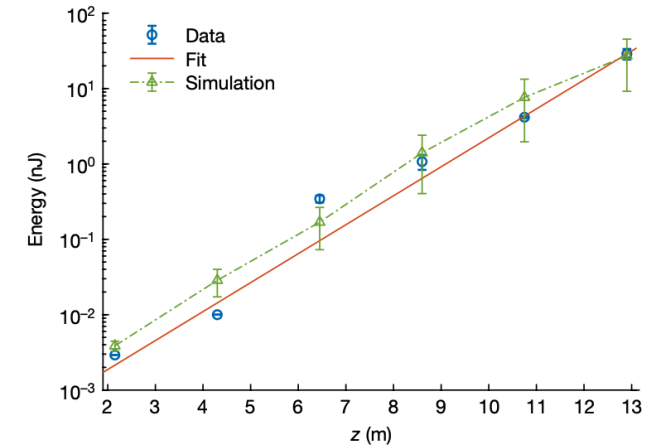
27 nm wavelength – still a ways from soft X-rays



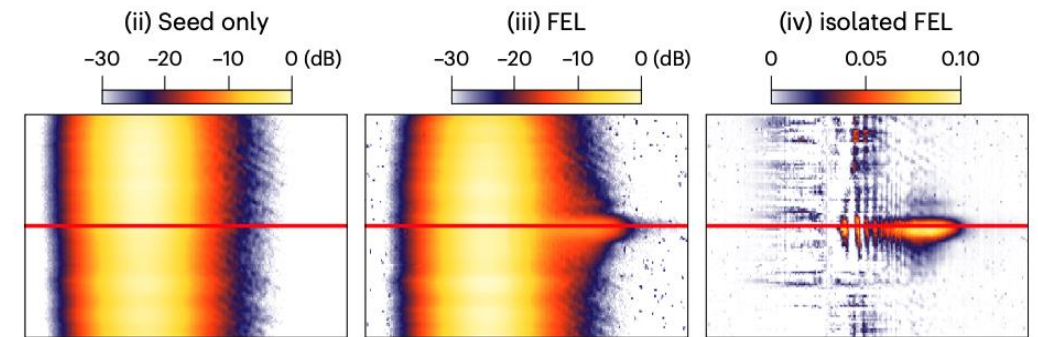
Many results followed soon after



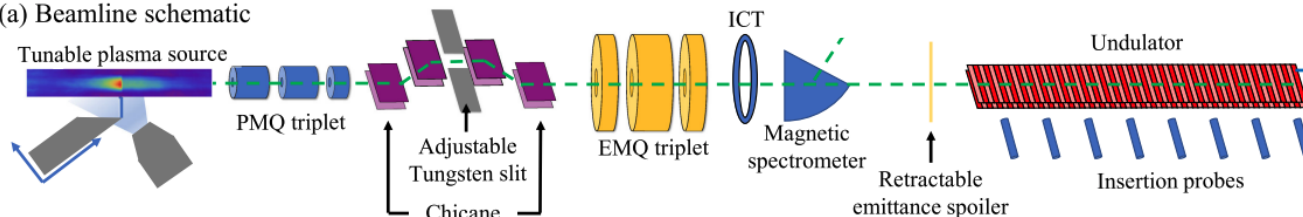
Pompili, R., et al. *Nature* 605.7911 (2022): 659-662.



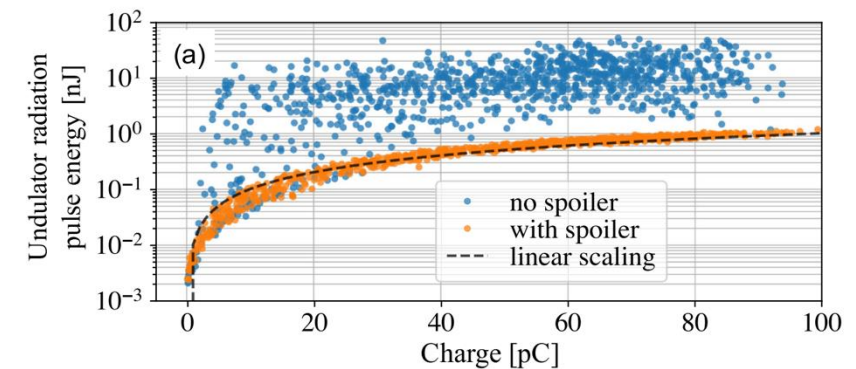
Labat, Marie, et al. *Nature Photonics* 17.2 (2023): 150-156.



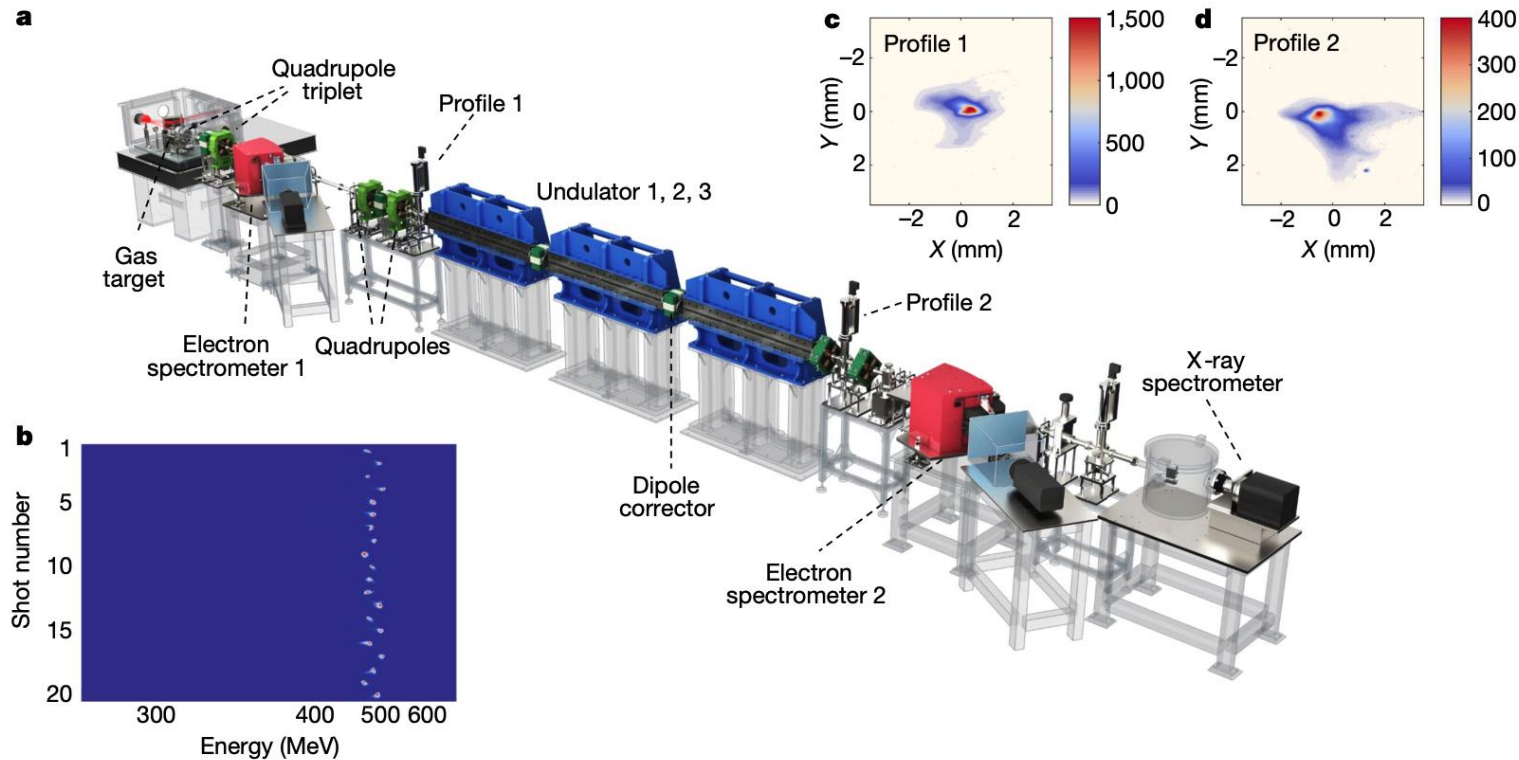
(a) Beamline schematic



Barber, S. K., et al. *Physical Review Letters* 135.5 (2025): 055001.

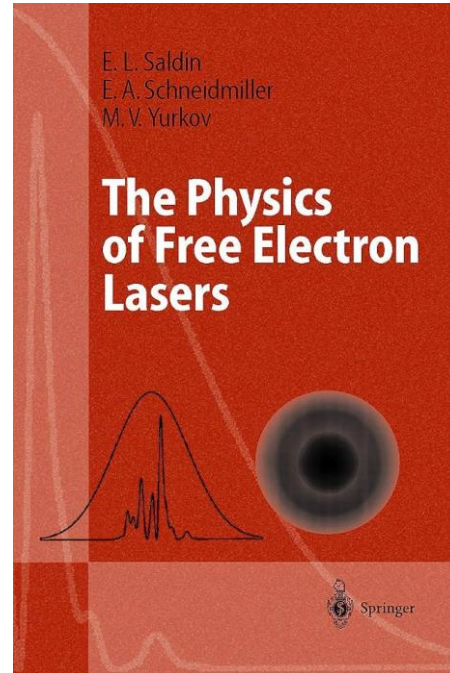
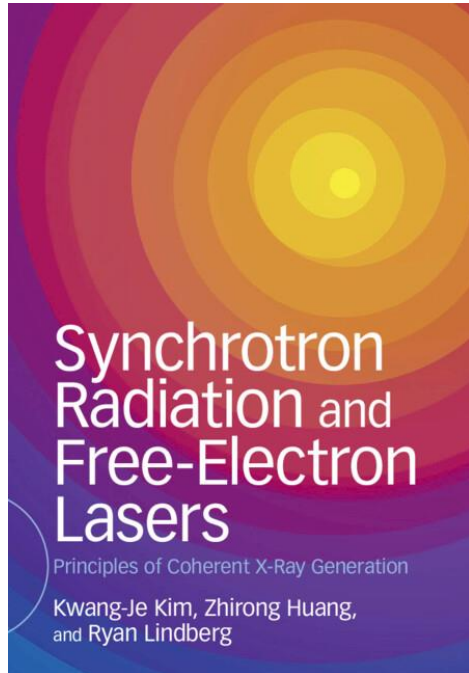


Work still to be done



- No PWFA FEL has reached saturation
- “Long term stability” still measured in hours
- Shortest wavelength still 27 nm
- Proof-of-principle experiment using the radiation not yet done

Further resources



OPEN ACCESS

Review of x-ray free-electron laser theory

[Zhirong Huang](#)

[Kwang-Je Kim](#)

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DOI: [10.1140/epjh/e2012-20064-5](https://doi.org/10.1140/epjh/e2012-20064-5)

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Phys. Rev. ST Accel. Beams **10**, 034801 – Published 12 March, 2007

DOI: <https://doi.org/10.1103/PhysRevSTAB.10.034801>

The history of X-ray free-electron lasers

C. Pellegrini^{1,2,a}

¹ University of California at Los Angeles, Los Angeles, 90095-1547 California, USA

² SLAC National Accelerator Laboratory, Menlo Park, 94025 California, USA

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Published online 19 June 2012

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The physics of x-ray free-electron lasers

[C. Pellegrini](#), [A. Marinelli](#), and [S. Reiche](#)

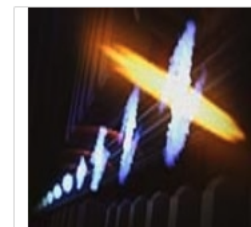
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Rev. Mod. Phys. **88**, 015006 – Published 9 March, 2016

DOI: <https://doi.org/10.1103/RevModPhys.88.015006>

For attosecond XFELs in particular:

Advances in Optics and Photonics Vol. 17, [Issue 3](#), pp. 623-725 (2025) • <https://doi.org/10.1364/AOP.540527>



Attosecond X-ray sources, methods, and applications at present and future free-electron lasers: tutorial

Nora Berrah, James Cryan, River Robles, Taran Driver, Agostino Marinelli, and Philip Bucksbaum

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